MIL-HDBK-414

23 MARCH, 1981

MILITARY HANDBOOK

TECHNICAL CONTROL FACILITIES AND EQUIPMENT FOR LONG HAUL COMMUNICATIONS

VOLUME 1 OF 2 VOLUMES

SLHC

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DEPARTMENT OF DEFENSE
WASHINGTON DC

MIL-HDBK-414
Technical Control Facilities and Equipment for Long Haul Communications
23 March 1981

1. This standardization handbook was developed by the Department of Defense in accordance with established procedures.

2. This document provides basic and fundamental information on equipment application and characteristics for technical control facilities and equipment for long haul communications. It will provide valuable information and guidance to personnel concerned with the preparation of specifications and the procurement of electrical and electronic equipment for the Defense Communication System. The handbook is not intended to be referenced in purchase specifications except for informational purposes, nor shall it supersede any specification requirements.

3. Every effort has been made to reflect the latest information on the interrelation of considerations for technical control facilities and equipment for long haul communications. It is the intent to review this handbook periodically to insure its completeness and currency. Beneficial comments (recommendations, additions, deletions) and pertinent data which may be used in improving this document should be addressed to: HQ AFCC/EPES, Scott AFB, IL 62225.

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FOREWORD

This volume is one of a two-volume series which sets forth the Technical Control Facilities and Equipment for Long Haul Communications. The subjects are interrelated and involve considerations of wide range of topics from DC Equipment Patch Facilities, miscellaneous equipments, including alarms, power supplies and intercommunications and orderwire equipments. This series of volumes reduces these varied considerations into a usable set of principles and practices which can be used by all concerned with and responsible for the safety and effective operations of complex electronic systems. Where possible, the principles are reduced to specific steps. Because of the inherently large number of interrelated factors, specific steps cannot be set forth for every possible situation. However, once the requirements and constraints of a given situation are defined, the appropriate steps for solutions of the problem can be formulated utilizing the principles set forth.

The second volume describes detailed test procedures for verifying the electrical characteristics of equipment used in Technical Control or Patch and Test Facilities. As such, the second volume complements the first by providing a common baseline for parameter definition.

VOLUME II CONTAINS:
2A--Paragraphs 1-11
2B--Paragraphs 12-21
2C--Paragraphs 22-23

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Chapter I

EQUIPMENT APPLICATION AND ELECTRICAL CHARACTERISTICS

1. INTRODUCTION.

1.1 PURPOSE. This handbook provides pertinent information regarding the application and electrical characteristics of equipment employed in Technical Control or Patch and Test Facilities in the Defense Communication System. Electrical requirements which are common to more than one equipment category are discussed in Section 2. The remainder of the Handbook is divided into seven sections, each of which includes specific application information and electrical characteristics for a single category of equipment items.

1.2 APPLICABILITY. This handbook is intended for use as a technical information guide to Technical Control or Patch and Test Facility equipments. As such, this handbook provides an informational description of functions and characteristics of equipment referenced in MIL-STD-188-310, Subsystem Design and Engineering Standards for Technical Control Facilities.

1.3 REFERENCED DOCUMENTS.

1.3.1 Military Standards.

MIL-STD-188-100	Common Long Haul and Tactical Communication System Technical Standards
MIL-STD-188-114	Electrical Characteristics of Digital Interface Circuits
MIL-STD-188-310	Subsystem Design and Engineering Standards for Technical Control Facilities

1.3.2 Handbook.

MIL-HDBK-232	Facility Design for RED/BLACK Application

1.3.3 Federal Standard.

FED-STD 1037	Vocabulary for Telecommunications

1.3.4 Other Publications.

DCA 310-50-6

DCAC 370-V165-1

DCAC 370-V175-6

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1.4 TERMS AND DEFINITIONS. Terms and definitions applicable to this handbook are located in Federal-Standard 1037, Vocabulary for Telecommunications.

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Chapter II

2. GENERAL EQUIPMENT REQUIREMENTS.

2.1 SCOPE AND APPLICABILITY. General requirements which are applicable to a number of different types of equipment are discussed in this section for the purpose of avoiding unnecessary repetition of detailed requirements in the following sections of this appendix. The applicability of material appearing in this section is indicated by specific paragraph references in Sections 3 through 9. In such cases, the referenced paragraphs shall be regarded as a mandatory part of the individual equipment requirement unless specifically stated to the contrary.

2.2 SURGE PROTECTION.

2.2.1 All technical control or patch and test facility equipment designed to interface with external transmission lines should be capable of withstanding voltage and current surges of magnitude commonly experienced at the out of the primary protectors.

2.2.2 For design and testing purposes, both metallic and longitudinal surges should be assumed to rise to a peak of 1000 volts (positive or negative) at a rate of 500 volts per microsecond and then decay to half voltage in 1000 microseconds. Repeated application of either metallic or longitudinal surges at the equipment-to-transmission-line interface shall not expose personnel engaged in operating the equipment to electrical or thermal hazards and shall not result in permanent deviation from equipment characteristics as stated in Sections 3 through 9.

2.3 POWER SOURCES.

2.3.1 All equipment intended for use in Technical Control Facilities, should be designed to require minimum input power. This is especially important for VF circuit conditioning equipment (Section 3), DC circuit equipment (Section 4) and VF circuit signaling equipment (Section 5).

2.3.2 Equipment requiring external power shall be designed to operate from one or more of the following power sources (facility supplies):

a. 120, 220 or 240 V ac.

b. -48 V dc.

c. $\pm$6 V dc.

2.3.3 All equipment performance requirements stated in Sections 3 through 9 shall be satisfied with power source characteristics (at the equipment interface) given in Sections 2.3.5.1(a), 2.3.6.1, and 2.3.7.1.

2.3.4 In cases in which an intermediate power supply (regulator, inverter, etc) is used

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between a facility supply and one or more pieces of equipment, all such equipments plus the intermediate power supply shall be considered a subsystem and all equipment performance requirements shall be satisfied with the power source characteristics given in Sections 2.3.5.1(a), 2.3.6.1, and 2.3.3.7.1 applicable at the subsystem-to-facility supply interface. Similarly, the requirements of Sections 2.3.5.1(b), 2.3.6.1, and 2.3.7.1 will apply at the subsystem-to-facility supply interface with the entire subsystem considered as a single piece of equipment.

2.3.5 Equipment utilizing 120, 220 or 240 V ac power.

2.3.5.1 Equipment which derives its power from the facility 120 V ac distribution system or from a commercial 120, 220, or 240 V ac source shall satisfy all performance requirements under the following supply source conditions at the equipment-to-facility supply interface:

a. Voltage regulation: ±10 percent.

b. Frequency: 47 to 63 Hz.

c. Waveform:

(1) Maximum deviation factor: 6 percent.

(2) Maximum individual harmonic: 3 percent.

d. Noise: 1.0 V peak-to-peak maximum at all frequencies other than the supply frequency.

e. Source impedance: Up to 0.1 ohm resistance in series with 120 uH inductance.

f. Single phase except 48 volt rectifiers (see 9.2.1.2.1).

2.3.5.2 All equipment connected to the facility ac distribution system shall conform to the following requirements:

a. Equipment generated noise: The equipment shall not generate noise (at frequencies other than the supply frequency) with amplitude greater than 0.2 V peak-to-peak at the equipment power input terminals when operated from a noiseless ac power source with an output impedance comprising of 0,1 ohm resistance in series with 120 uH inductance.

b. Transient conditions: The equipment shall be capable of withstanding deviations from nominal line voltage of up to ±30 percent for periods up to 4 seconds without damage and without exhibiting unstable behavior (e.g., spurious oscillation).

2.3.6 Equipment utilizing -48 V dc power.

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2.3.6.1 Equipment which derives its power from the facility -48 V dc supply shall satisfy all performance requirements under the following supply source conditions at the equipment-to-facility supply interface:

a. Voltage regulations: ±10 percent.

b. Ripple and noise: 200 mV peak-to-peak maximum at all frequencies.

c. Source impedance: Up to 0.1 ohm resistance in series with 120 uH inductance.

2.3.6.2 All equipment connected to the facility -48 dc V distribution system shall conform to the following requirements:

a. Equipment generated noise: The equipment shall not generate noise with amplitude greater than 40 mV peak-to-peak at the equipment power input terminals when operated from a noiseless -48 V dc power supply with an output impedance consisting of a 0.1 ohm resistance in series with a 120 uH inductance.

b. Reversed polarity protection: The equipment shall sustain no damage if connected to the power supply with reversed polarity.

2.3.7 Equipment utilizing +6 V dc power. It is intended that the ±6 V dc facility source be used only for low-level data, clock timing, and mark-hold applications where the minimum load resistance for any single loop is 6000 ohms.

2.3.7.1 Equipment that utilizes the facility ±6 V dc supply shall satisfy all performance requirements under the following supply source conditions at the equipment-to-facility supply interface:

a. Voltage regulation: ±10 percent.

b. Ripple and noise: 2 mV peak-to-peak maximum at all frequencies.

c. Source impedance: up to 0.1 ohm resistance in series with 120 uH inductance.

2.3.7.2 All equipment connected to the facility ±6 V dc distribution system shall conform to the following requirements:

a. Equipment generated noise: The equipment shall not generate noise with amplitude greater than 1 mV peak-to-peak at the equipment power input terminals when operated from a noiseless ±6 V dc power supply with an output impedance consisting of a 0.1 ohm resistance in series with 120 uH inductance.

b. DC current: 1 mA, maximum, for any single keying loop.

c. Reversed polarity protection: The equipment shall sustain no damage if connected to the power supply with reversed polarity.

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2.4 VF TRANSMISSION-LINE IMPEDANCE.

2.4.1 Unless otherwise indicated, equipment which must interface with voice-frequency (VF) transmission lines shall be designed for a VF transmission-line impedance which is design 600 ohms ±10% balanced to ground and nonreactive. It must, however, be recognized that the actual impedance of a voice-frequence transmission line may deviate substantially from the nominal value. In such cases, it is essential that the equipment stability and linearity not be significantly degraded. The nominal line impedance does, nevertheless, provide an acceptable basis for specifying many equipment characteristics which are adequately and conveniently defined for a fixed, nonreactive line impedance.

2.4.2 For design and testing purposes, the range of VF transmission-line impedance over which equipment performance requirements shall be satisfied as follows unless other conditions are specifically stated:

2.4.2.1 Equipment noise and harmonic distortion requirements shall be satisfied when the magnitude of the VF transmission-line impedance is between 50 percent and 200 percent of the design value with a phase angle ranging from -45° to $+45^{\circ}$ C. (Note: VF impedances are 600 ohms for 4-wire lines and 900 ohms for 2-wire lines.)

2.4.2.2 Spurious oscillation or regenerative instability shall not occur nor shall any form of permanent performance degradation occur as a result of extreme transmission-line impedance deviations ranging from short-circuit to open-circuit and from pure inductance to pure capacitance.

2.4.2.3 All other equipment performance requirements (including input and output impedance, return loss, longitudinal balance, gain or attenuation accuracy or adjustability, frequency response, envelope delay, etc) shall be satisfied when the VF transmission-line impedance is equal to the design value (600 or 900 ohms) at all relevant transmission line interfaces.

2.5 LONGITUDINAL BALANCE AND LONGITUDINAL-INPUT-TO-METALLIC-OUTPUT-VOLTAGE RATIO.

2.5.1 Interference caused by differences in ground potential or by coupling to adjacent power and communication circuits is an important consideration in the design of noncoaxial telephone cable transmission systems. The required insensitivity to this type of interference is commonly achieved by constructing the transmission facility in such a manner that the two conductors of each transmission circuit have nearly equal distributed coupling to all significant sources of interference and to ground. If the transmission circuit is perfectly balanced, ground currents and induced interference appear as purely longitudinal signals; i.e., as identical potentials on the two conductors and as identical currents flowing in the two conductors toward the same end of the transmission line. In this case, the interfering signal is totally distinguishable from the transmitted signal which ideally appears as a purely metallic signal; i.e., as the potential difference between the two conductors and as equal currents flowing in opposite directions through the two conductors.

2.5.2 If the transmission circuit balance is imperfect at any point in the system, the interfering signals on the two conductors will differ to some extent. The result is an

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interfering signal which is partially metallic and therefore not totally distinguishable from the transmitted signal. Because the longitudinal interference level is often high, a relatively small transmission circuit unbalance can seriously degrade the system output signal-to-noise ratio. It is for this reason that stringent balance requirements must be imposed on all equipment which may be inserted into a cable transmission circuit.

2.5.3 Equipment balance specifications. Equipment balance requirements are commonly expressed in terms of longitudinal balance specifications. A precise interpretation of longitudinal balance specifications is possible only when the applicable measurement procedures are specified. The longitudinal balance requirements appearing in this handbook are based on the measurement technique illustrated in figure 1 and described briefly in 2.5.2.

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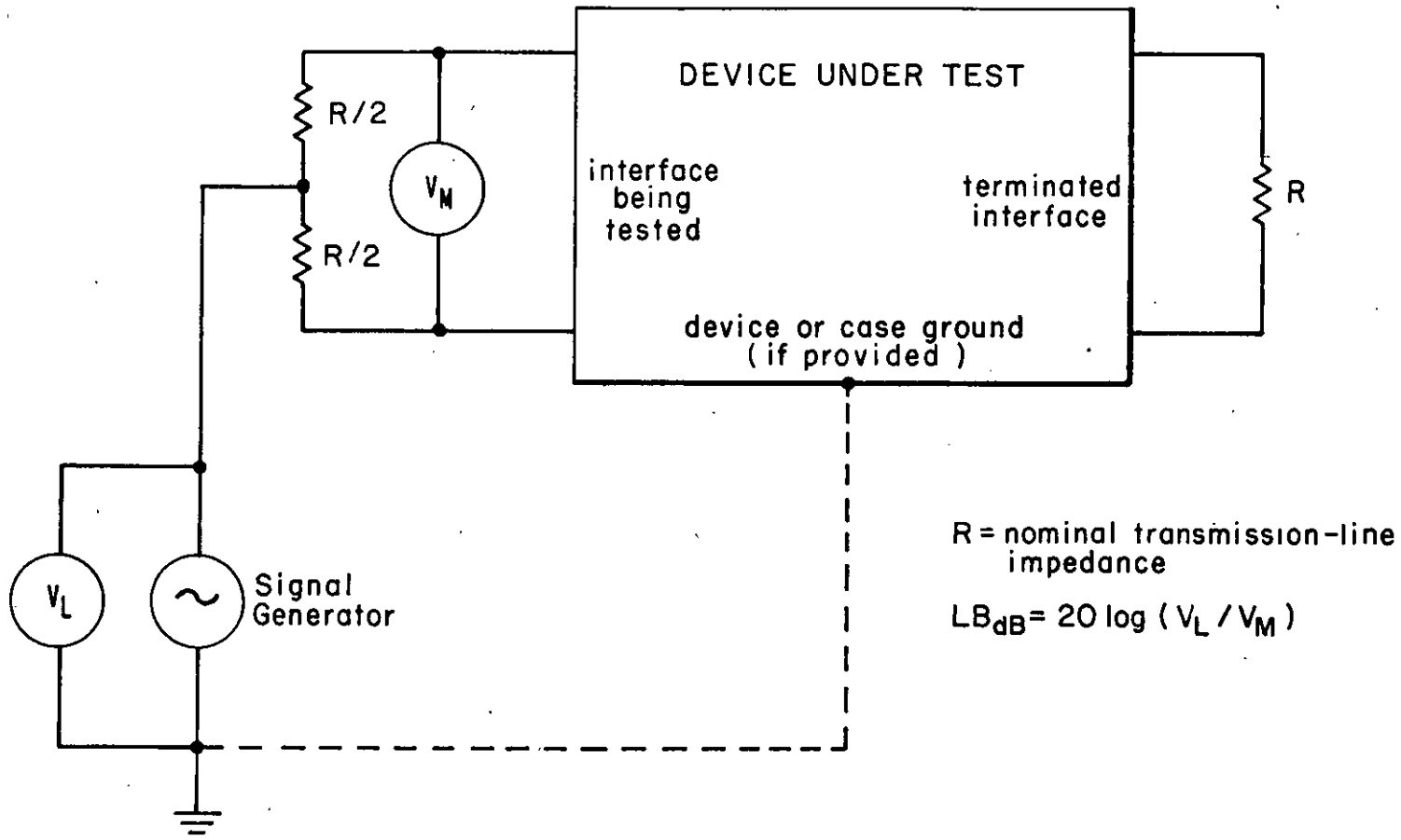


Figure 1. Basic Circuit for Measuring Longitudinal Balance.

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2.5.3.1 Longitudinal balance measurements (as specified in 2.5.4) are sensitive only to the internal impedance balance between ground and the transmission line terminals of the equipment. In some cases, the internal coupling between two sets of transmission line terminals is also an important equipment balance consideration. This is clearly illustrated, for example, in the case of an ungrounded resistive attenuator network which is inserted directly into a transmission line. Longitudinal signals appearing on the two conductors of the transmission circuit are coupled from one interface to the other by the series resistors of the attenuator network. If the corresponding resistances in the two conduction paths are not matched, the coupling between transmission-line interfaces becomes unbalanced, and the longitudinal balance of the transmission line would be degraded by the resistance imbalance in the attenuator network. However, the longitudinal balance of the attenuator by itself would not appear to be degraded because the attenuator network provides no internal ground return path for a longitudinal test signal. When a longitudinal signal is unintentionally introduced into an internal ground-referenced circuit by a deficient coupling device at one of the equipment-to-transmission-line interfaces, unwanted sensitivity to external longitudinal signals may also occur. Examples of such a case could be a poorly shielded coupling transformer or a differential input amplifier with a low common-mode rejection factor. For the purpose of restricting this type of unbalance, a minimum longitudinal-input-to-metallic-output-voltage ratio is specified as part of the appropriate equipment standards within this handbook. These specifications are based on the measurement technique illustrated in figure 2 and described briefly in 2.5.5.

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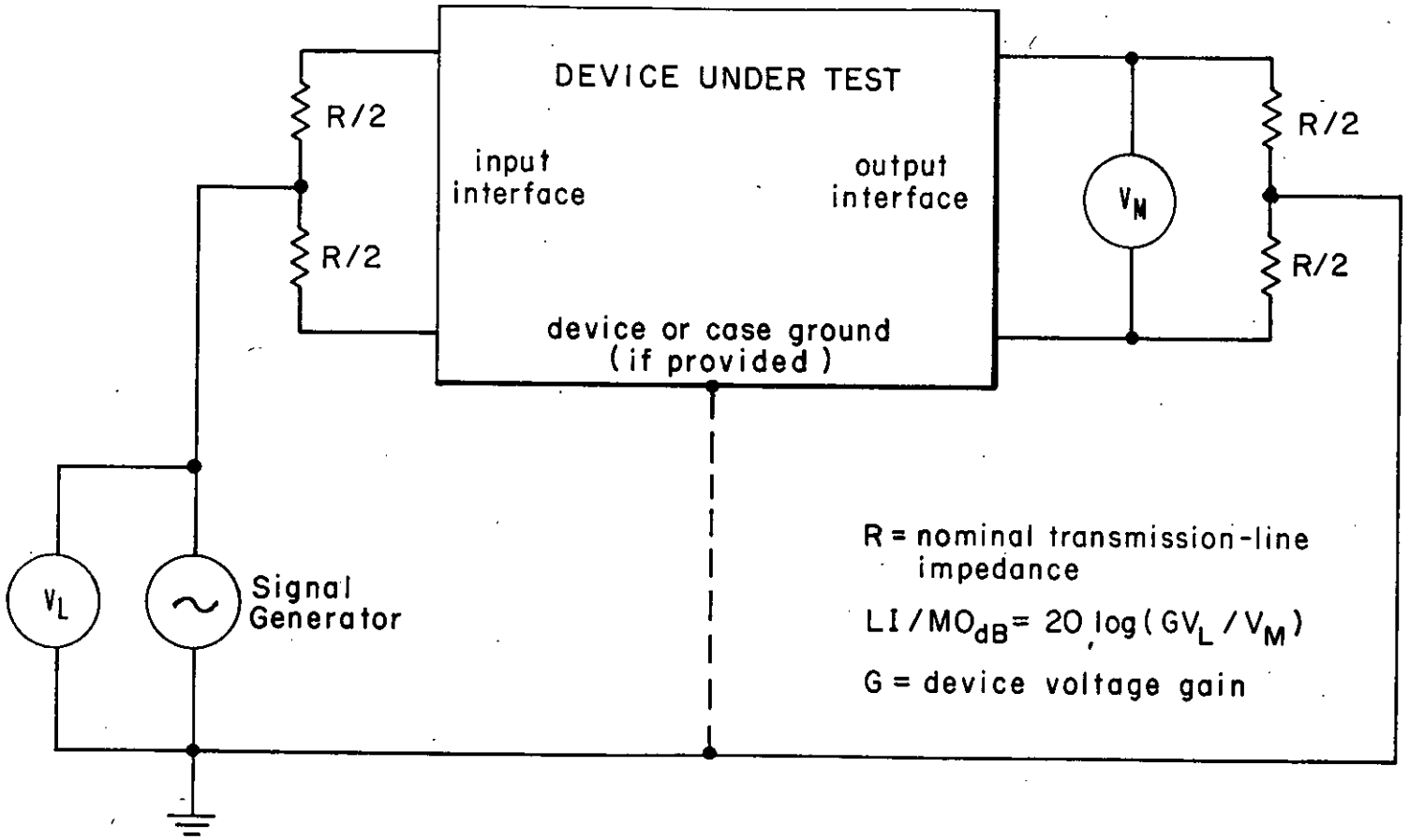


Figure 2. Basic Circuit for Measuring Longitudinal-Input-to Metallic-Output-Voltage Ratio.

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2.5.3.2 The unacceptable longitudinal nonlinearities are rarely apparent at practical metallic test signal levels, linearly equivalent test circuits in which the test signal is metallic rather than longitudinal shall not be used as a substitute for the basic test configurations illustrated in figures 1 and 2.

2.5.4 Conductor-to-ground impedance balance. As illustrated in figure 1 longitudinal balance (LB) is a direct measurement of the rms metallic voltage (V_M) resulting from an unbalance in the conductor-to-ground impedance at the designated interface when a longitudinal test voltage (V_L) is applied from a precisely balanced source. Transmission-line interfaces other than the one being tested are each terminated in an ungrounded (longitudinally isolated) impedance. The metallic impedances (measured from conductor to conductor) of both the source and termination are equal to the nominal transmission-line impedance (R). The longitudinal impedance measured from the interface of the source to ground, through both conductors in parallel, is equal to one-quarter of the nominal transmission-line impedance. If, in normal operation, a conduction path is provided from the device to ground, an equivalent conduction path is provided to the test-circuit ground. Because certain design deficiencies are sensitive to the amplitude and frequency of the longitudinal test signal (V_L), the amplitude and frequency range over which the balance shall be maintained is generally stated as part of each longitudinal balance requirement in Sections 3 through 9. (This technique is similar to that of ANSI IEEE Standard 455-1976 and may be used.)

2.5.5 Coupling balance between interfaces. As illustrated in figure 2, the longitudinal-input-to-metallic-output-voltage ratio (LI/MO) is a measure of the rms metallic voltage (V_M) appearing across a precisely balanced and grounded terminal at one equipment interface (normally an output interface) when a longitudinal test voltage (V_L) is applied from a precisely balanced source at a second equipment interface (normally an input interface). Both the source and termination have a metallic impedance equal to the nominal transmission-line impedance (R) and a longitudinal impedance equal to one-quarter of the nominal transmission-line impedance. If, in normal operation, a conduction path is provided from the device-to-ground, an equivalent conduction path is provided to the test-circuit ground. The longitudinal test signal (V_L) amplitude and frequency range over which the required minimum LI/MO shall be achieved is specified as part of the LI/MO requirement in Sections 3 through 9. Although LI/MO is normalized with respect to the metallic voltage gain (G) of the device (note the equation in figure 2), the value of LI/MO is not necessarily independent of the device gain. Where equipment specifications require that the voltage gain (or attenuation) be adjustable, the stated minimum LI/MO is applicable over the full specified range of the adjustment.

2.6 ALTERNATIVES TO TRANSFORMER COUPLING. It is recognized that advances in electronic technology will continue to provide superior alternatives to established equipment design and fabrication techniques. For this reason, unnecessary design constraints which might deter progressive innovation have been avoided wherever possible in the equipment standards of the appendix. In conformance with this policy, transformer coupling is not specifically required in many cases in which transformers have traditionally been used at the equipment-to-transmission-line interface. As a general rule, however, an alternative to transformer coupling is acceptable only if the equipment in which it is employed is functionally equivalent to the transformer-coupled counterpart. The following paragraphs provide supplementary specifications for equipment with transmission-line interface and dc signaling requirements as indicated by the individual

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paragraph headings. Excluding explicitly stated exceptions, the applicability of 2.6.1, 2.6.2 depending upon the indicated interface and dc signaling requirements shall be regarded as a mandatory part of the basic equipment specification when an alternative to transformer coupling is employed at any equipment-to-transmission-line interface.

2.6.1 Balanced terminating interface with no provision for dc signaling.

2.6.1.1 DC resistance (all transmission-line interfaces):

a. Metallic (conductor-to-conductor): 110 percent of the nominal transmission-line impedance, maximum (300-ohms maximum if a loop-holding path for dc supervision is specifically required).

b. Longitudinal (both conductors in parallel to ground): 100,000 ohms minimum.

2.6.1.2 DC attenuation (between all transmission-line interfaces):

a. Metallic input to metallic output (with the metallic source and terminating resistances equal to the nominal transmission-line impedance): 40 dB minimum.

b. Longitudinal input to longitudinal output (with the longitudinal source and terminating resistances equal to one quarter of the nominal transmission-line impedance): 40 dB minimum.

c. Longitudinal-input-to-metallic-output-voltage ratio: 60 dB minimum for longitudinal potentials between ±70 V dc (see 2.5).

2.6.1.3 DC bias tolerance: All transmission-line interfaces shall tolerate the following externally applied dc bias conditions without deviating from equipment performance specifications:

a. Metallic current (either polarity): Up to 5 mA (up to 120 mA if a loop-holding path for dc supervision is specifically required).

b. Longitudinal voltage (either polarity): Up to 70 volts.

2.6.1.4 DC offset voltage (originating in the equipment; all transmission-line interfaces):

a. Metallic (all transmission-line interfaces terminated in a metallic resistance equal to the normal transmission-line impedance): 100 mV maximum.

b. Longitudinal (all transmission-line interfaces terminated in a balanced longitudinal resistance equal to one-quarter of the nominal transmission-line impedance): 100 mV maximum.

2.6.1.5 AC impedance (all transmission-line interfaces):

a. Metallic: As required by the applicable equipment specification.

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b. Longitudinal: 30,000 ohms minimum, 50 to 4000 Hz, balanced as required by the applicable equipment specification.

2.6.1.6 AC attenuation (between all transmission-line interfaces):

a. Longitudinal input to longitudinal output (with longitudinal source and terminating impedances equal to one-quarter of the nominal transmission-line impedance): 40 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms.

b. Longitudinal-input-to-metallic-output-voltage-ratio: 40 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms unless otherwise indicated in the applicable equipment specification.

2.6.1.7 AC longitudinal dynamic range: All transmission-line interfaces shall tolerate ac longitudinal potentials up to 50 V rms from 50 to 4000 Hz without deviating from any equipment performance specifications.

2.6.2 Balanced terminating interface with provision for simplex operation (alternative to transformer coupling with externally accessible center tap at the equipment-to-transmission-line interface). It will be acceptable if each individual interface is provided with strappable means to allow for connection as either dc receiver interface, dc transmitter interface or center tap not used configuration.

2.6.2.1 Center tap not used. In this configuration all requirements of 2.6.1 shall be satisfied.

2.6.2.2 DC receiver interface (alternative to transformer with center tap connected to -48 V dc through a relay or other device). In this configuration the requirements of 2.6.1 shall be satisfied with the exception of 2.6.1.1.2.6.1.2 and 2.6.1.3. In addition, the following requirements shall apply:

a. DC resistance, metallic (between the two transmission-line terminals): 110 percent of the nominal transmission-line impedance maximum.

b. DC resistance longitudinal:

(1) Both transmission-line terminals in parrallel-to-ground (center-tap terminal open): 100,000 ohms minimum.

(2) Both transmission-line terminals in parallel to center-tap terminal (center-tap terminal open): 2000 ohms, maximum.

c. DC current capability at the center-tap terminal (both transmission-line terminals return to ground and center-tap terminal connected to -48 V dc through an appropriate resistive load): 150 mA minimum.

d. DC transmission-line currents (transmission-line terminals returned to ground equal resistances of any value between 0 and 6000 ohms and any value of dc load

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current at the center-tap terminal between 10 and 150 mA): 2.5 mA minimum, 5 mA maximum, each side of the transmission line, balanced to within 1 percent.

e. Dc offset voltage, metallic (transmission-line terminals terminated with a metallic resistance equal to the nominal transmission line impedance): 100 mV, maximum.

f. DC offset voltage between center-tap terminal and either transmission-line terminal: 2.5 V maximum.

g. Power derived from the equipment power source(s) (i.e., other than through the load connected to the center-tap terminal) for any line and load condition as in (d) above: 0.25 W maximum.

h. Transient protection.

(1) The transmission-line terminals at each transmission-line interface shall be protected in accordance with the requirements of 2.2.

(2) Repeated transient voltages at the center-tap terminal due to a largely inductive load such as a relay coil shall not result in damage to the equipment and shall not introduce more than 5% telegraph distortion at any keying speed up to 100 baud. This requirement shall be satisfied without reliance on external transient suppression (e.g., suppressors fitted at the relay).

i. Reverse polarity protection: Connecting each transmission-line terminal to -60 V dc through a resistance of 300 ohms and returning the center-tap terminal directly to ground shall not result in permanent damage to the equipment.

2.6.2.3 DC transmitter interface (alternative to transformer with center tap returned to ground through a keying contact). In this configuration the requirements of 2.6.1 shall be satisfied with the exception of 2.6.1.2, 2.6.1.3, and 2.6.1.4. In addition, the following requirements shall apply:

a. DC resistance, metallic (between the two transmission-line terminals): 110 percent of nominal transmission-line impedance, maximum.

b. DC resistance, longitudinal of both transmission-line terminals in parallel to ground, center-tap terminal open): 100,000 ohms, minimum.

c. DC current capability at the transmission-line terminals (center-tap terminal returned to ground, both transmission-line terminals in parallel and connected to -48 V dc through an appropriate resistive load): 75 mA, minimum, each terminal (i.e., 150 mA total), balanced to within 5 mA for any total transmission-line current between 10 and 150 mA.

d. DC center-tap current (center-tap terminal returned to ground, any value of total transmission-line current between 10 and 150 mA): 5 mA minimum, 10 mA maximum.

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e. DC offset voltage, metallic (center-tap terminal returned to ground, transmission-line terminals conncected to -48 V dc through equal resistances of any value between 640 and 10,000 ohms): 3 V maximum.

f. DC offset voltage between center-tap terminal and either transmission-line terminal (center-tap terminal returned to ground, transmission-line terminals connected to -48 V dc through equal resistances of any value between 640 and 10,000 ohms): 5 V maximum.

g. Power derived from the equipment power source(s) (i.e., other than through the load connected to the transmission-line terminals) for any value of total transmission-line current up to 150 mA: 0.25 W maximum.

h. Transient protection. The transmission-line terminals at each transmission-line interface shall be protected in accordance with the requirements of 2.2.

i. Reverse polarity protection: Connecting the center-tap terminal to -60 V dc through a resistance of 300 ohms and returning both transmission-line terminals directly to ground shall not result in permanent damage to the equipment.

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Chapter III

3. VF CIRCUIT CONDITIONING EQUIPMENT. The VF circuit conditioning equipment includes the following major components:

a. Strappable pad.

b. Line amplifier.

c. Echo suppressor.

d. Line equalizer.

e. Amplitude equalizer.

f. Six-way, four-wire bridge.

g. Four-wire terminating set and repeat coil "after" set.

h. Pilot make busy extension relay unit (PMBERU).

i. Passive peak limiters.

3.1 STRAPPABLE PAD.

3.1.1 Application. The strappable pad is a passive attenuator designed to be inserted in series with a balanced 600-ohm voice frequency line. Strapping options provide for attenuation in 0.5 dB increments from 0 to 31.5 dB.

3.1.2 Equipment characteristics. The strappable pad shall conform to the following characteristics:

a. Impedance (input and output): 600 ohms balanced, resistive, from 300 to 3400 Hz.

b. Return loss (input and output): 40 dB minimum from 300 to 3400 Hz.

c. Longitudinal balance (input and output): 60 dB minimum from 50 Hz to 4000 Hz for longitudinal potentials up to 50 V rms.

d. Longitudinal-input-to-metallic-output-voltage ratio: 60 dB minimum from 50 to 4000 Hz. For longitudinal potentials up to 50 V rms (See 2.5).

e. Attenuation accuracy and frequency response: $\pm$0.25 dB of strapped value from 300 to 3400 Hz referred to 1000 Hz.

f. Maximum power level (input): +15 dBm shall not cause deviations from the above requirements.

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g. Transformerless coupling: (See 2.6).

h. Surge protection: (See 2.2).

i. Transmission line impedance: Nominally 600 ohms (See 2.4).

j. Power requirements: None.

3.2 LINE AMPLIFIER.

3.2.1 Application. The line amplifier is employed in applications requiring voice-frequency amplification of up to 35 dB on balanced 600-ohm lines. In the TCF, line amplifiers are inserted between the equal level and the primary patch bay. Electrostatically-shielded, center-tapped input and output transformers (or the equivalent thereof; See 2.6) allow the use of simplex dc signaling. Provision is also made for monitoring the VF signal levels at the input and output of the line amplifier.

3.2.2 Equipment characteristics. The line amplifier shall conform to the following characteristics:

a. Impedance (input and output): 6600 ohms (nonreactive), balanced, from 300 to 3400 Hz.

b. Return loss (input and output): 40 dB minimum from 50 Hz to 4000 Hz for longitudinal potentials up to 50 V rms with the transformer center taps (or the equivalent thereof) in both the grounded and ungrounded states for VF frequencies and with dc currents (See 2.5).

d. Longitudinal-input-to-metallic-output-voltage ratio: 40 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms (See 2.5).

e. Gain: 0 dB to +35 dB at 1000 Hz, continuously variable by means of an operator-accessible control.

f. Gain stability: $\pm$0.3 dB maximum deviation at 1000 Hz under all normal operating conditions over a 24-hour period.

g. Frequency response:

(1) 300 to 3400 Hz: $\pm$0.25 dB referred to 1000 Hz.

(2) Above 200 kHz: at least 20 dB below the gain at 1000 Hz.

h. Envelope differential delay:

(1) 600 to 3400 Hz: 75 μs maximum.

(2) 300 to 3400 Hz: 300 μs maximum.

i. Distortion, total harmonic: 40 dB minimum from 300 to 3400 Hz under all

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normal operating conditions (including specifically those conditions stated in 2.3 and 2.4 and in requirements).

j. Noise output (for all gain settings from 0 dB to +35 dB:

(1) C-message weighted: 20 dBrnc maximum.

(2) 10 Hz to 10 MHz, unweighted: 75 mV rms maximum.

k. Interamplifier isolation: 70 dB minimum below 10 kHz for any two amplifiers in the same assembly or mounting shelf.

l. Operating signal level (output): Deviation from the above requirements shall not occur as a result of continuous operation at any gain setting from 0 dB to +35 dB with the output signal level within the following limits:

(1) VF power output, average: -30 to +15 dBm.

(2) VF power output, peak: up to +18 dBm.

m. Simplex signal level (input and output):

(1) DC current, balanced: 100 mA maximum.

(2) DC current, unbalanced: 5 mA maximum.

n. Surge protection: (See 2.2).

o. Transmission line impedance: Nominally 600 ohms (See 2.4).

p. Power requirement: 1.2 Watts maximum (See 2.3).

.3.3 ECHO SUPPRESSOR.

3.3.1 Application. Echo suppressors are used to reduce the degrading effects of echo signals on telephone circuits which are made up of two-wire terminal circuits connected via four-wire trunk circuits, and where the total one-way propagation delay exceeds 25 ms. For a complete circuit two echo suppressors are required--one at each end of the trunk circuit. The echo suppressors are inserted at the four-wire side of the four-wire terminating sets. To obtain a satisfactory user-to-user circuit, further requirements are that the end delay at each end (which equals the round trip delay between the local side of the echo suppressor and the near telephone set) is less than 25 ms, and minimum loss in the echo path at each end is at least 6 dB. Each echo suppressor has two separate signal paths in opposite directions; the send path (also referred to as suppression path) and the receive path (also referred to as the sense path). When signal is present at the receive "in" port and no signal is detected at the send "in" port (i.e., far end user talking, near end user silent), the echo suppressor operates in its unidirectional mode, providing essentially complete echo suppression by introducing very high attenuation (50 dB minimum) in the send path. When both users are talking simultaneously (duplex transmission), signals are present on both the send and the receive paths "in" ports. A differential detector in the

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echo suppressor determines if the signal at the send path "in" port is stronger than the expected echo from the received signal, and if so, switches the echo suppressor to the bidirectional or break-in mode. In this mode the suppression in the send path is removed and 6 dB attenuation is switched into the receive path. Detector threshold levels and switching time constraints (active and hangover times) for this mode are important to prevent false operation due to impulse noise and to ensure sufficient echo attenuation during duplex transmission while avoiding chopping of the speech. Under duplex transmission conditions, both echo suppressors at each end of the trunk will be in the bidirectional mode. When signal is present at the send path "in" port a no signal is detected at the receive path "in" port (i.e., near end user talking, far end user silent), the near-end echo suppressor is also operated in the bidirectional mode. In this case, however, the far-end echo suppressor will detect signal on its receive path "in" port and no signal on its send path "in" port and thus be in the unidirectional mode, providing essentially complete echo suppression on the circuit. By application of appropriate control signals, the echo suppressors can be disabled, resulting in no attenuation in either the send or receive path. This data-bypass mode of operation allows use of the circuit for data transmission.

3.3.2 Equipment characteristics. The echo suppressor shall conform to the following characteristics:

3.3.2.1 General. The following requirements shall be satisfied with the echo suppressor in any of its operational modes:

a. Impedance (input and output, send and receive ports): nominal value of 600 ohms nonreactive from 300 to 3400 Hz. The impedance unbalance to earth of each port shall not be less than 50 dB over the frequency range 300 to 3400 Hz.

b. Return loss (input and output, send and receive ports): 26 dB minimum from 300 to 3400 Hz.

c. Longitudinal balance (input and output, send and receive port): 40 dB minimum from 50 Hz to 4000 Hz for longitudinal potentials up to 50 V rms.

d. Longitudinal-input-to-metallic-output-voltage ratio (either direction): 40 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms (See 2.5).

e. Signal path frequency response (either direction): $\pm$0.5 dB from 300 to 3400 Hz referred to 1000 Hz.

f. Differential delay (output; each direction):

(1) 30 us maximum for any two frequencies between 1000 and 2400 Hz.

(2) 60 us maximum for any two frequencies between 500 and 3000 Hz.

(3) 100 us maximum for any two frequencies between 300 and 3400 Hz.

g. Noise (output): -70 dBm maximum from 250 to 4000 Hz (flat weighting).

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h. Spurious outputs: The zero to peak voltage of any transient output produced in the receive or transmit paths (terminated in 600 ohms due to echo suppressor operation causes by signals in the opposite path should not exceed 20 mV after first filtering the transient to a 500 to 3400 Hz bandwidth. Additionally, the duration of such transient should be such that it is not audible in the presence of normal levels of noise (e.g., -48 dBm).

i. Distortion, total harmonic: 40 dB minimum under all normal operating conditions from 300 to 3400 Hz.

j. Crosstalk: Crosstalk attenuation between the send path and receive path (and conversely) shall be such that the signal power in the disturbed path due to crosstalk from the disturbing path shall not exceed -65 dBm for any sinusoidal signal in the disturbing path having a power of +5 dBm or less and within the band 300-3400 Hz.

k. Operational level: The echo suppressor shall be designed to operate at 0 Transmission Level Point (TLP) for both the send (suppress) and receive (sense) path.

l. Overload protection: A signal of +10 dBm applied to either "in" port shall not adversely affect the operation of the unit after removal of the signal.

m. Transmission line impedance: Nominally 600 ohms (See 2.4).

n. Control path frequency response: The frequency response of both the suppression and the break-in control paths shall be $\pm$2 dB from 500 to 3000 Hz, referred to 1000 Hz. Below 500 Hz and above 4000 Hz the response shall roll off at a rate of 12$\pm$2 dB per octave.

o. Surge protection: (See 2.2).

p. Power requirements: 6.5 watts maximum (See 2.3).

(1) Overvoltage: 50 percent, 5 sec maximum duration every 15 minutes without equipment damage.

(2) Undervoltage: 100 percent, 5 sec maximum duration every 15 minutes without equipment damage.

q. Intermodulation: Intermodulation, for frequencies f_1=900 Hz and f_2=1020 Hz applied simultaneously each at a level of -5 dBm, the difference between the output levels of either frequency f_1 or f_2 and the level of either of the intermodulation products at ($2f_1-f_2$) or ($2f_2-f_1$) should be at least 45 dB. When speech compressors are used to provide loss during talking this requirement is reduced during the double talking mode to 26 dB.

3.3.2.2 Unoperated state (signals less than -38 dBm on either send or receive "in" ports). The insertion loss (at 1000 Hz) of an echo suppressor (both send and receive paths) in an unopened condition shall be 0 $\pm$0.3 dB, for tone levels less than or equal to 0 dBm.

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3.3.2.3 Unidirectional (suppress) state (signal at receive "in" port, no signal at send "in" ports).

a. Insertion loss: Send (suppress) path: 50 dB minimum. Receive (sense) path: 0 ±0.3 dB.

b. Suppression activate point: The level of a 1000 Hz sinusoid at the receive (sense) "in" port at which the echo suppressor switches from the unoperated state to the undirectional state shall be -31 dBm ±2 dB.

c. Suppression release point: The level of a 1000 Hz sinusoid at the receive (sense) "in" port at which the echo suppressor switches from the unidirectional state to the unoperated state shall be 1.5 ±1.5 dB below the suppression activate point.

d. Suppression operate (attack) time: The time interval between the instant that the signal level at the receive "in" port exceeds the suppression activate point and the instant when the suppression loss is introduced in the send path shall be between 1 and 5 ms for any value of applied signal level at the receive "in" port between -26 dBm and -5 dBm and with both slow rising and fast rising wave-fronts.

e. Suppression release (hangover) time: The time interval between the instant when the voice frequency signal is removed from the receive "in" port and the instant that the suppression loss is removed from the sne dpath shall be 60 ±20 ms, for any value of initial applied signal level at the receive "in" port between -26 dBm and -5 dBm.

3.3.2.4 Bidirectional (break-in) state (signal level, L_S, at send "in" port exceeding signal level, L_R, at receive "in" port):

a. Insertion loss:

(1) Send (suppress) path: 0 ±0.3 dB.

(2) Receive (sense) path: 6 ±1 dB.

b. Break-in activate point: The minimum level of a 1000 Hz sinusoid at the send "in" port at which the echo suppressor switches to the bidirectional state shall be:

(1) For L_R below -40 dBm: -36 L_L -29 dBm.

(2) For L_R between -26.5 dBm and +4.5 dBm: L_S L_R ±1.5 dB.

c. Break-in release point: The level of a 1000 Hz sinusoid at the send "in" port at which the echo suppressor switches out of the bidirectional state shall be:

(1) For $L_$ below -40 dBm: Between 0 and 3 dB below the corresponding break-in activate point (See 3.3.2.4(b)(1)).

(2) For L_R between -26.5 dBm and +4.5 dBm: Between C and (C3) dB

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below the corresponding break-in activate point (See 3.3.2.4(b)(2)), where C is the value of dB of the actual loss inserted in the receive path during break-in.

d. Break-in operate (attack) time. The time interval between the instant that the break-in activate point is exceeded and the instant that the echo suppressor has switched to the break-in state shall be:

(1) With L_R constant and L_S increasing: Between 1 and 10 ms.

(2) With L_S constant and L_R decreasing: Between 25 and 50 ms.

e. Break-in release (hangover) time: The time interval between the instant that the level at the send "in" port falls below the break-in release point and the instant that the echo suppressor has reverted to the unidirectional (or the unoperated) state shall be between 200 and 300 ms.

3.3.2.5 Data bypass state.

3.3.2.5.1 Insertion loss: 0 $\pm$0.3 dB for both send and receive paths.

3.3.2.5.2 Disabling control signals. The echo suppressor shall switch to the data bypass state upon receipt of any of the following control signals:

a. A ground placed on one control lead terminal on the echo suppressor for the duration of the data bypass state.

b. Removal of ground from another control lead terminal on the echo suppressor for the duration of the data bypass state.

c. A continuous tone at the receive "in" port having the following characteristics:

(1) Frequency: Between 2000 and 2250 Hz.

(2) Level: -28 dBm minimum.

(3) Duration: 200 ms minimum.

3.3.2.5.3 Guard bands. The presence of voice frequency energy outside the disabling tone channel shall be sensed to prevent false operation on speech signals. Disabling of the echo suppressor by the application of a 2100 Hz control signal at a level of -28 dBm or higher shall be inhibited in the presence of white noise in the band of 300 to 3400 Hz having an energy level no greater than 0 dB and no less than -5 dB with respect to the level of the control signal.

3.3.2.5.4 Holding characteristics. After disabling by means of a 2100 Hz control signal, the echo suppressor shall remain in the data bypass state in the presence of a single frequency sinusoid at either the send or receive "in" ports having a level of:

a. -27 dBm or greater in the band from 390 to 700 Hz.

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b. -31 dBm or greater in the band from 700 to 3000 Hz.

3.3.2.5.5 Disabling release characteristics. When the echo suppressor had been disabled by means of a 2100 Hz control signal, the disabling shall be removed when the signal level in the band of 300-3400 Hz at both the send and receive "in" ports drops to -36 dBm or less.

3.3.7.5.6 Operate time. The interval between the application of the 2100 Hz control signal and the disabling of the echo suppressor shall be 300 ±100 ms for levels of the control signal between -28 dBm and 0 dBm.

3.3.2.5.7 Release time. The interval between the instant that the holding signal level drops below the release threshold (See 3.3.2.5.5) and the instant that the echo suppressor disabling is removed shall be 250 ±150 ms.

3.4 LINE EQUALIZER.

3.4.1 Application. As voice frequency signals are transmitted through equipment and transmission channels having restricted bandwidth characteristics, various frequency components are delayed in time and distorted in amplitude with respect to each other; this may cause unsatisfactory transmission. The line equalizer is designed to produce an output envelope delay spread of less than 250 ms when the input envelope is distorted by as much as 3 ms. In addition, the line equalizer provides for a 6 dB range of amplitude equalization of the affected frequencies. The line equalizer consists of various sections of equalization from 600 to 2800 Hz. Both amplitude and frequency adjustments are accomplished by front panel controls on each individual section. Any of these sections can be bypassed whenever correction is not required. See figure 3 for an illustration of delay equalization.

3.4.2 Equipment characteristics. The line equalizer shall conform to the following characteristics:

a. Impedance input and output: 600 ohms balanced, from 300 to 3400 Hz.

b. Return loss (input and output): 26 dB minimum from 300 to 3400 Hz.

c. Longitudinal balance (input and output): 40 dB minimum from 50 Hz to 4000 Hz for longitudinal potentials up to 50 V rms (See 2.5).

d. Longitudinal-input-to-metallic-output-voltage radio: 40 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms (See 2.5).

e. Envelope delay equalization: 250 us envelope delay of the output signal shall be obtainable for an input signal with a maximum delay of 3 ms. This delay shall be adjustable from 300 to 3400 Hz.

f. Operating signal level (input): -35 to +10 dBm from 300 to 3400 Hz.

g. Gain: -10 to +30 dB, continuously variable.

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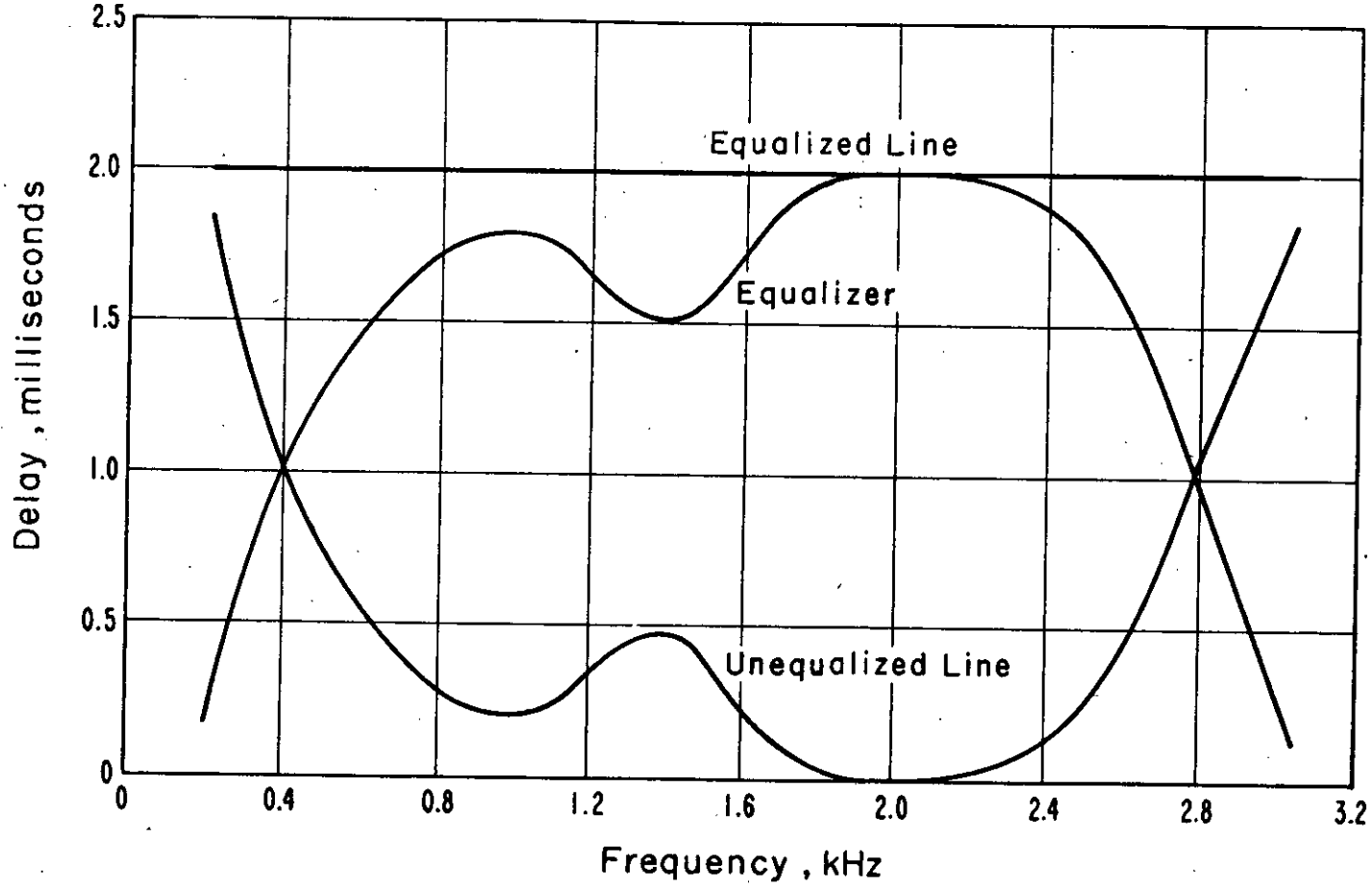


Figure 3. Idealized Illustration of Delay Equalization.

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h. Gain stability: +0.3 dB maximum at 1000 Hz under normal operating conditions over a 24-hour period.

i. Noise (output): -70 dBm maximum from 300 to 3400 Hz (flat weighting).

j. Amplitude equilization: ±3 dB, continuously variable from 300 to 3400 Hz.

k. Operating signal level (output): -5 to 0 dBm from 300 to 3400 Hz.

l. Distortion, total harmonic: 40 dB minimum under all normal operating conditions from 300 to 3400 Hz.

m. Transmission line impedance: Nominally 600 ohms (See 2.3).

n. Surge protection: (See 2.2).

o. Power requirements: 1 watt maximum (See paragraph 2.3).

3.5 AMPLITUDE EQUALIZER.

3.5.1 Application. The amplitude equalizer will produce output frequency responses over the range of 200 to 5000 Hz as specified below for circuits that employ up to 50 miles of H-44 (19 gauge, 0.083 uF per mile) cable or length of 16, 19, 22, and 24 gauge nonloaded cable that produce losses up to 8.5 dB at 5000 Hz relative to 1000 Hz. The equalizer will compensate for losses over the 200 to 3500 Hz range for lengths of H-88 cable (19, 22, 24, and 26 gauge; 0.083 µF per mile) up to those that produce 8.5 dB loss at 3500 Hz relative to 1000 Hz. The equalizer also matches line impedances of 300, 600, 900, and 1200 ohms to a 600 ohms load. A center tapped primary winding on the input transformer allows for simplex signaling.

3.5.2 Equipment characteristics. The amplitude equalizer shall conform to the following characteristics:

a. Impedance (for any setting of the equalization controls):

(1) Input: 300, 600, 900, and 1200 ohms from 200 to 500 Hz, impedance selectable by means of primary winding taps on input transformer.

(2) Output: 600 ohms ±10 percent from 200 to 5000 Hz with input correctly terminated.

b. Return loss (input and output): 26 dB minimum from 200 to 5000 Hz with output and input terminated respectively, as in (1) and (2) above.

c. Longitudinal balance (input and output): 40 dB minimum from 50 Hz to 5000 Hz for longitudinal potentials up to 50 V rms with the transformer center tap (or the equivalent thereof) in both the grounded and ungrounded states.

d. Longitudinal-input-to-metallic-output-voltage ratio: 40 dB minimum from

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50 to 5000 Hz for longitudinal potentials up to 50 V rms (See 2.5).

e. Amplitude equalization. The frequency response shall be adjustable to ±1 dB from 200 to 3000 Hz for input signals with relative response shown by all curves in figures 3-2a, 3-2b, and 3-2c. The response shall be adjustable to ±2 dB from 3000 to 3500 Hz for input signals with responses shown by curves in figures 4, 5, and 6. The response shall be adjustable to ±2 dB from 3000 to 3500 Hz for input signals with responses shown by curves 1, 2, 3, and 4 and to ±2 dB from 3000 to 5000 Hz for input signals with responses given by curves 5, 6, 7, and 8.

f. Distortion, total harmonic (output): 40 dB minimum under all normal operating conditions from 200 to 5000 Hz.

g. VF transmission line impedance: (See 2.4).

h. Surge protection: (See 2.2).

i. Insertion loss: 5 dB maximum at frequency of minimum attenuation.

j. Out-of-balance dc signaling current: 10 mA shall not cause deviations from the above requirements.

k. Maximum power (input): +10 dBm from 200 to 5000 Hz shall not cause deviation from the above requirements.

l. Power requirements: 1.5 watts maximum (See paragraph 2.3).

3.6 SIX WAY, FOUR-WIRE BRIDGE ASSEMBLY.

3.6.1 Application. The six-way, four-wire bridge assembly consists of a six-way, four-wire resistive hybrid and six line amplifiers that can be externally connected to either the bridge inputs or outputs. The bridge provides conference capability for six each four-wire telephone circuits. The hybrid provides high loss between transmit and receive circuits going to and from the same telephone station and a minimum loss between an input circuit from any station to an output circuit to the other five stations. Strapping provision is included for optional sidetone for each four-wire circuit.

3.6.2 Equipment Characteristics. The six-way, four-wire bridge assembly shall conform to the following characteristics:

3.6.2.1 Six-way, four-wire resistive hybrid (without amplifiers):

a. Impedance (input and output): 600 ohms balanced for the six-way, four-wire resistive hybrid from 300 to 3400 Hz.

b. Return loss (input and output): 26 dB minimum from 300 to 3400 Hz.

c. Longitudinal balance (input and output; all circuits): 40 dB minimum from 50 Hz to 4000 Hz for longitudinal potentials up to 50 V rms.

d. Longitudinal-input-to-metallic-output voltage ratio (all circuits): 40 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms (see 2.5).

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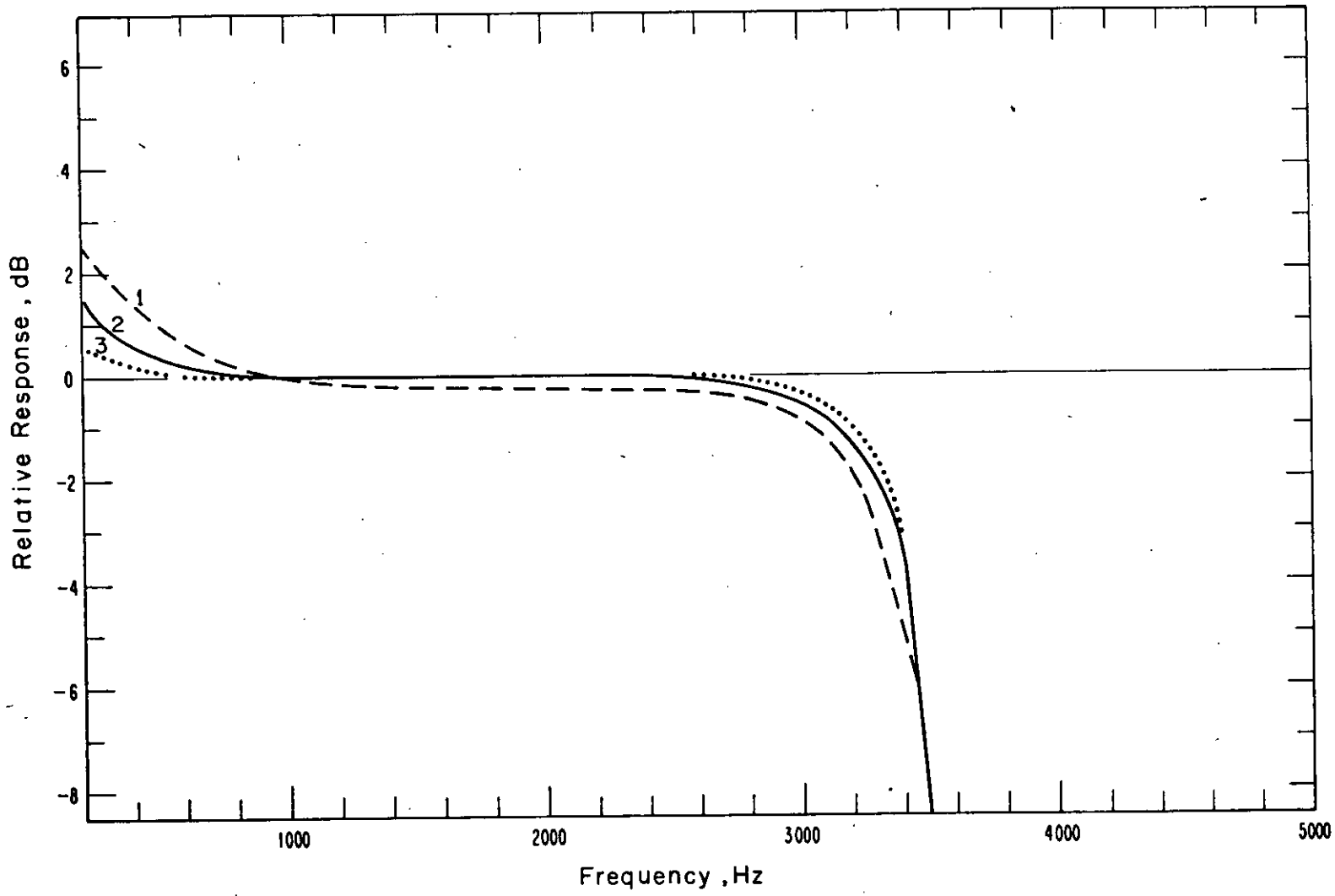


Figure 4. Input Signal Frequency Response Curves.

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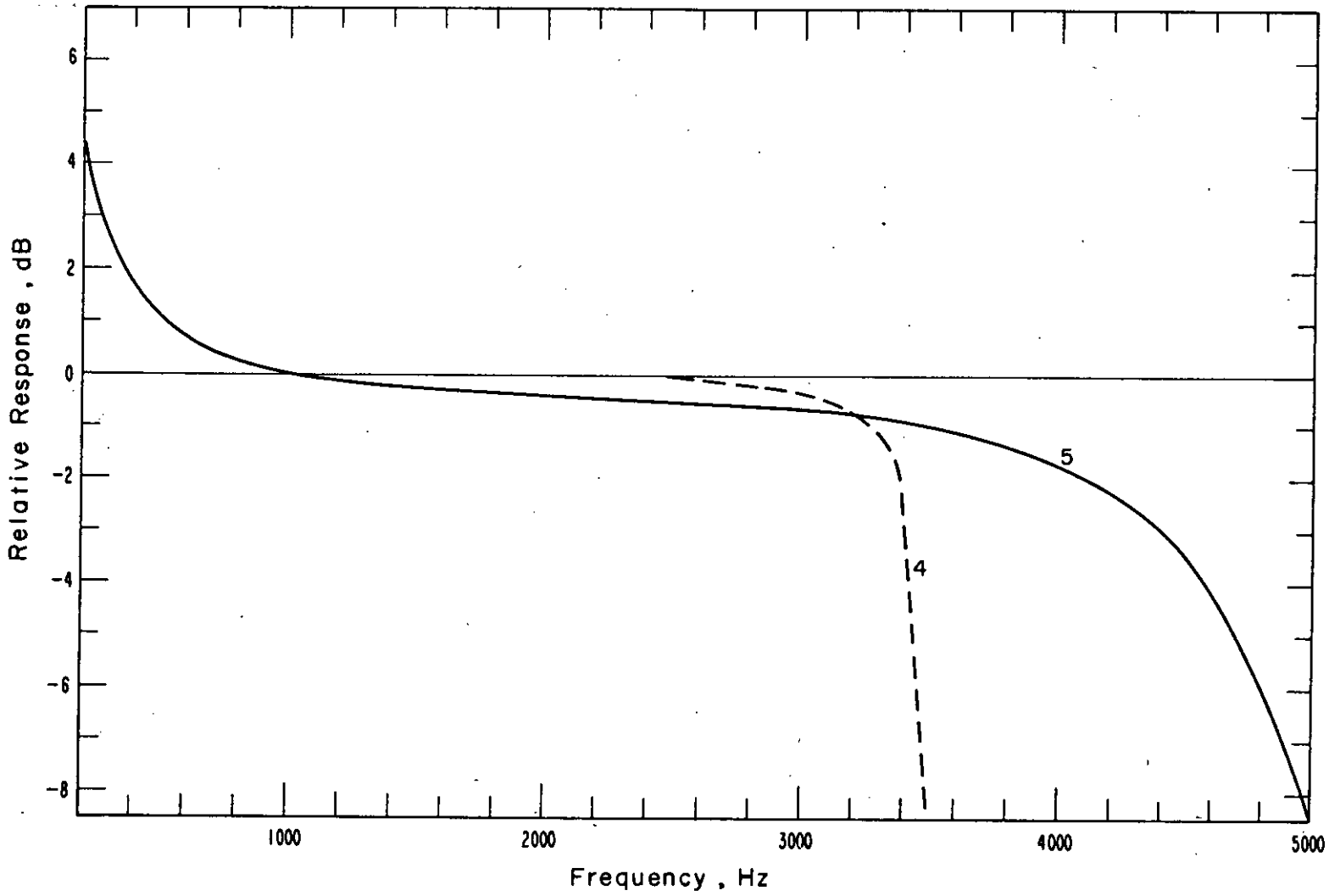


Figure 5. Input Signal Frequency Response Curves.

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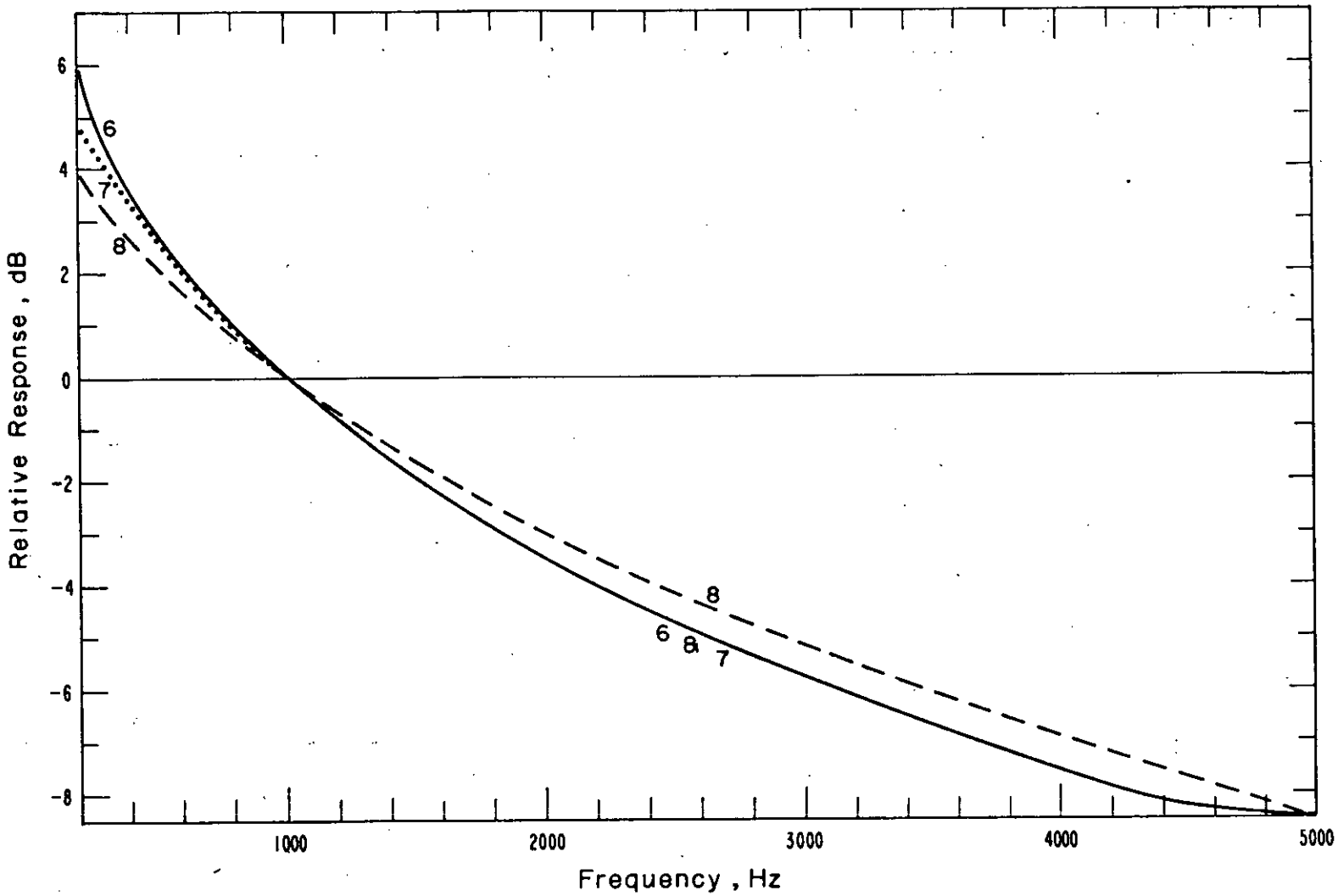


Figure 6. Input Signal Frequency Response Curves.

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e. Frequency response: ±0.25 dB from 300 to 3400 Hz and ±0.5 dB from 200 to 3500 Hz referred to 1000 Hz.

f. Return path loss: 65 dB minimum between any input and its associated output, when all inputs and outputs are terminated into a 600-ohm resistive load.

g. Insertion loss: The loss measured between any input terminal and each output terminal not associated with the selected input shall be 20 dB maximum at 1000 Hz.

h. VF transmission line impedance: Nominally 600 ohm (see 2.4).

i. Surge protection: (see 2.2).

j. Isolation: 70 dB minimum below 10 kHz for any two bridge circuits mounted on the same or adjacent shelves.

k. Noise (output): -95 dBm maximum from 300 to 3400 Hz (flat weighting).

l. Distortion, total harmonic (output): 40 dB minimum from 300 to 3400 Hz.

m. Maximum power level (input): +20 dBm peak shall not cause deviations from the above requirements.

n. Power requirements: None.

3.6.2.2 Amplifier requirements: (see 3.2).

3.7 FOUR-WIRE TERMINATING SET AND REPEAT COIL.

3.7.1 Application. The equipment may be used to provide interconnection compatibility between balanced four-wire to two-wire circuits or it may be used as a repeat coil between balanced four-wire to four-wire circuits. The selection is made by strapping. The four-wire terminating set commonly consists of two balanced transformers with a balancing network including build-out capacitance arranged in the hybrid circuit. The transformers are electrostatically shielded to minimize longitudinal effect. The shield is brought out to a terminal isolated from the transformer case.

3.7.2 Equipment Characteristics. The four-wire terminating set and repeat coil shall be strappable to conform to the following characteristics:

3.7.2.1 Four-wire terminating set (two-wire to four-wire hybrid) operation:

a. Insertion loss: 4 dB maximum at 1000 Hz, as measured from the four-wire terminal to the two-wire terminal in either direction of transmission.

b. Frequency response: ±0.25 dB from 300 to 3400 Hz referred to 1000 Hz.

c. Signaling lead terminal: A and B signaling lead terminals are provided connected to the split-center taps of the two-wire side of the hybrid. A 2 uF capacitor is

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provided in parallel with the A and B terminals. A and B terminals can be strapped together when required.

d. Signaling current: 100 mA maximum unbalanced.

e. Trans-hybrid loss: 50 dB minimum from 300 to 3400 Hz with two-wire terminals and the network terminals terminated in a 600- or 900-ohm resistor in series with 2.2 uF capacitor.

f. Balancing networks. A 300 to 1800-ohm linearly variable resistor in series with a 2.2 uF ±10 percent to provide values of 0 to 0.03 uF in 0.003 uF steps shall be included. These build-out capacitors are connected, by strapping, in parallel with the entire balancing network.

g. Operating signal level (input): -30 to +10 dBm.

h. Longitudinal-input-to-metallic-output voltage ratio: 60 dB minimum from 50 to 4000 Hz for longitudinal potentials up 50 V rms (see 2.5). With four-wire receive circuit as input and two-wire circuit as output, and with two-wire circuit as input and four-wire send circuit as output.

i. Four-wire side:

(1) Impedance: 600 ohms balanced, from 300 to 3400 Hz.

(2) Return loss (send and receive circuits): 26 dB minimum from 300 to 3400 Hz when the two-wire side is terminated in any impedance given in j(1) below ±1 percent and the four-wire input terminals are terminated in 600 ohms ±1 percent.

(3) Longitudinal balance (send and receive circuits): 60 dB minimum from 50 Hz to 4000 Hz for longitudinal potentials up to 50 V rms.

(4) Surge protection: (see 2.2).

(5) VF transmission line impedance: Nominally 600 ohms (see 2.4).

j. Two-wire side:

(1) Impedance: 300, 450, 600, 900, 1200 and 1800 ohms balanced, from 300 Hz to 3400 Hz. Impedance adjustment is by strapping with taps on both transformers.

(2) Return loss: 26 dB minimum from 300 to 3400 Hz when the four-wire terminals are terminated in 600 ohms ±1 percent resistors.

(3) Longitudinal balance: 60 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms with split center tap grounded and unbalanced signaling current.

(4) Surge protection: (see 2.2).

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(5) VF transmission line impedance: Nominal 600 and 900 ohms (see 2.4).

k. Power requirements: None.

3.7.2.2 Four-wire-to-four-wire repeat coil operation.

a. Impedance of 600 to 900 ohms to an impedance of 600 ohms.

b. Signaling: D. C. signaling shall be provided for on the line side from the electrical center point of either or both of the send and receive windings.

c. Frequency response: (see 3.7.2.1(b)).

d. Signaling current: 100 uA maximum unbalanced.

3.8 PILOT MAKE BUSY EXTENSION RELAY UNIT (PMBERU).

3.8.1 Application. The pilot make busy feature is designed to protect the AUTOVON switch from mass seizure when a transmission path experiences a failure or fade condition. This feature, in addition to protecting the swtich, provides alarm conditions to alert technical control and AUTOVON switch personnel of transmission path failure. The group pilot receivers are normally adjusted to the lowest receive level sensitivity possible. The loss of signal from the MUX equipment is sensed by a group pilot control (GPC) in the MUX which causes the BPC alarm lead to the Pilot Make Busy Relay Unit (PMBRU) to become grounded. The contracts closure (12) in the PMBRU causes a ground to be placed on the PMB1 thru PMB12 leads which provides the following function (see figure 7):

a. A ground is placed on the PMB_2 lead going to the AUTOVON switch for alarm indication and busy lamp at the secondary test board flashes at 60 IPM, indicating a lock-out.

b. The E lead from the pilot make busy extension relay unit (PMBERU) remains in the normal straight-through condition until a predtermined time delay has elapsed. Upon completion of the delay, the E lead indicates an open circuit to the AUTOVON switch.

c. If the ground condition is removed from the GPC alarm lead (MUX signal restored) prior to the completion of the time delay period, the E lead will remain in the normal straight-through state and the delay uint will reset.

d. The Pilot Make Busy circuitry consist of:

(1) Group pilot control unit alarm circuit (MUX).

(2) CAR PMB and EQUIP PMB, jacks located in the miscellaneous fields of the equal level patch panel (VF).

(3) Pilot make busy relay unit (one per associated group).

(4) Pilot make busy extension relay unit (PMBERU). One PMBERU circuit

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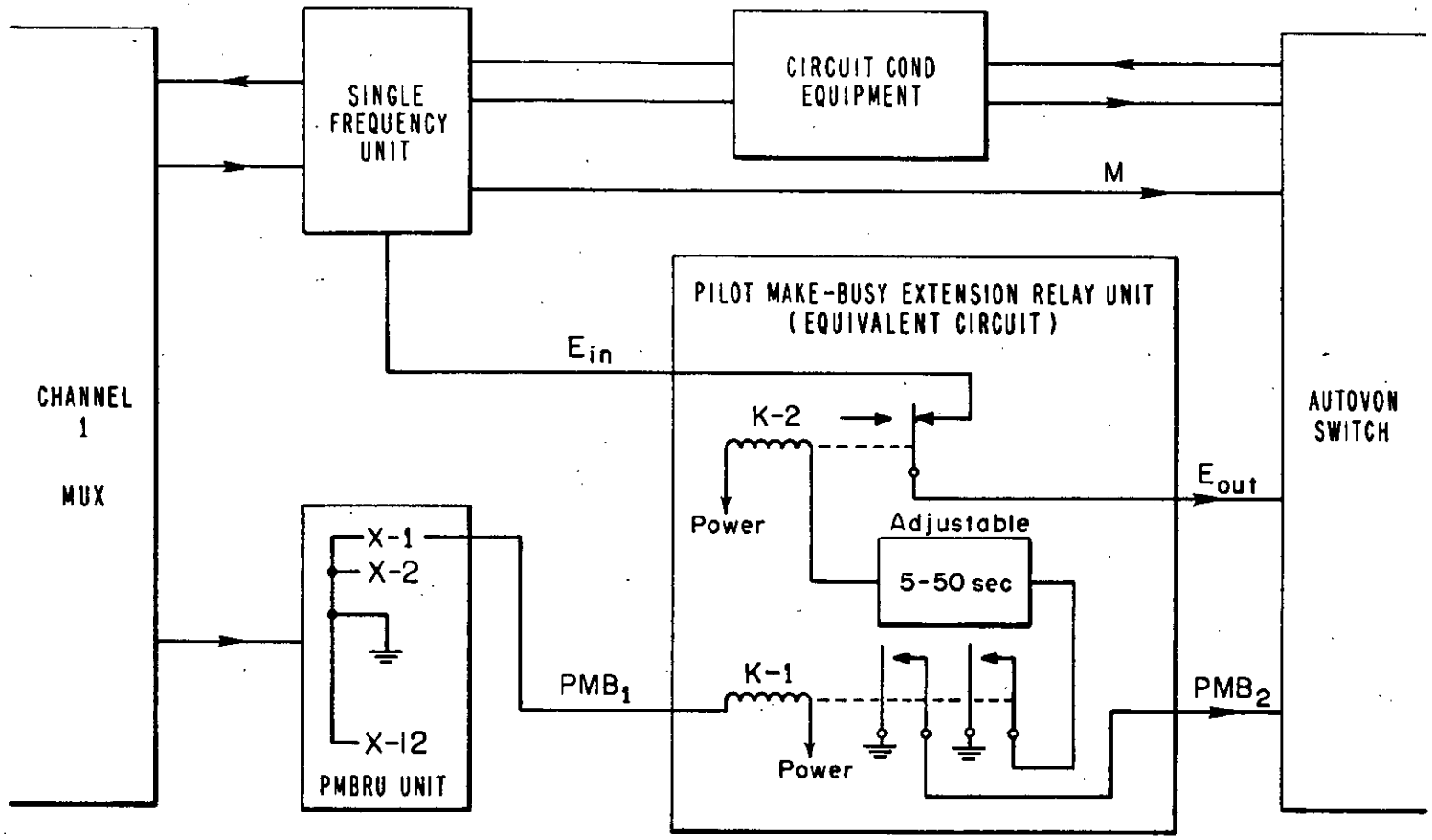


Figure 7. Pilot Make Busy Extension Relay Unit (PMBERU).

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is used for each voice channel of the MUX equipment (up to 12 total per associated group).

(5) Audible/visual alarms.

(6) Busy lamp at secondary test board.

(7) Associated AUTOVON circuitry and equipment.

(8) Circuit patch jacks (circuit patch panel).

3.8.1.1 The pilot make busy extension relay unit (PMBERU) controlled by the PMBRU, provides a ground for the AUTOVON switch alarm indication and the time delay opens E lead circuit between the AUTOVON switch and SF units.

3.8.2 Equipment Characteristics.

a. Response time: 5 ms maximum for action describer in 3.8.1.(a).

b. Delay time (see 3.8.1(b)): Adjustable from 5 to 50 seconds in steps not greater than 5 seconds. Overall accuracy shall be better than 5 percent of selected value or 1 second maximum, whichever is greater.

c. Alarm signal (PMB_1 to PMBERU) current: 15 mA maximum in alarm condition.

d. Power requirements: 2.5 watts maximum (see 2.3).

3.9 PASSIVE PEAK LIMITERS.

3.9.1 Application. Passive peak limiters are two-terminal devices that are connected across balanced 600-ohm voice frequency circuits to prevent possible excessive signal levels and to alleviate effects of noise transients introduced in a multiple link system. Two types, the standard passive peak limiter and the optional passive peak limiter are used in the TCF.

3.9.1.1 The standard limiter acts essentially as an open circuit when shunted across a 600-ohm circuit operating at levels below about 0 dBm. Its typical limiting action at levels above 1 V peak is shown by the heavy solid line for the standard limiter in figure 8. The standard limiter is connected across the send side of multiplex channels between the multiplex equipment and the equal-level patch bay.

3.9.1.2 The optional limiter acts essentially as an open circuit when shunted across a 600-ohm circuit operating at levels below about -7 dBm. Its typical limiting action for signals greater than 0.5 V peak is shown by the heavy solid line for the optional limiter in figure 8. If needed to control excessive levels from users, the optional limiter is connected across the send side of the multiplex channel between the equal-level patch bay and the conditioning equipment.

3.9.2 Equipment Characteristics. The passive peak limiter shall conform to the

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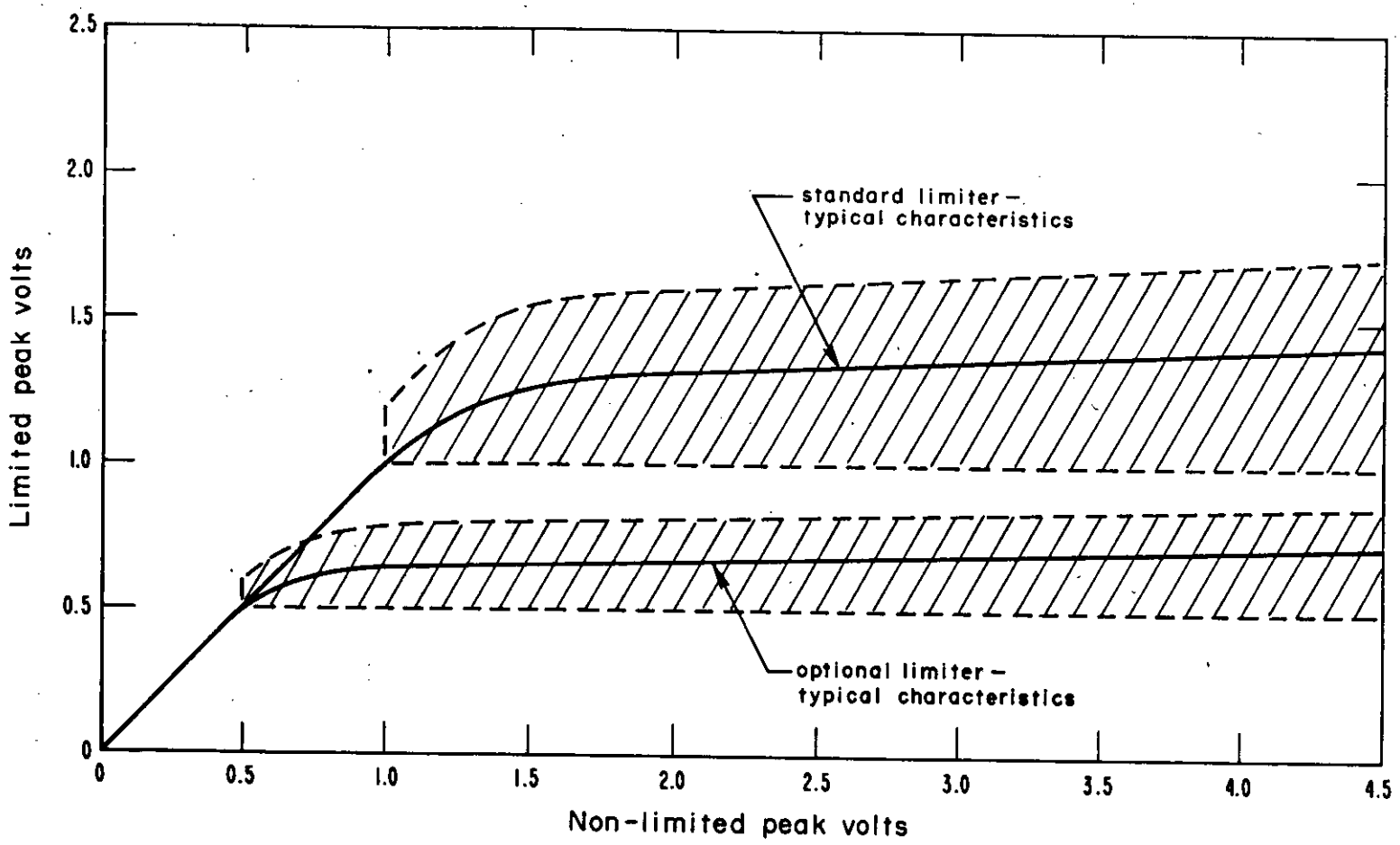


Figure 8. Limiting Characteristics of Standard and Optional Peak Limiters.

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following characteristics when shunted across a 600-ohm line (or across a 600-ohm resistive load driven by a 600-ohm resistive source). The preferred locations of such devices are at the 0dBr Test Level Points (TLPs) but location at other TLPs is permissible. All such devices installed at 0 TLPs will be linear to at least 1.10 peak volts with the non-linear portion of the characteristic falling anywhere between 1.10 peak volts and 1.74 peak volts. As a result of the limiting action, the output voltage shall not exceed 1.74 volts peak. At TLPs other than 0dBr, each of the voltage limits are multiplied by the following factors:

a. For TLPs above 0dBr, multiply by $(1.2222)^n$, where n = number of dB above 0dBr. Ex: $1.222^2 = 1.258 \times 1.10\text{-}1.38$ pk or +2 dBr.

b. For TLPs below 0dBr, multiply by $(0.8913)^n$, where n = number of dB below 0dBr. Table I shows the limits of the non-linear range of such devices when applied at any TLP from -16 through +10 dBr. The linear range will extend to the lower limit of the non-linear range in each case. Ex: $0.8912^2 = .7944 \times 1.10 = 0.873$ pk v or -2 dBr.

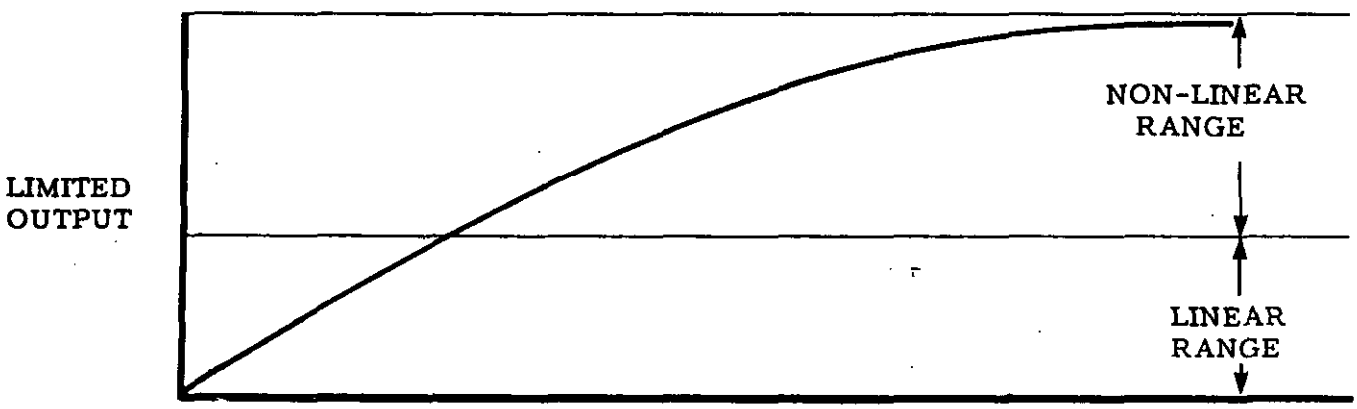
c. See Table I for additional TLP values.

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TABLE I

STANDARD CHARACTERISTICS OF PASSIVE PEAK LIMITING DEVICES INSTALLED AT FOUR-WIRE, 600 OHM INTERFACE POINTS

Test Level Point (dBr)	Non-Linear Range (peak volts)
+10	3.48 - 5.51
+9	3.10 - 4.90
+8	2.76 - 4.33
+7	2.48 - 3.90
+6	2.20 - 3.48
+5	1.95 - 3.10
+4	1.74 - 2.76
+3	1.55 - 2.48
+2	1.39 - 2.20
+1	1.23 - 1.96
0	1.10 - 1.74
-1	0.98 - 1.55
-2	0.87 - 1.39
-3	0.78 - 1.23
-4	0.69 - 1.10
-5	0.62 - 0.98
-6	0.55 - 0.67
-7	0.49 - 0.78
-8	0.44 - 0.69
-9	0.39 - 0.62
-10	0.35 - 0.55
-11	0.31 - 0.49
-12	0.28 - 0.44
-13	0.25 - 0.39
-14	0.22 - 0.35
-15	0.20 - 0.31
-16	0.17 - 0.28



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3.9.2.1 Standard Limiter.

a. Linear range: 0 to 1.1 V peak. Over the frequency range of 300 to 3400 Hz, total harmonic distortion at different temperatures and levels shall be in agreement with Table II.

b. Limiting characteristics: Insertion of the limiter at levels above 1 V peak shall not result in limited output voltages outside the cross-hatched area as shown in figure 3-4 for the standard limiter. The slope of the characteristic shall not be negative at any point.

c. Frequency response (of resistive circuit with limiter connected): ±0.25 dB from 300 to 3400 Hz at any level up to 0 dBm. Frequency response (of resistive circuit with limiter connected) +/-0.25db from 300 to 3400 Hz a level above 0dBm to 7 dBm (2.5V peak/600 ohms).

d. Power requirements: None.

3.9.2.2 Optional limiter.

a. Linear range: 0 to 0.5 V peak. Over the frequency range of 300 to 3400 Hz, total harmonic distortion at different temperatures and levels shall be in agreement with Table II.

b. Limiting characteristics: Insertion of the limiter at levels above 0.5 V peak shall result in limited output voltages outside the corss-hatched area as shown in figure 8 for the optional limiter. The slope of the characteristic shall not be negative at any point.

c. Frequency response (of resistive circuit with limiter connected): ±0.25 dB from 300 to 3400 Hz at any level up to -7 dBm.

d. Power requirements: None.

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TABLE II MINIMUM VALUES OF TOTAL HARMONIC DISTORTION FOR THE STANDARD LIMITER

Temperature (T) °C	Level V Peak	Minimum Total Harmonic Distortion dB
0 to 25	1.0	40
0 to 25	1.1	34
25 to 50	1.0	40-1/2 (T-25)
25 to 50	1.1	34-1/2 (T-25)

TABLE III MINIMUM VALUES OF TOTAL HARMONIC DISTORTION FOR THE OPTIONAL LIMITER

Temperature (T) °C	Level V Peak	Minimum Total Harmonic Distortion dB
0 to 25	0.43	40
0 to 25	0.48	34
25 to 50	0.43	40-1/2 (T-25)
25 to 50	0.48	34-1/2 (T-25)

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CHAPTER IV

4. DC CIRCUIT EQUIPMENT. The dc circuit equipment includes the following major electronic units:

a. Digital line-level converters.

b. Digital modulation-rate converter.

c. Mode selection regenerative repeater.

4.1 DIGITAL LINE-LEVEL CONVERTERS.

4.1.1 Application. The digital line-level converters are solid-state modular devices, used to provide isolation and signal-level compatibility between lines and equipments operating on different signal levels at modulation rates up to 150 baud.

4.1.1.1 There are two types of level converters. One type (high-level to low-level) converts high-level signals to low-level signals and the other type (low-level to high-level) converts low-level signals to high-level signals.

4.1.1.1.1 Isolation between input and output signals is provided with signal and logic battery externally supplied. The high level to low-level converter is also equipped to operate from low-level to low-level for isolation purposes. Current adjustment capability is provided on the high-level input side of the high-level to low-level converter.

4.1.2 Equipment Characteristic. The digital line-level converters shall conform to the following characteristics:

4.1.2.1 High-level to low-level converter.

a. Input characteristics (high-level to low-level operation).

(1) Signal levels: High-level (signal power external) neutral or polar by option. Neutral 130v dc at optionally available 20 or 60 ma. Polar $\pm$ 60 V dc at 20 ma in accordance with MIL-STD-188-100 (see 4.3.1.4.1 and 4.3.1.4.2).

(2) Impedance: 1500 to 3000 ohms for polar signals; 1500 to 6500 ohms for neutral signals. Any value in ranges shall be available by adjustment of controls and strapping.

(3) Sensitivity: In accordance with MIL-STD-188-100 (4.3.1.4.6).

b. Input characteristics (low-level to low-level operation):

(1) Signal levels: Low-level, in accordance with MIL-STD-188-114 (5.2.2.1).

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(2) Impedance: In accordance with MIL-STD-188-114 (5.2.2).

(3) Sensitivity: In accordance with MIL-STD-188-114 (5.2.2).

c. Output characteristics:

(1) Signal levels: Low-level, in accordance with MIL-STD-188-114 (5.2.1.2 and 5.2.1.3).

(2) Impedance: In accordance with MIL-STD-188-114 (5.2.1).

(3) Waveshaping: In accordance with MIL-STD-188-114 (5.2.1.7).

(4) Sense: Convertible by strapping to positive or negative mark with input of either sense.

(5) Distortion, telegraph: The distortion of the output signal shall not exceed the distortion of the input signal by more than 2 percent.

d. Modulation rate: 0 to 150 baud.

e. Power requirements: 1.5 watts maximum (not including external signal battery power) (see 2.3).

4.1.2.2 Low-level to high-level converter.

a. Input characteristics:

(1) Signal levels: Low-level, in accordance with MIL-STD-188-114 (5.2.2.1).

(2) Impedance: In accordance with MIL-STD-188-114 (see 5.2.2).

(3) Sensitivity: In accordance with MIL-STD-188-114 (see 5.2.2.2).

b. Output characteristics:

(1) Signal levels: High level (signal power external) neutral or polar by option. Neutral 130 V dc at optionally available 20 or 60 ma. Polar $\pm$60 V dc at 20 ma in accordance with MIL-STD-188-100 (see 4.3.1.4.1 and 4.3.1.4.2).

(2) Impedance: In accordance with MIL-STD-188-100 (see 4.3.1.4.3).

(3) Waveshaping: Provide standard resistiance and capacitance values in accordance with MIL-STD-188-100, (see 4.3.1.4.4) with provisions for substituting other values by "buildout."

(4) Sense: Convertible by strapping to positive or negative mark with input of either sense.

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(5) Distortion, telegraph: The distortion of the output signal shall not exceed the distortion of the input signal by more than 2 percent.

c. Modulation rate: 0 to 150 baud.

d. Power requirements: 1.5 watts maximum (not including external signal battery power) (see 2.3).

4.2 DIGITAL MODULATION RATE CONVERTER.

4.2.1 Application. The modulation rate converter provides interface capability between start-stop and synchronous signals. The converter operates either in a "send" or "receive" mode with one unit required at each end of a simples circuit. The two modes of operation are as follows:

a. Mode one (send): The converter accepts as an input a start-stop (asynchronous) 5-unit code (7.0 or 7.42 units per character including start and stop pulses) or a start-stop, 8-unit code (10.0 or 11.0 units per character including start and stop pulses) at specified rates up to 300 baud and, in free running operation (without character release stepping), produces a synchronous output signal at any specified modulation rate up to 300 baud which is higher than the input rate. Character release pulses are provided to prevent loss of transmitted characters when the input modulation rate is equal to the output rate. The output is a regenerated (retimed and reshaped) signal.

b. Mode two: The converter accepts a synchronous input signal at specified rates not exceeding 300 baud and produces an output start-stop 5-unit code (7.0 or 7.42 units per character) or start-stop 8-unit code (10.0 or 11.0 units per character) at any specified modulation rate not exceeding the input rate and not lower than the modulation rate at the mode one "send" input. The output is a regenerated (reshaped and retime) signal.

c. For both modes, the input and output character rates are always equal.

4.2.2 Equipment Characteristics. The digital modulation rate converter shall conform to the following characteristics:

4.2.2.1 Mode one (send) characteristics.

a. Signal format (switch selectable).

(1) Input: Start-stop, 5-unit code with 7.0 or 7.42 units per character or start-stop, 8-unit code with 10.0 or 11.0 units per character.

(2) Output: synchronous.

b. Modulation rates.

(1) Input: 37.5, 45.45, 74.2, 75.0, 150 and 300 baud.

(2) Output: 75, 150, and 300 baud.

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c. Signal levels.

(1) Input: High-level or low-level.

(2) Output: Low-level.

d. High-level digital interface requirements (input):

(1) Signal levels: High-level (signall power external) neutral or polar by option. Neutral 130 V dc at optionally available 20 or 60 mA. Polar ±60 V dc at 20 mA in accordance with MIL-STD-188-100 (4.3.1.4.1 and 4.3.1.4.2).

(2) Impedance: 1500 to 3000 ohms for polar signals; 1500 to 6500 ohms for neutral signals. Any value in these ranges shall be available by adjustment of controls and strapping.

(3) Sensitivity: In accordance with MIL-STD-188-100 (4.3.1.4.6).

e. Low-level digital interface requirements (input): In accordance with MIL-STD-114, (5.2.2.1) selection of high- and low-level signals (input) is made with internal strapping or optional logic card substitution.

f. Sense: Convertible by strapping to positive or negative MARK.

g. False start protection: Rejects start pulse of less than 1/2 bit.

h. Character release pulse output: One cycle per character. Low-level DC polar. Output source resistance and voltage in accordance with MIL-STD-188-114 (5.2.1).

i. Modulation rate conversion: From any specified input rate to any higher output rate operating without character release. Equal input and output rates with charater release.

4.2.2.2 Mode two (receive) characteristic:

a. Signal format (switch selectable):

(1) Input: Synchronous as furnished by another unit operating in mode one.

(2) Output: Same as input format for mode one (see 4.2.2.1a(1)).

b. Modulation rates:

(1) Input: 75, 150, and 300 baud.

(2) Output: 37.5, 45.45, 74.2, 75.0, 150, 300 baud.

c. Signal levels:

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(1) Input: Low level.

(2) Output: High level or low level.

d. High-level digital interface requirements (output):

(1) Signal levels: High level (signal power external) neutral or polar by opion. Neutral 130 V dc at optionally available 20 or 60 mA. Polar ±60 vdc at 20 mA in accordance with MIL-STD-188-100 (4.3.1.4.1 and 4.3.1.4.2).

(2) Impedance: In accordance with MIL-STD-188-100 (4.3.1.4.3).

(3) Waveshaping: In accordance with MIL-STD-188-100 (4.3.1.4.4).

e. Low-level digital interface requirements: Input requirements in accordance with MIL-STD-188-114, (5.2.1.2 and 5.2.1.3). Selection of high- and low-level signals (output) is made with internal strapping or optional logic card substitution. Output requirements in accordance with MIL-STD-188-114, (5.2.1).

f. Sense: Convertible by strapping to positive or negative MARK.

g. Modulation rate conversion: From any specified input rate to a specified output rate that is equal or lower than the input but not lower than the input to the "send" unit operating in mode one.

4.2.2.3 Characteristics for both modes.

a. Distortion, telegraph: The converter shall accept signals with up to 48% distortion and provide a regenerated signal having no more than 2% distortion.

b. Timing signal: Externally provided. Input interface shall accept a low-level dc polar signal in accordance with MIL-STD-188-114, (5.2.2.1).

c. Power requirements (input): 20 watts maximum (not including external signal battery power (see paragraph 2.3).

4.3 MODE SELECTING REGENERATIVE REPEATER.

4.3.1 Application. The mode selecting regenerative repeater is a solid-state device that regenerates (retimes and reshapes) synchronous and asynchronous digital signals. There are three modes of operation of the repeater:

a. Synchronous mode: In this mode, the repeater accepts, regenerates and transmits a synchronous digital signal.

b. Asynchronous mode: In this mode, the repeater accepts an asynchronous signal, regenerates and transmits the signal at the same modulation rate with start and stop pulses of unit length.

c. Automatic mode: In this mode, the repeater automatically switches to

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either the synchronous or asynchronous mode by sensing the mode of the incoming signal. After switching, the repeater performs as described in 4.3.1a and 4.3.1b above.

4.3.2 Equipment characteristics. The mode selecting regenerative repeater shall conform to the following characteristics:

a. High-level digital interface requirements:

(1) Signal level (input and output): High level (signal power external) neutral or polar by option. Neutral 130 V dc at optionally available 20 or 60 mA. Polar ±60 Vdc at 20 mA in accordance with MIL-STD-188-100, (4.3.1.4.1 and 4.3.1.4.2).

(2) Impedance (input): 1500 to 3000 ohms for polar signals; 1500 to 6500 ohms for neutral signals. Any value in these ranges shall be available by adjustment of controls and strapping.

(3) Impedance (output): In accordance with MIL-STD-188-100 (4.3.1.4.3).

(4) Sensitivity: In accordance with MIL-STD-188-100 (4.3.1.4.6).

(5) Waveshaping: In accordance with MIL-STD-188-100 (4.3.1.4.4).

(6) Sense: Convertible by strapping to positive or negative MARK.

b. Low-level digital interface requirements (input and output): ±6 V positive MARK in accordance with MIL-STD-188-114 (5.2.1.2 and 5.2.1.3) selection of high- and low-level signals (output and input) is made with internal strapping or optional logic and substitution.

c. Distortion, telegraph: The repeater shall accept signals with up to 48 percent distortion and provide a regenerated signal having no more than 2 percent distortion.

d. Modulation rate: "(Switch selectable by separate switches for asynchronous and synchronous)".

(1) Asynchronous: 44.5, 45, 45.45, 55.6, 56.3, 74.2, 75 and 110 baud.

(2) Synchronous: 37.5, 61.12, 75, 150, 300, 600, 1200 and 2400, 4800 and 9600 baud.

e. Automatic mode selection. In the automatic mode and at preselected modulation rates, "(any rates in (d)(1) above for asynchronous, and (d)(2) above for synchronous)" the repeater shall be capable of switching to either the synchronous or asynchronous mode to conform with the input signal. Recognition and switching shall occur within 100 transitions of the input signal. Loss of signals or power shall not cause mode changes.

f. Signal codes:

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(1) 1 to 16-bit synchronous signals.

(2) 5 to 16-bit asynchronous signals with 1.0 to 2.0 element stop lengths.

g. False start protection: For asynchronous operation, when the incoming signal is less than 1/2 bit in duration.

h. Clock requirements:

(1) Internal Accuracy and stability is $1x10^{5}$ for 30 days or $1x10^{6}$ for 24 hours.

(2) Synchronization: Internal clock can synchronize to an external clock at 307.2 kHz. Input interface shall be $\pm$6 V in accordance with MIL-STD-188-114, (5.2.2 and 5.2.2.2).

(3) External (unique) transmission rate: Repeater capable of operation from an external clock of 1 Hz per bit of output signal. Input interface is given in (2) above.

i. Output timing signal: The repeater shall be capable of retransmitting a timing signal in synchronism with the regenerated data signal on a separate timing signal output with interface requirements given in 4.3.2(b) above.

j. Power requirements: 20 watts maximum. (See 2.3).

k. Synchronization: When the repeater is operating in the automatic mode, synchronization must be obtained within 100 transitions after mode selection has been made.

l. Skew and jitter: Less than 1 percent.

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Chapter V

5. VF CIRCUIT SIGNALING EQUIPMENT. The following types of equipment are commonly used in voice frequency signaling circuits:

a. DX signaling unit.

b. E & M to 20-Hz ringing converter.

c. Dial loop to E & M converter.

d. Single frequency (SF) signaling unit.

e. Station 20-Hz ringing supply.

f. Pulse link repeater.

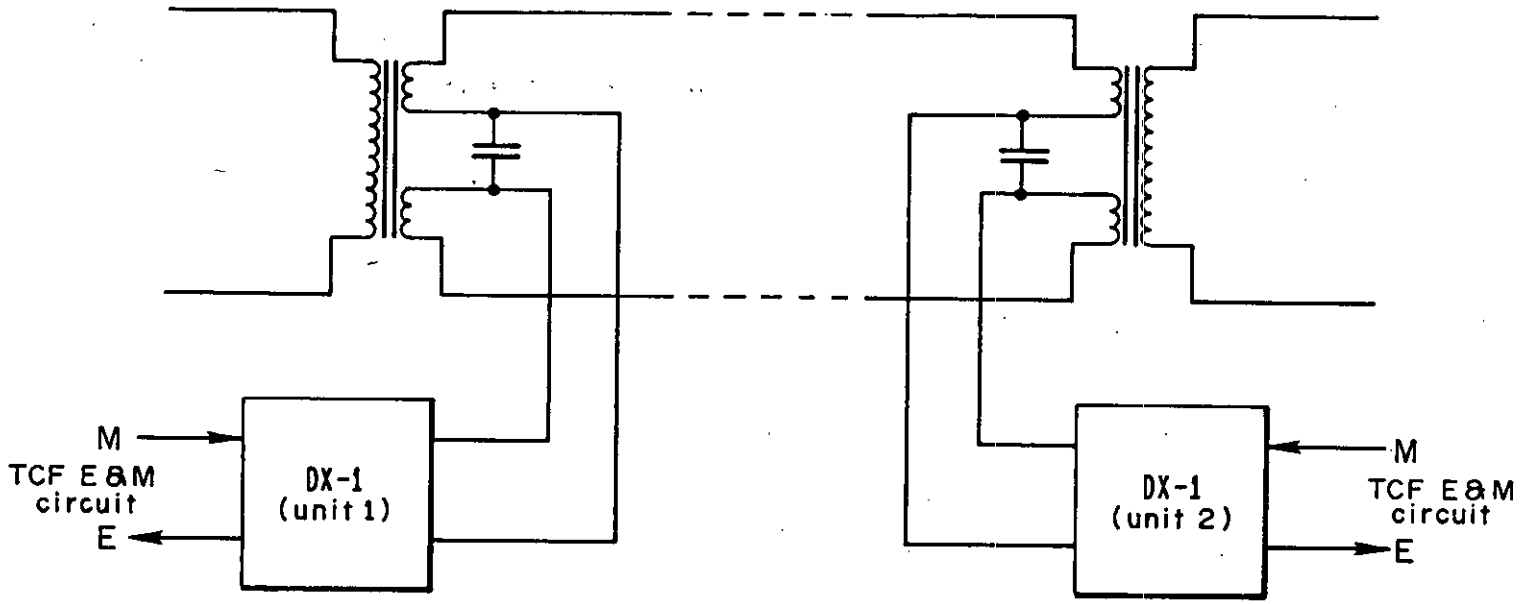
5.1 DX SIGNALING UNIT.

5.1.1 Application. DX units are used in pairs for two-way dc signaling over metallic voice frequency (VF) trunk circuits having a loop resistance up to 5000 ohms and where large and varying differences in earth potential may exist between the ends of the circuit. The DX signaling circuit normally uses the same cable pair as the voice frequency path with DX units connected to the voice frequency circuit through the split-center tap terminals of repeat coils. One conductor of the cable pair is used for signaling in both directions, the second conductor is required to permit the use of a balanced circuit which is self-compensating for earth potential differences, provides partial compensation for battery supply variations and is insensitive to induced longitudinal currents. DX signaling can be used with both 10 pps and 20 pps dialing. For speeds up to 12 pps, 47 to 67% break overseas and 49 to 72% break in CONUS with a receiving end distortion not exceeding +4% break, permitting the use of two DX circuits in tandem on one trunk. Two types of DX units are used. DX-1 units are designed to accept standard E & M signals, i.e., the M-lead terminal is the input accepting ground or -48 V and the E-lead terminal is the output presenting an open circuit or ground condition. DX-2 units are designed to provide a source of E & M signals, i.e., the M-lead terminal is the output and provides ground or -48 V and the E-lead terminal is the input accepting an open circuit or a ground condition. The most common DX signaling circuit consists of two DX-1 units as shown in figure 9.

5.1.1.1 For trunk circuits where the end-to-end loop resistance exceeds 5000 ohms a DX-1 to DX-2 extension circuit can be used in tandem with the DX-1 to DX-1 circuit. The DX-1 to DX-2 circuit serves as a single repeater extending the TCF E & M signals to the DX-1 unit as shown in figure 10.

5.1.1.1.1 DX-2 to DX-2 signaling circuits can be used when both ends of the circuit interface with equipments designed to accept E & M signaling. Figure 11 shows an example of such a case where the signal extension circuit of figure 10 is required to interface with a single frequency (SF) signaling unit rather than TCF E & M circuits.

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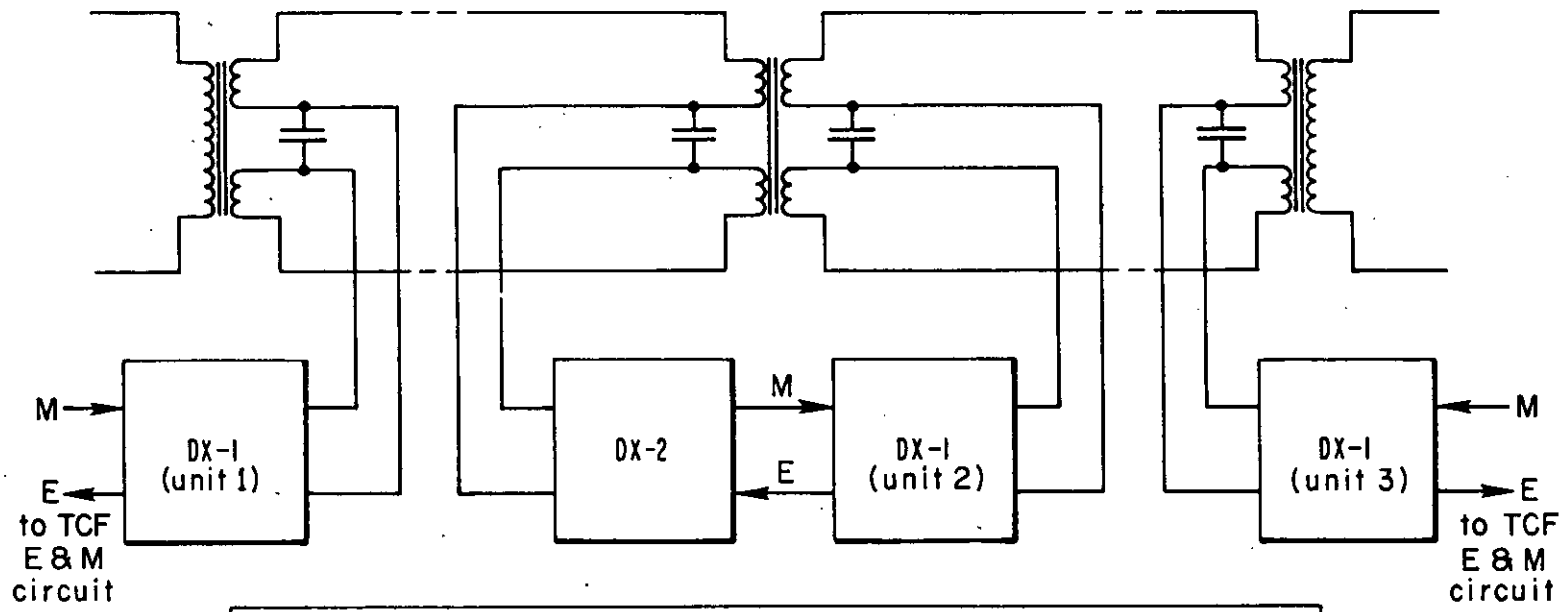


DX-1 (unit 1) input to M-lead terminal	Corresponding DX-1 (unit 2) E-lead terminal (output)
ground	open circuit
-48 V	ground

Corresponding DX-1 (unit 1) E-lead terminal (output)	DX-1 (unit 2) input to M-lead terminal
open circuit	ground
ground	-48 V

Figure 9. DX-1 to DX-1 Signaling for Trunk Circuits with Loop Resistance up to 5000 ohms.

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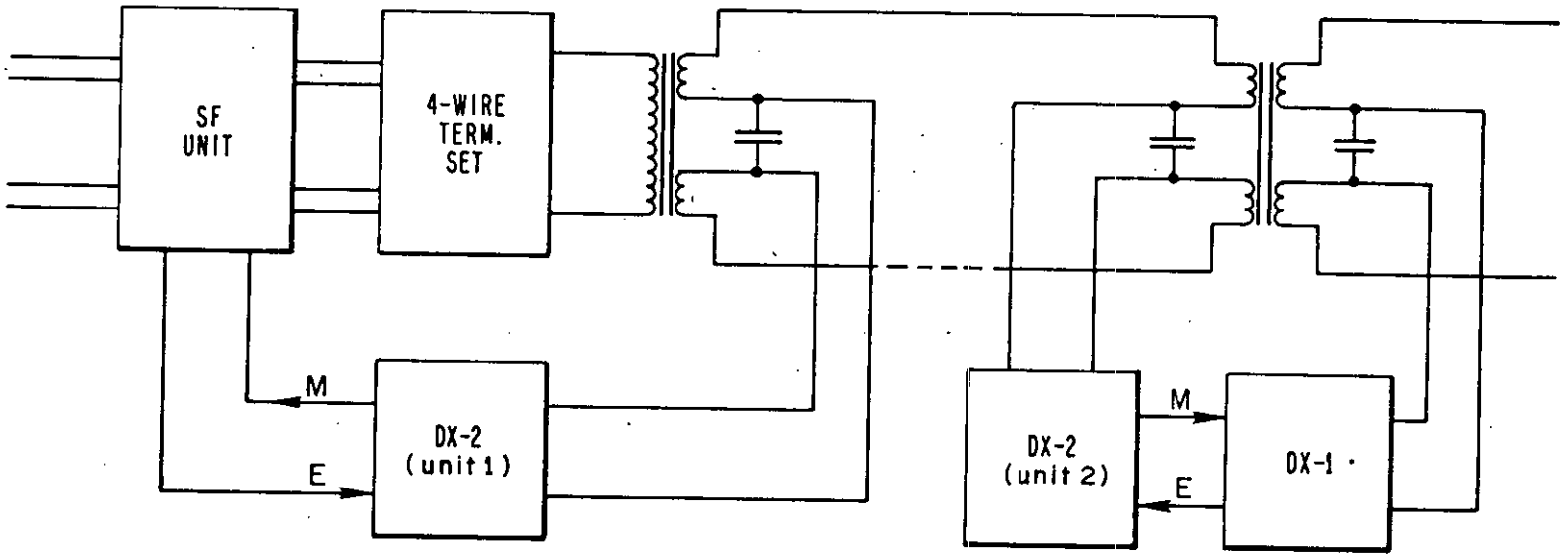


DX-1 (unit 1) input to M-lead terminal	Corresponding DX-2 M-lead terminal output
ground	ground
-48 V	-48 V

Corresponding DX-1 (unit 1) E-lead terminal (output)	DX-2 input to E-lead terminal
open circuit	open circuit
ground	ground

Figure 10. DX-1 to DX-2 Signaling Extension for Trunk Circuits with End-to-End Loop Resistance Exceeding 5000 ohms.

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DX-2 (unit 1) input to E-lead terminal	Corresponding DX-2 (unit 2) M-lead terminal (output)
open circuit	ground
ground	-48 V

Corresponding DX-2 (unit 1) M-lead terminal (output)	DX-2 (unit 2) input to E-lead terminal
ground	open circuit
-48 V	ground

Figure 11. DX-2 to DX-2 Signaling Circuits.

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5.1.2 Equipment characteristics. The DX signaling units shall conform to the following characteristics:

a. The units shall have the following terminals:

(1) M lead.

(2) E lead.

(3) A lead.

(4) B lead.

(5) Ground.

(6) -48 V supply.

b. Operational requirements. The following operational requirements shall apply from the input of one unit to the output of the second unit of a pair connected via a two-wire trunk circuit which may have a loop resistance between 0 and 5000 ohms and with an earth potential difference between the two terminals that may vary between 0 and $\pm$45 V. Each DX unit is coupled to the trunk via the split-center taps of a repeat coil with a 2 uf capacitor connected across these taps.

(1) Pulse rate: The AUTOVON dial pulse sender outpulses at 9.5 to 10.5 pps, 47 to 67% break overseas and 49 to 72% break in CONUS.

(2) Distortion: For pulse rates up to 12 pps the difference between the output signal break interval and input signal break interval shall not exceed 4% of the total pulse interval.

(3) DX-1 units shall accept M-lead input and provide E-lead output. DX-2 units shall accept E-lead input and provide M-lead output. See figures 9 to 11.

c. Equipment parameters. Unless otherwise specified, equipment parameters shall be measured on single units with the A and B terminals open and with a power source of -48 V dc having an internal impedance of less than 1 ohm.

V_A = voltage between A-lead terminal and ground.

V_B = voltage between B-lead terminal and ground.

Z_A = resistance between A-lead terminal and ground.

Z_B = resistance between B-lead terminal and ground.

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d. DX-1 unit parameters.

(1) M-lead conditions	V_A	V_B
ground	0 V max -2 V min	-17 V max -20 V min
-48V	-44 V max -48 V min	-19 V max -22.5 V min

In either condition of the M-lead Z_A and Z_B shall be 600 ohms $\pm$20% with A_A and Z_B matched to better than 1%.

(2) M-lead terminal resistance to ground (A- and B-lead terminals grounded): 550 ohms minimum, 5000 ohms maximum.

(3) E-lead terminal resistance to ground (B-lead open, M-lead to -48 V):

A-lead condition	E-lead terminal resistance to ground
gnd through 600 ohms	1 ohm max
-48V through 600 ohms	100,000 ohms min

e. DX-2 unit parameters.

(1) E-lead condition	V_A	V_B
open	0 V max -2 V min	-17 V max -20 V min
ground	-44 V max -48 V min	-19 V max -22.5 V min

In either condition of the E-lead, Z_A and Z_B shall be 600 ohms $\pm$20% with Z_A and Z_B matched to better than 1%.

(2) E-lead terminal resistance to -48 V: 750 ohms minimum, 2000 ohms maximum.

(3) M-lead parameters (B-lead open, E-lead to ground):

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A-lead condition	M-lead voltage to gnd	M-lead resistance to gnd
gnd thru 600 ohms	OV	1 ohm max
-48V thru 600 ohms	-48V	25 ohms max (for load current up to 100mA)

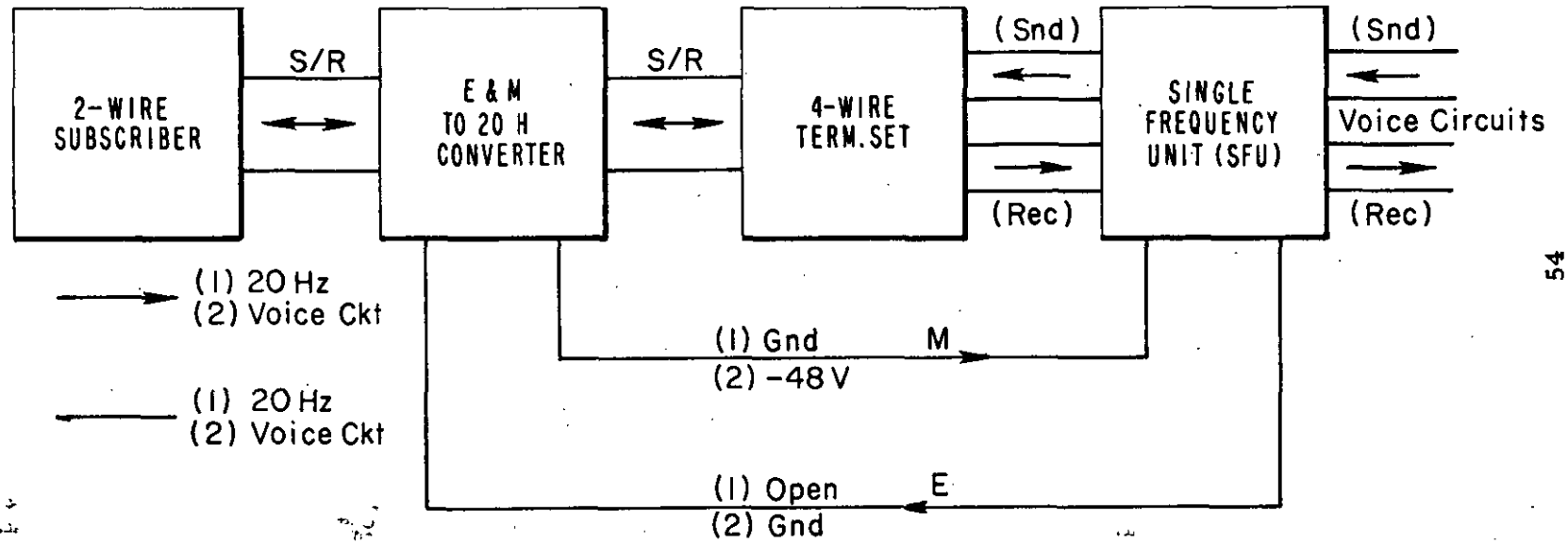
(4) No damage to the equipment shall result from a short circuit from the M-lead terminal to ground of up to 2 minutes duration with the A-lead connected to -48 V through 600 ohms.

f. Power requirements (A, B, M and E leads open): -48 V (See 2.3) 3 W maximum per unit.

5.2 E AND M TO 20-HZ RINGING CONVERTER.

5.2.1 Application. (See figures 12 through 17). The E&M to 20-Hz ringing converter is used to interface subscribers with the telephone system where the telephone system utilizes E&M signaling and the subscriber utilizes 20-Hz ringing signals. When not signaling, the converter provides a normal through voice path connection. The converter provides a 20-Hz ringing signal to a two- or four-wire circuit of 600-ohms impedance upon receiving a signal on the E-lead and produces a signal on the M-lead upon receiving a 20-Hz ringing signal from a two- or four-wire circuit of 600-ohms impedance. When the converter provides the 20-Hz ringing signal to the subscriber, the line side of the converter is disconnected and terminated. This is necessary to prevent the 20-Hz ringing signal from going out on the line. This action is referred to as a line split.

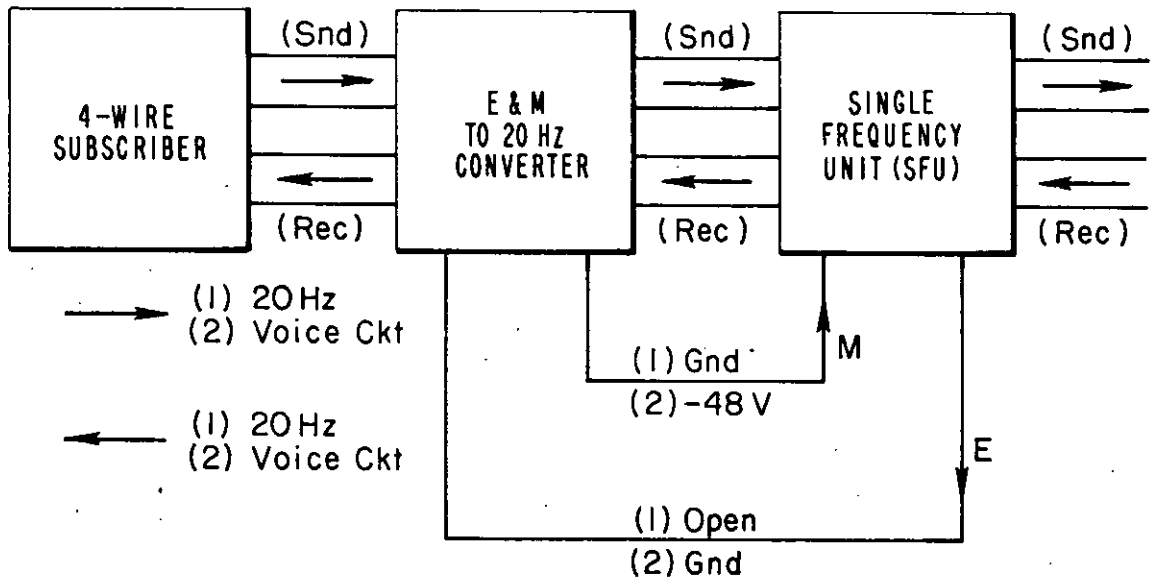
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Note: The senses of the E & M leads are reversed for option 2; i.e., a ground appearing on the E lead will produce 20-Hz signaling and, upon detection of a 20-Hz signal from the subscriber, the converter will produce -48 V on the M lead.

Figure 12. Two-Wire Application—Option 1.

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Note: The senses of the E & M leads are reversed for option 2; i.e., a ground appearing on the E lead will produce 20-Hz signaling, and upon detection of a 20-Hz signal from the subscriber, the converter will produce -48 V on the M lead.

Figure 13. Four-Wire Application—Option 1.

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5.2.2 Equipment characteristics. The E&M to 20-Hz ringing converter shall conform to the following characteristics:

a. Frequency response, voice channel (output): ±0.25 dB from 300 to 3400 Hz referred to 1000 Hz with 600-ohm source and termination impedances.

b. Impedance, voice channel (input and output): 600 ohms from 300 to 3400 Hz when the opposite side is terminated with a 600-ohm ±1% resistor.

c. Return loss, voice channel (input and output): 26 dB minimum from 300 to 3400 Hz when the opposite side is terminated with a 600-ohm ±1% resistor.

d. Longitudinal balance, voice channel (input and output): 50 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms.

e. Longitudinal-input-to-metallic-output-voltage ratio, voice channel: 50 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms (see 2.5).

f. Envelope differential delay, voice channel (output): 100 microseconds maximum from 300 to 3400 Hz with 600-ohm source and termination impedances.

g. Noise output voice channel: -70 dBm (flat weighting) with input and output terminated to 600 ohms.

h. Distortion, total harmonic (voice channels): 40 dB minimum from 300 to 3400 Hz with 600-ohm source and termination impedances at levels up to 10 dBm.

i. Insertion loss for 1 kHz tone: 0 ±0.1 dB for either send or receive circuits.

j. VF transmission line impedance: Nominally 600 ohm (See 2.4).

k. Surge protection. (See 2.2)

l. Strappable options:

Option 1 (See figures 5-5, 5-6, 5-7, 5-9).

(1) The ringing converter shall cause a 20-Hz ringing signal to be impressed on a two- or four-wire circuit of 600-ohms impedance when an open circuit is impressed on the E lead.

(2) The ringing converter shall cause a normal through voice path connection when a ground is impressed on the E lead.

(3) The ringing converter shall produce a ground on the M lead upon detection of a 20-Hz ringing signal from a two- or four-wire circuit of 600-ohms impedance (see figures 13 and 14).

(4) The ringing converter shall produce nominal -48 V dc (from an external source) on the M lead when a 20-Hz ringing signal is not detected.

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Option 2 (See figures 15 and 17).

(1) The ringing converter shall cause a 20-Hz ringing signal to be impressed on a two- or four-wire circuit of 600-ohms impedance when a ground is impressed on the E lead.

(2) The ringing converter shall cause a normal through voice path connection when an open circuit is impressed on the E lead.

(3) The ringing converter shall produce nominal -48 V dc (from an external source) on the M lead upon detection of a 20-Hz ringing signal from a two- or four-wire circuit of 600-ohms impedance.

(4) The ringing converter shall produce a ground on the M lead when a 20-Hz ringing signal is not detected.

m. Split line termination (both options): During periods when the voice circuit is split, the line side of the split shall be terminated into a 600 ±10% ohm resistor in series with a 1.0 ±10% uF capacitor.

n. 20-Hz signal source: 35 to 105 V rms, 16-2/3 to 50 Hz from an external source.

o. Sensitivity, 20-Hz detector: Open-circuit ringing voltage 35 to 120 V rms, 16 to 50 Hz with ringing voltages from 1 to 2 seconds duration and ringing cycle between 4 and 8 seconds.

p. VF isolation from signal circuits: Transmitted VF levels up to 10 dBm in the frequency range of 300 to 3400 Hz shall not change the state of the M lead.

q. Power requirements: 4 watts maximum at 48 V ±10% and 1.0 watt maximum to operate 20-Hz detector (See 2.3).

5.3 DIAL LOOP TO E AND M CONVERTER SET.

5.3.1 Application. The dial loop to E&M converter set provides for two-way dial loop interface between a central office and a dial telephone subscriber over an access line which somewhere requires E&M signaling. The set consists of two converters. One converter is used for signal conversion at the central office and is referred to as a central office dial loop to E&M converter. The other converter is used for signal conversion at the subscriber terminal and is referred to as a subscriber dial loop to E&M converter.

a. The subscriber dial loop to E&M converter accepts on-hook and off-hook indications (including dial pulse information) from the subscriber instrument and converts them to M signals. The converter also accepts E signals and converts them to ringing signals which are applied to the subscribers instrument.

b. The central office dial loop to E&M converter accepts ringing signals from the central office and converts them to M signals. The converter also accepts E-lead signals and converts them to on-hook and off-hook indications (including dial pulse information) for use by the central office.

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5.3.2 Equipment characteristics (See figure 18, table III). The dial loop to E&M converter set shall conform to the following characteristics: The following refers to the operation of a dial loop to E&M converter set in a typical application interfacing a dial telephone subscriber with a dial central office over a radio carrier.

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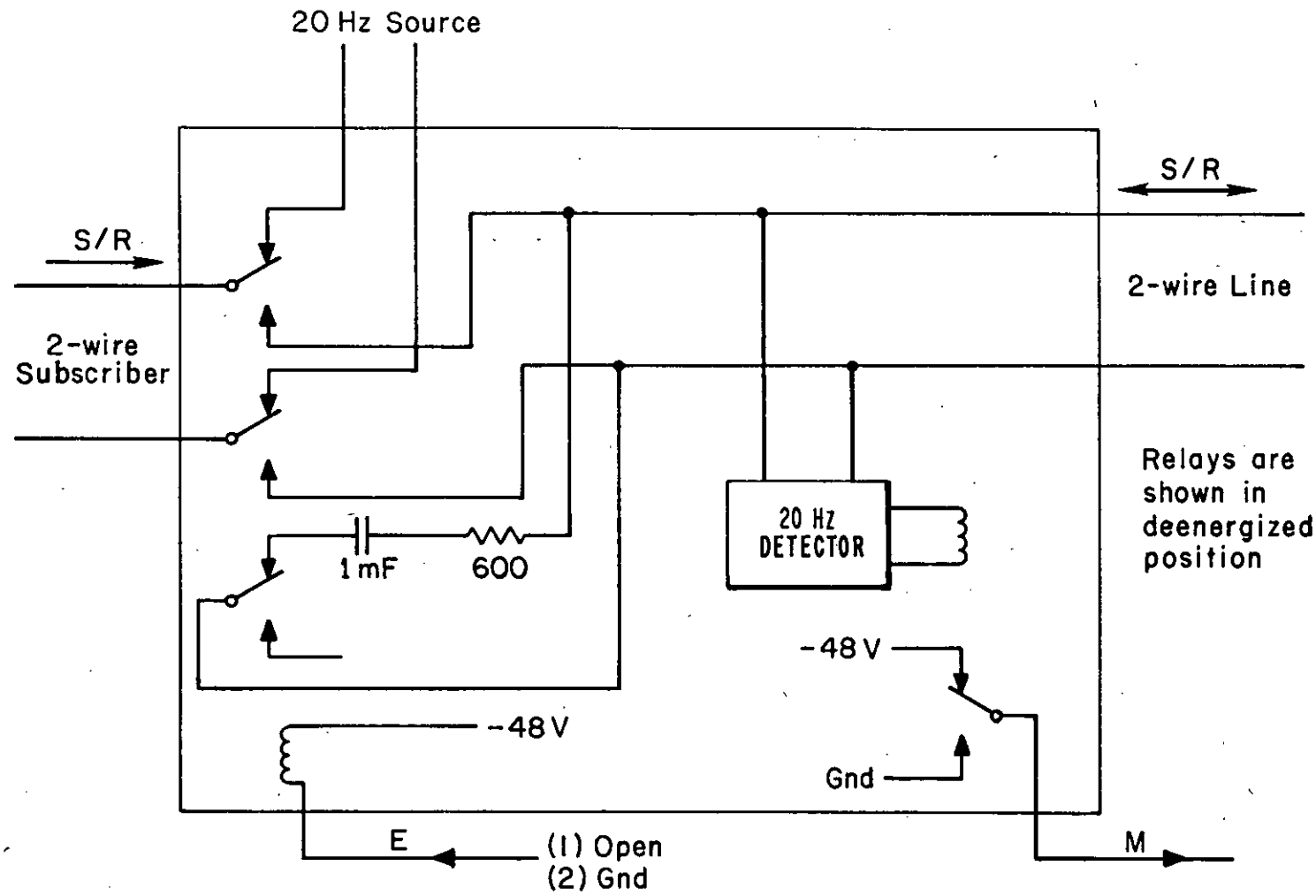


Figure 14. E & M to 20-Hz Converter—Two-Wire Application—Option 1.

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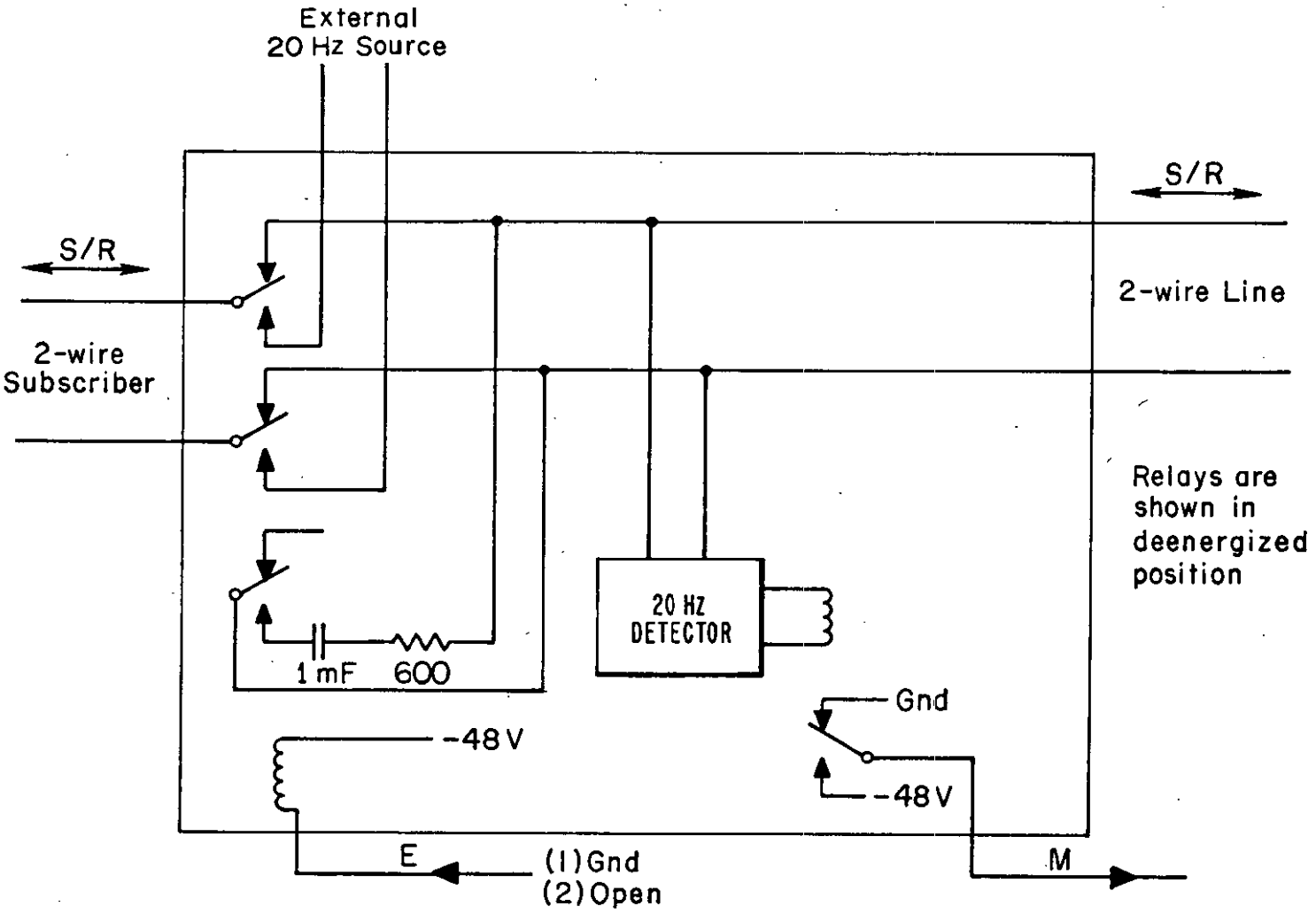
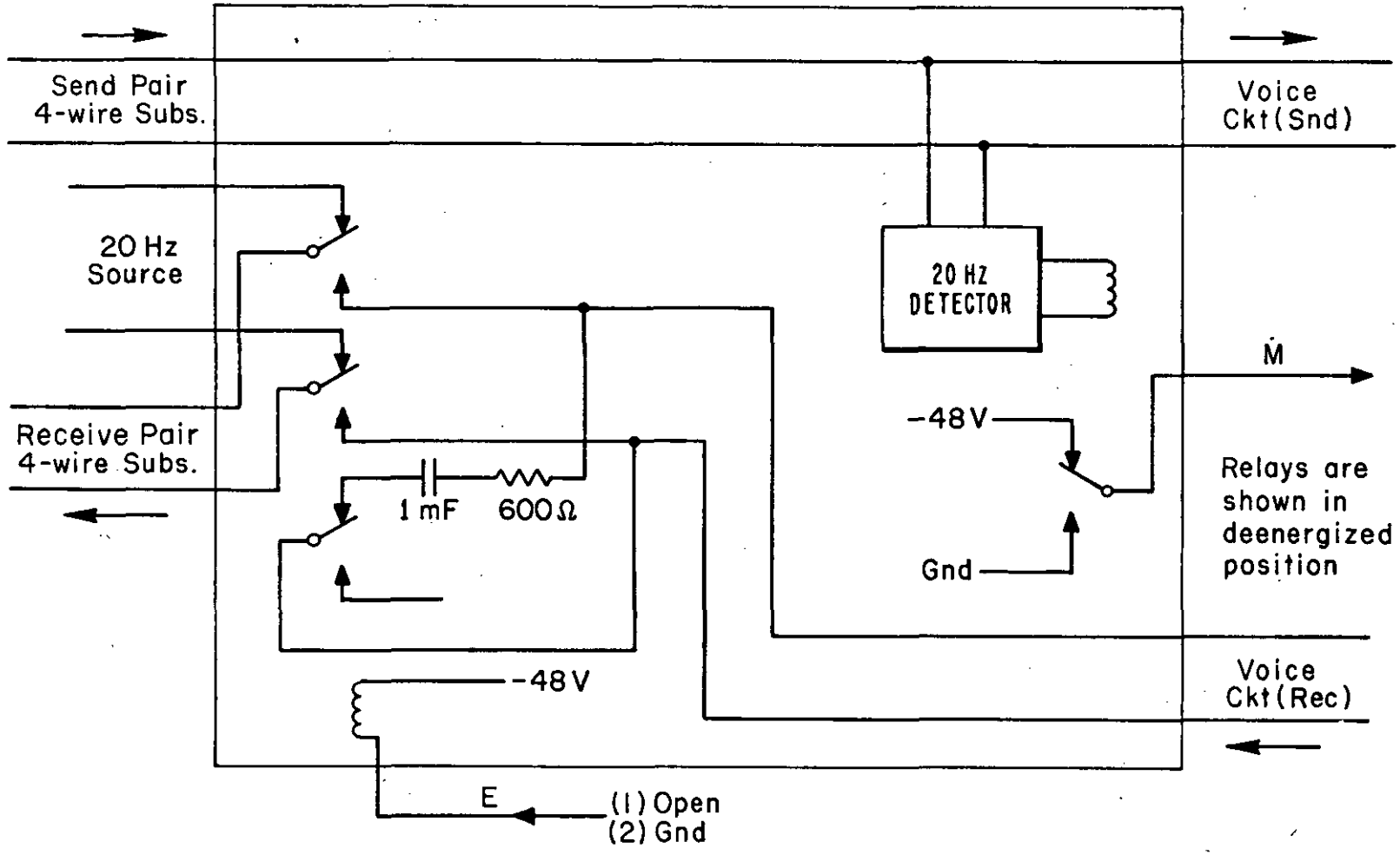


Figure 15. E & M to 20-Hz Converter—Two-Wire Application—Option 2.

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Figure 16. E & M to 20-Hz Converter—Four-Wire Application—Option 1.

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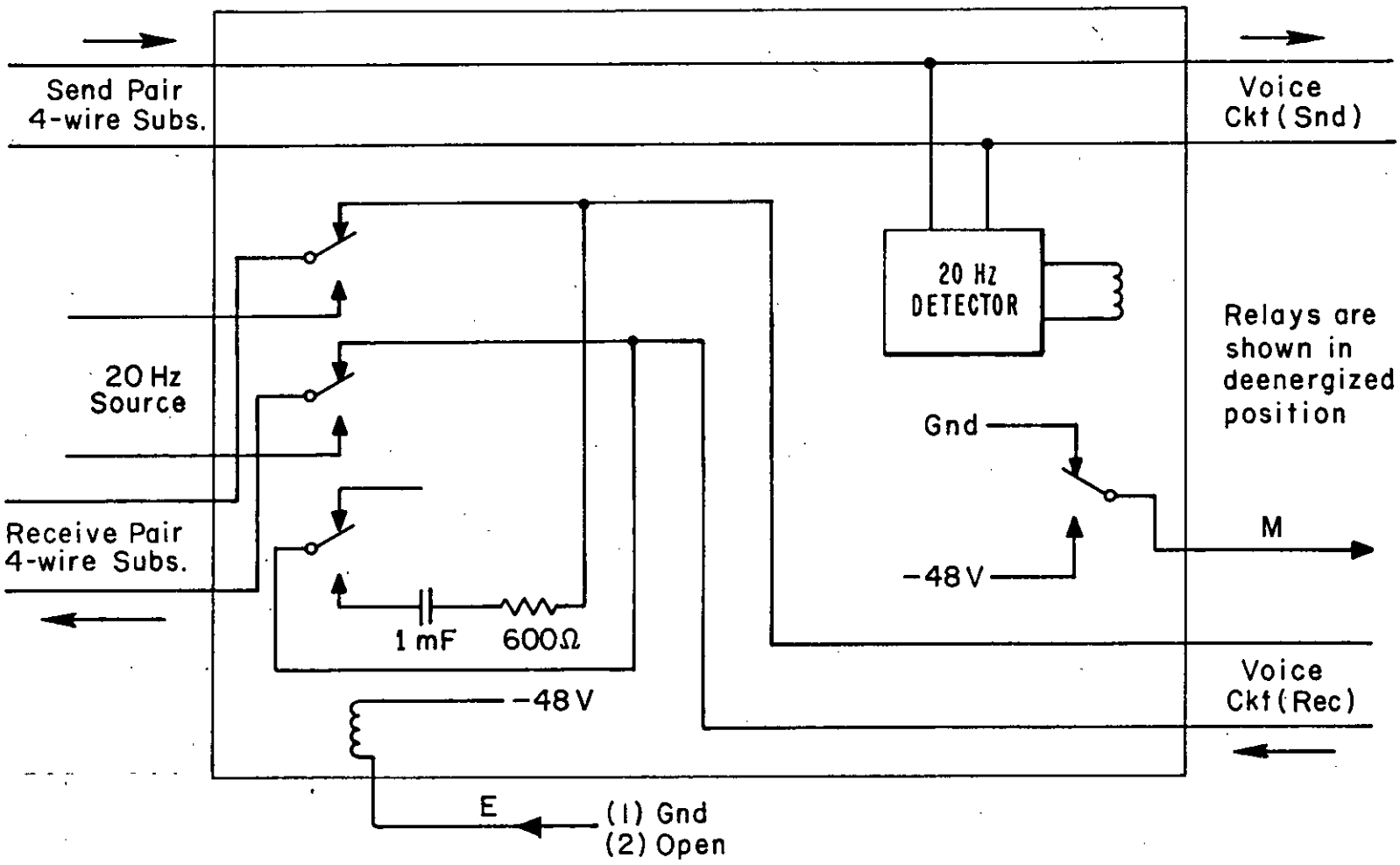


Figure 17. E & M to 20-Hz Converter—Four-Wire Application—Option 2.

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TABLE IV CONVERTER OPERATION

	CONDITION	SUBSCRIBER END OF DIAL LOOP	CENTRAL OFFICE END OF DIAL LOOP
1.	Normal System-"On-Hook" and no Incoming Call	M Lead Grounded*	E Lead Open-Circuited
		E Lead Grounded*	-48 V dc (±5 V) on M Lead
2.	Ringing Condition "On-Hook" and Incoming Call	M Lead Grounded	E Lead Open-Circuited
		E Lead Open-Circuited	M Lead Grounded
3.	Non-ringing Condition. Call answered and "Off-Hook"	-48 V (±5 V) on M Lead	E Lead Grounded
		E Lead Grounded	-48 V dc (± 5 V) on M Lead
4.	Initiating Call "Off-Hook"	-48 V (±5 V) on M Lead	E Lead Grounded
		E Lead Grounded	-48 V dc (±5 V) on M Lead
5.	During Dialing	M Lead Pulses	E Lead Pulses
		E Lead Grounded	-48 V dc (±5 V) on M Lead

*E-lead conditions are imposed on the converter by an external circuit (SFSU unit in figure 10).

The converter imposes (transmits) M-lead conditions on an external circuit (the SF converter in figure 17).

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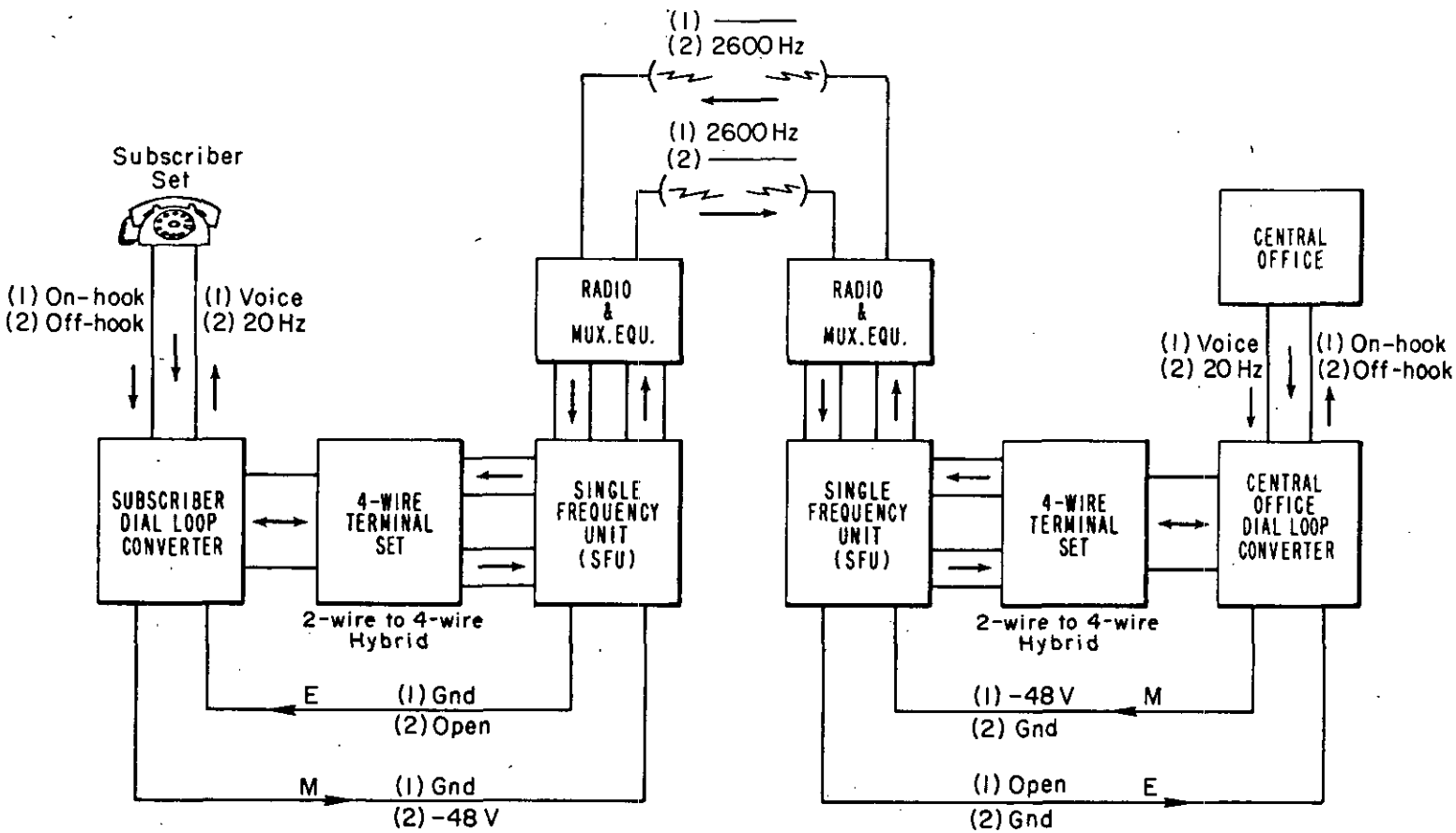


Figure 18. Dial Loop to E & M Converter Set.

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a. The central office dial loop to E&M converter shall cause an off-hook indication (closed circuit) to appear at the central office end of the circuit upon receipt of a ground on the E lead from a single frequency signaling unit (SFSU). The converter shall cause an on-hook indication (open circuit) to appear at the central office end of the circuit upon receipt of an open circuit on the E lead from the SFSU unit. In addition, the converter accepts 20-Hz ringing signals from the central office and shall cause a ground to appear on the M output lead going to the SFSU. When no ringing signal is provided from the central office, the converter shall cause a nominal -48 V dc (from an external source) to appear on the M-output lead going to the SFSU. Other supervisory signals such as ringback tone, dial tone and busy tone shall be transmitted through the converter without conversion. During dialing, the E lead of the SFSU provides alternate ground and an open circuit at 9 to 12 pulses per second, with 57 $\pm$10% break to the converter. Duration of dial pulses applied to the central office equipment shall be within 4 milliseconds of the duration of the pulses received from the E lead of the SFSU. When a ringing signal is provided from the central office, the voice path toward the subscriber is terminated into a resistor of 600 ohms $\pm$10% in series with a fixed 1 to 2 uF capacitor. A monitor jack shall be provided for monitoring the talking circuit without interruption and a test jack for testing of the impulse signal circuit.

b. The subscriber dial loop to E&M converter shall cause a -48 V dc (from an external source) to appear on the M-output lead going to an SFSU when an off-hook (closed circuit) condition exists at the subscribers telephone instrument. The converter shall cause a ground to appear on the M-output lead going to the SFSU when an on-hook (open circuit) condition exists at the subscribers telephone instrument. In addition, the converter shall cause a 20-Hz ringing voltage to appear at the subscribers telephone instrument upon receipt of an open circuit on the E lead coming from the SFSU, and terminates the voice path toward the central office into a resistor of 60 ohms $\pm$10% in series with a 1 to 2 uF fixed capacitor. The converter shall cause the subscriber instrument to be connected to the voice path upon receipt of a ground on the E lead coming from the SFSU. Other supervisory signals such as ringback tone, dial tone, and busy tone shall be transmitted through the converter without conversion. During dialing, the converter shall cause an alternate condition of -48 V dc and ground to be transmitted to the M lead of the SFSU at a dialing rate of 9 to 12 pulses per second, with nominal 57% break. Duration of dial pulses applied to the M lead is within 4 milliseconds of the duration of pulses received from the subscribers telephone instrument. A monitor jack shall be provided for monitoring the talking circuit without interruption and a test jack for testing of the impulse signal circuit.

c. 20-Hz signal source: 35 to 105 V rms, 16-2/3 to 50 Hz, from an external source.

d. Power requirements: For both converters, 6 watts maximum at -48 V dc $\pm$10%, and 1 watt maximum to operate the 20-Hz detector.

e. Insertion loss for 1-kHz tone: 0 $\pm$0.1 dB.

f. Frequency response: Flat within $\pm$0.1 dB from 300 to 3400 Hz.

g. Impedance (input and output): 600 ohms with all other inputs and outputs terminated in 600 ohms.

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h. Return loss: 26 dB minimum.

i. Longitudinal balance, voice channel (input and output): 40 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms.

j. Longitudinal-input-to-metallic-output-voltage ratio (voice channel: 40 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms (See 2.5).

k. Crosstalk, voice channel: 70 dB minimum coupling between VF channels within the equipment or adjacent equipment, from 300 to 3400 Hz with 0 dBm tone levels.

l. Noise, voice channel (outputs): -70 dBm maximum from 300 to 3400 Hz (flat weighting) with input and outputs terminated in 600 ohms.

m. Distortion, total harmonic, voice channel (output): 40 dB minimum from 300 to 3400 Hz.

n. Signal level, maximum (input): A signal level of +10 dBm shall not cause deviation from the above requirements. The insertion loss at 1 KHz shall not increase by more than 0.2 dB referred to the loss at 0 dBm with a test tone of +10 dBm.

o. VF transmission line impedance: Nominally 600 ohms (See 2.4).

p. Surge protection: (See 2.2).

5.4 SINGLE FREQUENCY SIGNALING UNIT (SFSU).

5.4.1 Application. The SF signaling unit is designed to transmit dial trunk supervisory and address signals and AUTOVON supervisory signals over four-wire voice frequency channels. The signaling information is accepted and delivered across the drop interface over E&M signal leads. The signaling information is transmitted over the four-wire line by means of 2600-Hz keyed tone signals. Although the signaling system is fairly complex in detail, it is simple in principle. (See figure 19). Normally, when the circuit is idle, a steady tone is transmitted over the line holding certain relays in the operated condition at the receiving end. Signaling is accomplished by removing and reapplying this tone, which in turn releases and reoperates the distant relays. The same signaling frequency can be used for both directions if the four-wire channels have separate one-way transmission paths from terminal to terminal.

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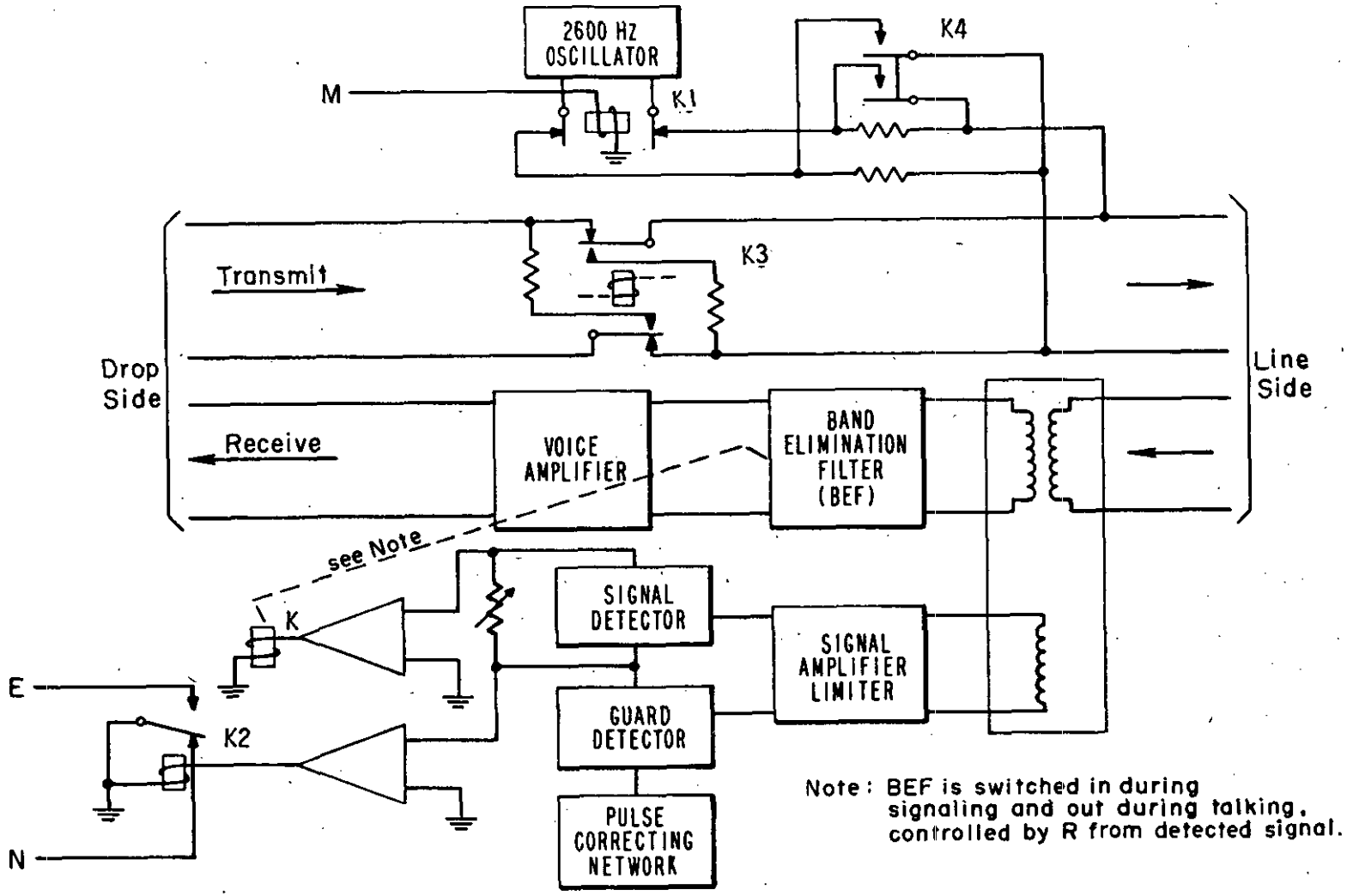


Figure 19. Functional Block Diagram Illustrating SF Signaling Unit Operation.

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a. The SFSU signaling system is provided as a separate entity. It is connected in series with the transmitting and receiving branches of the line circuit at each end of a trunk and is connected to the terminal relay circuit (trunk circuit) by two one-way signaling leads (E&M leads).

b. When a trunk equipped with 2600-Hz SF units is in idle condition, a 2600-Hz tone is present in both directions of transmission.

c. An off-hook signal originating at one terminal results in a change in state on the originating M lead from ground to battery (-48 V). As a result, a relay in the keying circuit (K1 in figure 19) operates the tone keying lead control which removes the 2600-Hz from the sending line. The loss of tone is detected at the distant SF-signaling unit which as a result switches its E lead from an open to a ground condition (relay K2). The grounded E lead results in a request for connection of an incoming sender. Since there may be a slight delay in securing a sender, the terminating trunk circuit changes the M-lead state at the terminating office from ground to battery, resulting in the removal of the 2600-Hz tone from the line facility, i.e., from the terminating to the originating office. This results in a signal on the originating office E lead to delay the transmission of the address information.

d. When an incoming sender register is connected, the terminating trunk circuit restores ground to the M lead of the SFSU unit, and the 2600-Hz tone is reapplied to the line. At the originating terminal the reapplied tone is detected and the resulting E-lead signal indicates that pulsing of the address information can begin.

e. If dial pulsing is used for addressing, the M-keyer relay (K1) is operated and released in accordance with the dial pulses being transmitted. This results in the transmission of pulses of 2600-Hz tone, one for each dial pulse. During the pulsing, the level of tone applied to the line is increased by 12 dB (relay K4) to improve signaling reliability. At the end of pulsing, the 2600-Hz tone is removed from the line in the forward direction but the 2600-Hz tone is still being received from the distant terminal. When the receiving party answers, the action of lifting the receiver results in an M-lead (ground to -48 V dc) signal from the terminal trunk circuit to the SF unit, and the tone is removed in the backward direction. There is no tone present in either direction during the talking interval and no band-restricting components present.

5.4.2 Equipment characteristics (overall). The single-frequency signaling unit shall conform to the following characteristics:

5.4.2.1 Back-to-back characteristics: The following requirements apply when two SFSU units are connected back-to-back at the line sides:

a. Insertion loss: 0 $\pm$0.1 dB at 1 kHz.

b. Frequency response: $\pm$0.5 dB from 300 to 3500 Hz referred to insertion loss at 1000 Hz.

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c. Distortion, envelope delay (output): 20 μs maximum spread from 1000 to 2500 Hz and 50 μs maximum spread from 600 to 3200 Hz.

d. Distortion, total harmonic (output): 40 dB minimum from 300 to 3400 Hz with 0 dBm0 test tone at the VF input and receive section adjusted to operate at +7 TLP.

e. Crosstalk: 70 dB minimum between adjacent units and transmit and receive sections when disturbed and disturbing units are strapped for 0 TLP.

f. Signaling performance: By a strapping option, the signaling unit shall be able to transmit either dial, AUTOVON or other supervisory signals.

(1) Dial Signaling: 9 to 11 pps with 47 to 67% break overseas and 8 to 12 pps with 49-72% break in CONUS.

(2) Dial Pulse Break and Make Durations: The percent break permissible is determined by the pulsing speed and a minimum of 27 ms make is required.

(3) Signaling distortion: The output signal of the SF equipment should not be distorted by more than ±5 ms or ±5% whichever is less.

g. Signaling pulse delay: Measured from M-lead input to E-lead output, the delay of off-hook to on-hook transition shall be between 30 to 100 ms.

5.4.2.2 Transmit section.

a. Impedance (input and output): 600-ohm balanced-to-ground when opposite side is terminated in 600 ohm ±1% resistor.

b. Return loss: 26 dB minimum from 300 to 3400 Hz when opposite side is terminated in 600 ohm ±1% resistor.

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c. Longitudinal balance (input and output): 40 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms.

d. Longitudinal-input-to-metallic-output voltage ratio: 40 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms (See 2.5).

e. Transmission level (transmit section): Adjustable by strapping option to +4, 0, -2, -4, and -16 TLP.

f. Signaling input: The M-lead signaling input controls the signaling unit as follows:

Signaling State	M-Lead Condition	VF Signal State
on-hook	grounded	tone on
off-hook	-48 V	tone off

g. M-lead sensitivity. Specified performance shall be obtained under the following conditions:

(1) M-lead keying circuit series resistance from 0 to 200 ohms.

(2) Application of ±3 V dc and ripple or noise voltage up to 12 V p-p shall not affect M-lead operation.

h. Signaling-tone levels.

(1) Initial signal tone level: -8 ±1.5 dBm0 at start of SF signal and maintained for at lease 70 ms.

(2) Steady signal tone level: Level reduces to -20 ±1.5 dBm0 within 300 to 550 ms after the start of the tone.

(3) Signal leak to line (when SF signal is not being sent): -75 dBm0 maximum.

i. Signaling distortion: 4 ms maximum; pulse correction may be applied to dial pulse. If dial pulse correction is provided, minimum pulse length of transmitted pulse is 45 ms and minimum interval between pulses is 25 ms.

j. Signaling tone purity: Harmonics and spurious frequency components -40 dB maximum with respect to signaling tone level.

5.4.2.2.1 Transmit line split (illustrated by relay R3 in figure 5-12): The VF transmitting path of the transmit section shall be disconnected on the drop side and both sides of the split terminated in 600 ohm ±10% under the following conditions:

a. At the start of an on-hook signal, transmit line split shall occur in the period of 20 ms before to 5 ms after tone is switched on. The split shall be maintained 350 to 750 ms.

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b. At the end of an on-hook signal the transmit line split shall occur in the period of 20 ms before to 5 ms after tone is switched off. The split shall be maintained 75 to 156 ms.

c. When the transmitted and received tones are simultaneously on, the split shall be maintained continuously.

d. After a change in signaling state, the split shall be removed as follows:

(1) 75 to 160 ms after transmit tone is switched off.

(2) 350 to 750 ms after received tone is terminated.

e. When signaling unit is transmitting a signaling tone but not receiving a tone, transmit line split shall occur less than 100 ms after the start of a signaling tone on the receive line.

f. When M-lead input to the transmit section is in the off-hook state, control of the transmit line split shall be independent of the associated receive section.

5.4.2.3 Receive section characteristics.

a. Impedance (input and output): 600 ohm balanced-to-ground.

b. Return loss: 26 dB minimum when opposite side is terminated in 600 ohm ±1% resistor.

c. Longitudinal balance (input and output): 40 dB minimum for 50 to 4000 Hz for longitudinal potentials up to 50 V rms.

d. Longitudinal-input-to-metallic-output voltage ratio: 40 dB minimum from 50 to 4000 Hz for longitudinal potentials up to 50 V rms (See 2.5).

e. Transmission level: Adjustable by strapping option to +7, +4, 0, -2, -8 TLP.

f. Signaling output: By means of form "C" contacts with the armature grounded. Signaling outputs as follows:

Signaling State	VF Signal State	E-Lead Condition	N-Lead Condition
on-hook	tone on	open	grounded
off-hook	tone off	grounded	open

g. Receive sensitivity: Accept signaling tones (steady state levels) -29 to -11 dBm0 and initial levels 12 dB higher than steady state levels, for the period specified in 5.4.2.2(h)(1) and (2).

h. Receiver selectivity: Receiver shall accept signal tone frequencies from 2585 to 2615 Hz without degradation.

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i. Recognition time.

(1) Dial operation: Responds to tone pulses of any duration greater than 38 ms; rejects tone pulses of durations less 28 ms.

(2) AUTOVON operation: Responds to tone pulses of any length greater than 50 ms; rejects tone pulses of duration less than 40 ms.

(3) Tone breaks: After signaling tone has persisted for more than 450 ms, the receiver shall not respond to breaks less than 25 ms (dial operation) or less than 40 ms (AUTOVON operation).

j. Noise input: With a flat noise at levels up to -33 dBm0 or speech weighted noise at levels up to 57 dBrnc0 applied to the VF input, the receiver shall meet all requirements.

k. Pulse correction (if provided): With pulse length 38 to 47 ms at receiver input, corrected output on E lead shall be 51 $\pm$4 ms.

l. Signaling distortion: 4 ms maximum with signaling tone pulse duration greater than 50 ms when terminated in nominal 600-ohm impedance.

m. False signal discrimination (on-hook conditions): When a valid signaling tone is applied to the VF input, the E lead shall be maintained in on-hook state with speech, data or other complex signals applied at the VF input at levels up to 0 dBm0 rms.

n. False signal discrimination (off-hook conditions): When no valid signaling tone is applied to the VF input, the receive signaling unit shall remain in the off-hook state under the following conditions:

(1) With single frequency tones at the VF input in the band from 300 to 2400 Hz and 2800 to 3400 Hz, at levels up to -1 dBm0 and in the band from 2400 to 2500 Hz and 2700 to 2800 Hz at levels up to -8 dBm0.

(2) With complex data or other tone signals applied to the VF input at power levels up to -1 dBm0 rms and the total power in the 800 to 2450 Hz band equal to or greater than the power in the 2500 to 2700 Hz band.

(3) With noise burst applied to the VF input at levels of -5 dBm0 maximum and with a maximum duration of 10 ms.

(4) With speech signals having an average talker volume of -12.5 VU and a standard deviation of 5 dB applied to the receiver VF input, the average occurrence of false signals at the E-lead output shall be one per 70 call hours maximum for false signals of more than 47 ms duration and one per 1500 call hours maximum for false signals of more than 150 ms duration.

o. Signaling pulse delay: As measured from the beginning of a signaling tone pulse applied to the input of the receiver section to the time at which the E-lead output is switched from the off-hook to the on-hook state, the absolute delay shall be 30 ms minimum.

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5.4.2.3.1 Receive line split.

a. When a signaling tone is present, a narrow band split (band elimination filter) is inserted in the receive VF input with the following attenuation characteristics:

(1) 35 dB minimum from 2585 to 2615 Hz referred to 1000 Hz loss before split occurs.

(2) 5 dB maximum from 2200 to 2400 Hz and from 2800 to 3000 Hz referred to 1000 Hz before split occurs.

(3) 0.5 dB maximum from 300 to 2200 Hz and from 3000 to 3400 Hz referred to 1000 Hz loss before split occurs.

b. Receive split control: Split inserted within 20 ms after signaling tone is applied, maintained for duration of tone and removed within 300 ms after tone is terminated.

5.4.2.4 General Requirements.

a. Signaling tone (adjustable).

(1) Accuracy, 2600 ±2 Hz.

(2) Stability, ±2 Hz, 90 days.

(3) Output level stability: Normal operating level ±1.5 dB, 90 days.

b. VF transmission line impedance (receive and transmit section): Nominal 600 ohms (See 2.4).

c. Surge protection (receive and transmit section): (See 2.2).

d. Power requirements: 5 watts maximum at -48 V dc ±10% (See 2.3).

5.5 STATION 20-HZ RINGING SUPPLY.

5.5.1 Application. The station 20-Hz ringing supply (20-Hz supply) is used to provide 20-Hz ringing power for telephone signaling. The basic 20-Hz supply consists of two identical 20-Hz, 25-watt generators, one ballast lamp panel, and one alarm and monitor panel. Provisions are included to expand the capability of the 20-Hz supply by the addition of 20-Hz generators, ballast lamp panels, and alarm and monitor panels as required. Functions of the individual units of the system are described below.

a. 20-Hz generators: The basic 20-Hz supply contains two identical 20-Hz, 25-watt generators. Each generator contains provisions for synchronizing for parallel operation to other 20-Hz generators connected to the same bus. An output disconnect relay is also provided in each generator to disconnect it from the bus in the event of input or output power failure. Visible fault alarm indication is provided on the generators in addition to relay contacts to transfer fult alarm information to the remote alarm and monitor panel.

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b. Ballast lamp panel: The ballast lamp panel is equipped with fuses and ballast lamps for distribution of the 20-Hz ringing power.

c. Alarm and monitor panel: The alarm and monitor panel performs the following functions:

(1) Monitors the 20-Hz bus and furnishes a visible and audible indication of failure.

(2) Monitors each 20-Hz generator and furnishes a visible and audible indication upon failure of any generator.

(3) Is capable of being connected to transfer alarm indications to the station common alarm system.

(4) Has capability for manually removing any generator connected to the bus without degrading the operation of the other generators remaining in service.

5.5.2 Equipment characteristics. The equipment connected as a system shall conform to the following characteristics:

a. Output characteristics.

(1) Voltage: 90 to 110 V rms, continuously adjustable.

(2) Frequency: 20 Hz $\pm$5%.

(3) Power: 25 watts per generator maximum.

(4) Voltage regulation: 15% maximum voltage change from no load to full load: $\pm$15% maximum output voltage change for $\pm$10% input supply voltage change.

(5) Harmonic distortion: 16.5 dB minimum total harmonic distortion; 26 dB minimum for harmonic distortion due to combined effects of second and third harmonics.

b. Internal protection: The individual 20-Hz generators (not operating through the ballast panel) shall be capable of withstanding a short circuit without internal damage or blown fuses.

c. Parallel operation:

(1) 20-Hz generators shall be capable of being synchronized with other identical generators connected across the bus.

(2) Output disconnect relays and control circuitry shall be provided in the 20-Hz generators for automatically disconnecting a paralleled generator in case of input power or 20-Hz output voltage failure. In addition, manual capability for disconnecting individual generators shall be provided in the alarm and monitor unit. Disconnection of a generator by these methods shall not cause degradation of the other generator outputs.

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d. Alarm and monitor: Audible and visible alarms shall be provided on the alarm and monitor panel to indicate failure of the bus or individual generator 20-Hz output voltages. In addition, visible alarms shall be provided on each 20-Hz generator. The alarm and monitor panel shall have provision (form C relay contacts) for transfer of alarm functions to the station common alarm system.

e. Output distribution: .The ballast panel shall contain 50 ballast lamps of the miniature filament type for distribution of the 20-Hz power. One fuse shall be used on conjunction with five ballast lamps. Each ballast lamp shall be capable of withstanding a 20-Hz voltage of 110 V without sustaining damage or dissipating more than 10 watts ow power.

f. Expandability: Capability shall be provided for increasing capacity of the 20-Hz supply by the addition of any of the following similar kinds of equipment:

(1) 20-Hz generator.

(2) Ballast lamp panel.

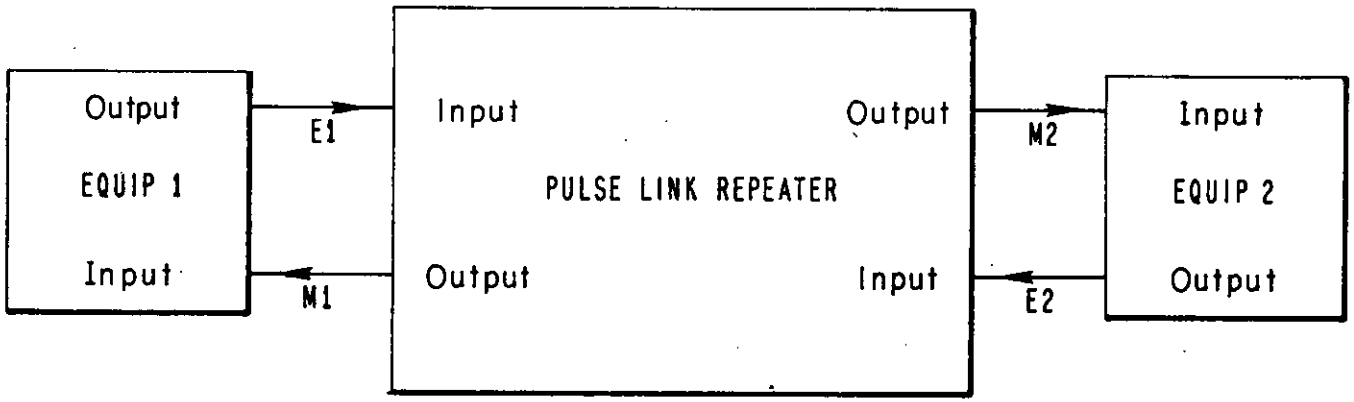
(3) Alarm and monitor panel.

g. Power requirements: 130 watts maximum for two each, 20-Hz generators only (See 2.3).

5.6 PULSE LINK REPEATER.

5.6.1 Application. The pulse link repeater (PLR) is a device which converts E signals (ground or open circuit) to M signals (-48 V or ground). It is used to interface two equipments both of which require an M-input signal and produce an E-output signal (See figure 20). Signals on the E&M leads (E1 and M1) on one side of the PLR affect or are affected by signals on the E&M leads (E2 and M2) on the other side of the PLR. Supervisory signals appearing on the E1 lead cause supervisory signals to appear on the M2 lead. Likewise supervisory signals appearing on the E2 lead cause supervisory signals to appear on the M1 lead. The PLR will operate in both directions simultaneously.

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EQUIP 1	EQUIP 2	E1	E2	M1	M2
Idle	Idle	Open	Open	Gnd	Gnd
Busy	Idle	Gnd	Open	Gnd	-48v
Idle	Busy	Open	Gnd	-48v	Gnd
Busy	Busy	Gnd	Gnd	-48v	-48v

Figure 20. Block Diagram and Function Table for a Pulse Link Repeater.

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5.6.2 Equipment characteristics. The equipment shall conform to the following characteristics:

a. The PLR shall produce -48 V ±5 V dc on the M2 lead when the E1 lead is grounded. This indicates that equipment 1 is off-hook (busy).

b. The PLR shall produce -48 V ±5 dc on the M1 lead when the E2 lead is grounded. This indicates that equipment 2 is off-hook (busy).

c. The PLR shall produce a ground on the M2 lead when the E1 lead is open circuited. This indicates that equipment 1 is off-hook (idle).

d. The PLR shall produce a ground on the M1 lead when the E2 is open circuited. This indicates that equipment 2 is on-hook (idle).

e. Pulse rate (input signal): 7 to 25 pulses per second.

f. Percent break (input signal): 47 to 75% break (normal 62%).

g. Distortion, signaling (output): 4% maximum additional over input).

h. Signal delay (output, M): 3 ms maximum, compared to input signal (E).

i. Power requirements: 4 watts maximum at -48 V dc ±10% (See 2.3).

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Chapter VI

6. DC TEST EQUIPMENT. The dc test equipment includes the following major electronic units:

a. High-level digital pattern generator.

b. Low-level digital pattern generator.

c. Low-level digital multiplier unit.

d. DC test cabinet.

6.1 HIGH-LEVEL DIGITAL PATTERN GENERATOR.

6.1.1 Application. The high-level pattern generator is used to generate standard test sentences and several other test signals as an aid in system, circuit or equipment adjustment.

The pattern generator is completely self-contained, but optional provisions are included to operate it from an external timing oscillator or to synchronize the internal oscillator with an external timing source.

The device is of modular construction and contains the following basic modules:

a. Message logic generator.

b. Time base generator.

c. Relay assembly (supplies five test signal outputs). Provisions for external character stepped control are provided.

6.1.2 Equipment characteristics. The equipment shall conform to the following characteristics:

a. Test message format. The 80-character baudot code FOX message shall contain all characters and machine functions shown in the following example: (Car ret)(Car ret)(Line feed) THE space QUICK space blank blank blank blank blank BROWN space FOX space JUMPS space OVER space A space LAZY space DOGS space BACK space (Fig shift) 1 2 3 4 5 6 7 8 9 0 (Ltr shift) space DE space (plus eight programmable characters for station identifier).

b. Modulation rates (selectable by front panel control): Transmits the 80-character FOX message test sentence in BAUDOT code at modulation rates of 45.45, 50 and 75 baud with a 7.0 unit code.

c. Output interface characteristics are as follows:

(1) Signal levels: High level (signal power external) neutral or polar by option. Neutral 130 V dc at optionally available 20 or 60 mA. Polar $\pm$60 V dc at 20 mA in accordance with MIL-STD-188-100 (4.3.1.4.2 and 4.3.1.4.3).

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(2) Impedance: In accordance with MIL-STD-188-100 (4.3.1.4.3).

(3) Waveshaping: In accordance with MIL-STD-188-100 (4.3.1.4.4).

(4) Sense: Optionally available positive or negative mark.

d. Input STEP interface requirements (conversion from free-running to STEP is made by strapping).

(1) Level: -60 V.

(2) Impedance: 2500 ohms ±10%.

(3) Step duration: 15 to 35 ms.

e. Distortion, telegraph (output): 1% maximum.

f. Internal oscillator stability and accuracy.

(1) Long term stability: 1 part in 10^4, maximum, for 30 days.

(2) Short term stability: 1 part in 10^5, maximum for 24 hours.

(3) Timing accuracy: 0.05% of output frequency.

(4) Basic clock frequency: 3.0 kHz, minimum.

g. External timing provisions.

(1) Synchronization capability: Capability shall be provided for synchronizing the internal clock at 3.0 kHz minimum frequency from an external reference source. INPUT interface requirements shall be low level ±6 V in accordance with MIL-STD-188-114, (5.2.2 and 5.2.4).

(2) External clock: The generator shall be capable of operating from an external clock with 1 Hz of clock input per bit output and with input interface requirements given in (1) above.

h. Power requirements: 15 watts maximum (not including external signal battery power). (See 2.3)

6.2 LOW-LEVEL DIGITAL PATTERN GENERATOR.

6.2.1 Application. The low-level digital pattern generator is a solid-state test instrument capable of generating asynchronous or synchronous telegraph and data signals. The instrument is used for system, circuit or equipment alignment. The unit is capable of generating either an 80-character five-level BAUDOT message for the testing of teletypewriter systems or an 80-character eight-level (parity check bit included) ASCII message for the testing of data systems.

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a. The pattern generator is completely self-contained, but optional provisions are included to operate it from an external timing oscillator or to synchronize the internal oscillator with an external source. The device is of modular construction and contains the following basic modules:

(1) Message logic generator.

(2) Time-base generator.

(3) Solid-state output relay.

b. Provisions for external character stepped control are provided.

6.2.2 Equipment characteristics. The low-level digital pattern generator shall conform to the following characteristics.

a. Test message format.

(1) BAUDOT code test message: The 80-character FOX message shall contain all characters and machine functions shown in the following example: (Car ret)(Car ret)(Line feed) THE space QUICK space blank blank blank blank blank BROWN space FOX space JUMPS space OVER space A space LAZY space DOGS space BACK space (Fig shift) 1 2 3 4 5 6 7 8 9 0 (Ltr shift) space DE space (plus eight programmable characters for station identifier).

(2) ASCII code test message: The 80-character FOX message shall contain all characters and machine functions shown in the following example: (Car ret)(Car ret)(Line feed) THE space QUICK space BROWN space FOX space JUMPS space OVER space LAZY space DOG space BACK space (DEL) space (DEL) space 1 2 3 4 5 6 7 8 9 0 (de109de10 space (plus eight programmable characters for station identifier).

b. Test message modulation rates (selectable by front panel switch).

(1) BAUDOT code: Asynchronous at modulation rates of 45, 45, 50, 56.8, 74.2, 75, and 150 baud.

(2) ASCII code: Synchronous or asynchronous at modulation rates of 75, 110, 150, 300, 600, 1200 and 2400 baud.

(3) Reversals at all modulation rates given in (1) and (2) above.

c. Character interval control:

(1) Asynchronous free running operation with stop bit intervals of 1.0, 1.5, and 2.0.

(2) External character stepped control with step interval equal to or greater than the character interval.

(3) Synchronous.

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d. Input STEP interface requirements.

(1) Level: 6 V ±1 V dc.

(2) Impedance: In accordance with MIL-STD-188-114 (5.2.2).

e. Pattern generator output requirements.

(1) Signal levels: Low-level, in accordance with MIL-STD-188-114 (5.2.2.1).

(2) Impedance: In accordance with MIL-STD-188-114 (5.2.1).

(3) Waveshaping: In accordance with MIL-STD-188-114 (5.2.1.7).

(4) Sense: Optional positive or negative mark.

(5) Distortion, telegraph: "0.5%" vice "1%".

f. Internal oscillator stability and accuracy.

(1) Long term stability: 1 part in 10^4 maximum for 30 days.

(2) Short term stability: 1 part in 10^5 maximum for 24 hours.

(3) Basic clock frequency: 614.4 kHz vice "48 kHz".

(4) Timing accuracy: 0.05% of output frequency.

g. External timing provisions.

(1) Synchronization capability: Capability shall be provided for synchronizing the internal clock at 614.4 kHz minimum frequency from an external reference source. INPUT interface requirements shall be low level ±6 V in accordance with MIL-STD-188-114, (5.2.2 and 5.2.2.2).

(2) External clock: The pattern generator shall be capable of operating from an external clock with 1 Hz of clock input per bit output and with input interface requirements given in (1) above.

h. Power requirements: 15 watts maximum (not including external signal battery power). (See 2.3)

6.3 LOW-LEVEL DIGITAL MULTIPLIER UNIT.

6.3.1 Application. The low-level digital multiplier is a solid-state instrument that accepts a single data or timing source as an input and provides twenty separate and isolated output sources of data or timing signals for multiple distribution. The multiplier shall be capable of operating in the following modes:

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a. Synchronous. Each multiplier output shall be individually selectable for operation either in phase with the input signal or 180 degrees out of phase with the input signal. The maximum delay between any two outputs shall not exceed 2% of the operating modulation rate.

b. Random. Each multiplier output signal shall be randomly keyed (individually) with respect to the input signal or any other output signal. Each multiplier output shall also have the capability of individual selection of positive or negative mark.

6.3.2 Equipment characteristics. The low-level digital multiplier unit shall conform to the following characteristics:

a. Output characteristics.

(1) Signal levels: Low-level with external keying battery, in accordance with MIL-STD-188-114 (5.2.1.2).

(2) Impedance: In accordance with MIL-STD-188-114 (5.2.1.2).

(3) Waveshaping: In accordance with MIL-STD-188-114 (5.2.1.7).

(4) Sense: Each individual output convertible by strapping to positive or negative mark with input of either sense.

(5) Distortion, telegraph: The distortion of the output signal shall not exceed the distortion of the input signal by more than 0.5%.

(6) Differential delay: 2% maximum between any two outputs.

b. Input characteristics.

(1) Signal levels: Low-level, in accordance with MIL-STD-188-114 (5.2.2.1).

(2) Impedance: In accordance with MIL-STD-188-114 (5.2.2).

(3) Sensitivity: In accordance with MIL-STD-188-114 (5.2.2.2).

c. Modulation rates.

(1) Asynchronous: 45.45 to 300 baud.

(2) Synchronous: 37.5 to 4800 baud.

d. Power requirements: 3 watts maximum (not including external signal battery power). (See 2.3)

6.4 DC TEST CABINET. The test cabinet consists of the following major electronic units:

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a. Data error rate measuring set.

b. Digital distortion analysis equipment.

c. Digital multimeter.

d. Dual-trace oscilloscope.

e. Patching equipment.

6.4.1 Application. The function of the dc test cabinet is to perform testing and evaluation of teletypewriter and data circuits. The data error rate measuring equipment has a transmit and a receive unit. The transmit section supplies a reference test pattern to the send terminal of the system under test. The receiver section accepts data from the system under test and makes a comparison between the received signal and its own locally generated perfect pattern. Any errors in the incoming pattern are detected, counted and recorded on a front panel counter. The digital distortion analyzer measures teletype distortion. DC voltage, current, and resistance measurements are made by means of a digital multimeter. A dual trace oscilloscope allows visual signal analysis. An interbay trunk module and a test-equipment patching module complete test cabinet components.

6.4.2 Equipment characteristics. The equipment items shall conform to the following characteristics.

6.4.2.1 Data error rate measuring set. This item is part of teletypewriter test set AN/GGM-16.

Unit accepts a 2047 bit pseudo-random test pattern and compares it with an internally generated pattern. Transitions occurring beyond a preset percent distortion threshold, adjustable in 1% intervals from 0 to 49%, are identified as errors.

6.4.2.2 Digital data analysis equipment. This item is part of teletypewriter test set AN/GGM-16.

Unit measures teletype distortion in either the synchronous or asynchronous mode. The unit is also capable of being operated as peak distortion monitor with a percent distortion threshold variable in 1% increments from 0 to 49%.

6.4.2.3.1 Multimeter measurement capabilities.

6.4.2.3.1.1 DC voltages. 150 mV to 150 V with input resistance greater than 60,000 ohm.

6.4.2.3.1.2 DC current: 150 μA to 0.5 A.

6.4.2.3.1.3 Resistance: 0.1 ohm to 50 megohm.

6.4.2.3.2 Accuracy: Better than 1.

6.4.2.3.3 Multimeter features: Three-digit, digital readout, automatic ranging, overrange indicator, automatic polarity indicator, remote access to multimeter inputs and overrange protection.

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6.4.2.3.4 Power requirements: 15 watts, maximum (See paragraph 2.3).

6.4.2.4 Dual-trace oscilloscope.

a. Vertical amplifiers. Equipment shall contain two identical amplifiers with the following characteristics:

(1) Bandwidth: DC to 5 MHz with dc coupling; 2 Hz to 5 MHz with ac coupling.

(2) Rise time (10 to 90%): 0.07 us maximum.

(3) Sensitivity: 5 mV per cm to 20 V per cm deflection sensitivity, calibrated and with verniers for continuous uncalibrated adjustment between range steps and to extend the deflection range to 50 V per cm for uncalibrated operation.

b. Presentation: Dual-trace, alternate or chopped.

c. Time base: 1 µsec per cm to 5 sec per cm.

d. Triggering shall occur for the following signals:

(1) Internal: Selected from either channel, 50 Hz to 5.0 MHz or from line voltage, with amplitude sufficient to cause 0.5 cm minimum vertical deflection.

(2) External: 50 Hz to 5.0 MHz with 0.5 peak-to-peak minimum amplitude.

e. Trigger point: Continuously variable from -10 to 10 V on both positive or negative slope of external trigger signal.

f. Signal delay: Sufficient to allow leading edge of nonrecurrent waveform to be displayed.

g. Power requirements: 80 watts maximum (See paragraph 2.3).

6.4.1.5 Patching equipment.

a. Interbay trunk module: Interbay trunk terminations shall be provided in a module containing at least 24 lamps and a comparable number of jacks (compatible with 310-type telephone plug). An example of a module with 26 lamps and jacks is illustrated in figure 21. A functional wiring diagram is depicted in figure 22.

b. *Test equipment patching module (not illustrated):* A module containing at least 48 jacks (compatible with 310-type telephone plug) shall provide patching capability for the input and output terminations at the test equipment units mounted in the test cabinet.

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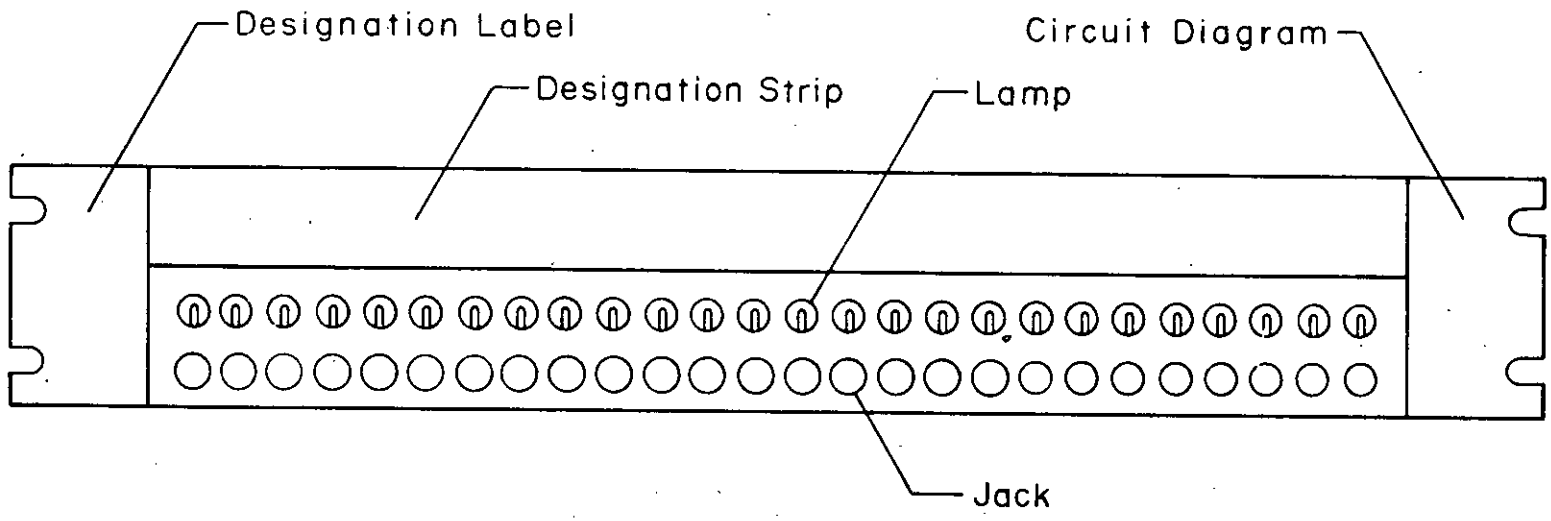


Figure 21. Interbay Trunk Module (DC Test Cabinet).

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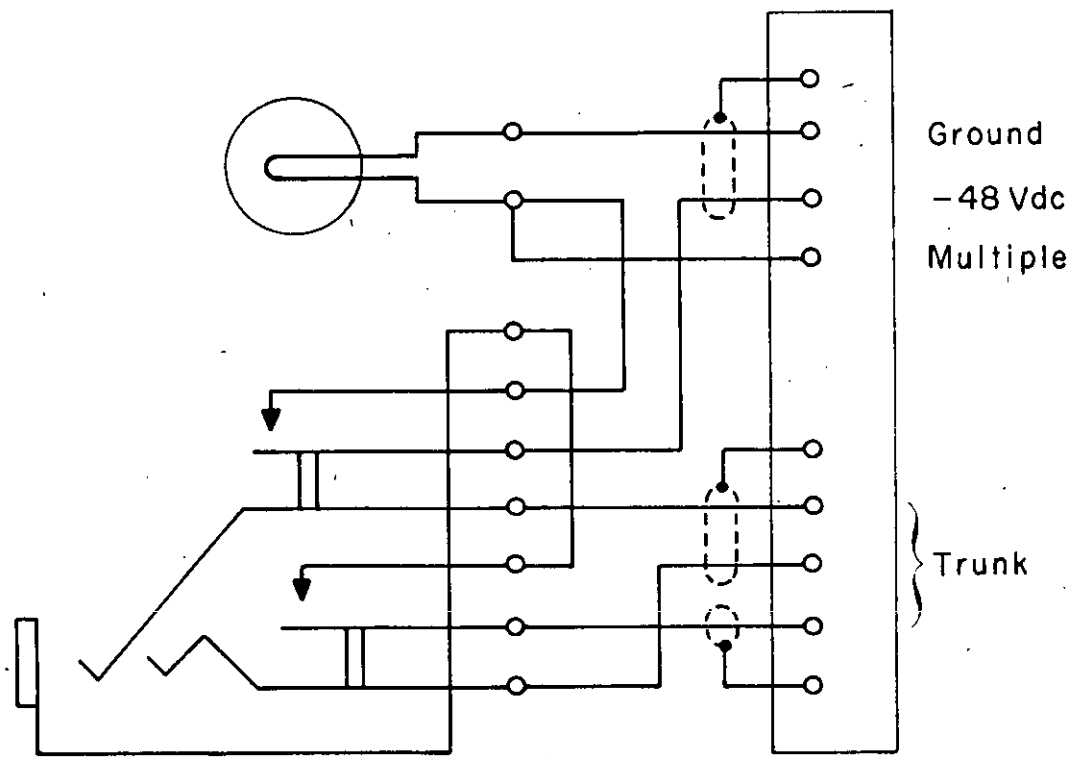


Figure 22. Jack Wiring of Interbay Trunks (DC Test Cabinet).

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Chapter VII

7. VF TEST EQUIPMENT. The VF test equipment includes the following major items:

a. Quality assurance (QA) test center.

b. Cable test bay.

7.1 QUALITY ASSURANCE (QA) TEST CENTER.

7.1.1 Application. The quality assurance (QA) test center permits the measurement of various parameters necessary to achieve and maintain the high standards of circuit quality required in the Defense Communications System. QA tests are performed when new circuits are activated, during scheduled maintenance operations or when operational difficulties arise. In-service tests are measurements that can be made without interruption of normal service, using high-impedance bridging techniques. In-service tests include, signal level, visual display of spectrum, visual display of eye patterns and visual comparison of two signals. Out-of-service tests are measurements that necessitate temporary removal of the circuit from normal service and may need coordination with other TCF's for end-to-end circuit tests. Out-of-service tests include frequency response, test-tone level, longitudinal balance, harmonic distortion, intermodulation distortion, single-tone interference, stability of net loss, impulse noise count, input impedance, output impedance, crosstalk, phase jitter and return loss. Envelope delay distortion and idle circuit noise level.

7.1.2 Equipment characteristics. The equipment in the QA test center shall conform to the following characteristics:

a. Frequency response.

(1) Measurement at discrete points in the frequency range 300 Hz to 4 MHz with frequencies accurate to 0.1% ±1 Hz, and levels of -90 to +10 dBm accurate to ±0.2 dB.

(2) Provisions for obtaining data given in (1) in digital prinout form. Capability shall be included for printing out signal levels at a minimum of 10 equally-spaced frequencies across the band being measured. Capability for recording circuit and time identifying information shall be included.

(3) Automatic plotting of response curves over any band in the range 50 to 5000 Hz with frequencies accurate to ±5 Hz and levels of -40 to +10 dBm accurate to ±0.2 dB. X-axis deflection sensitivity range of 0 to 100 Hz per inch minimum; Y-axis deflection sensitivity range of 0 to 1 dB per inch, minimum. Grid size of 80 square inches, minimum with height to width ratio of 0.8.

b. 1 kHz test-tone levels: -90 to +10 dBm, accurate to ±0.1 dB, at 1004 Hz to 1020 Hz.

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c. Envelope delay.

(1) Measurement in the range 0 to 20 ms with an accuracy of ±5 usec at discrete frequencies measured with an accuracy of 0.1% ±1 Hz in the range of 300 to 4 kHz.

(2) Automatic plotting of a curve of delay vs frequency with frequency range, X-deflection sensitivity range and grid size the same as in (a) above. Y-deflection sensitivity range shall be 0 to 100 usec per inch minimum.

d. Frequency translation error: 0 to 10 Hz, with an accuracy of ±0.5 Hz from 300 Hz to 4 kHz.

e. Longitudinal balance (input and output): From 0 to 60 dB with an accuracy of ±1 dB, from 50 to 4000 Hz.

f. Signal levels: From -90 to +10 dBm with an accuracy of 0.1 dB, from 100 Hz to 4 MHz.

g. Impedance: From 10 to 5000 ohms with an accuracy of ±5%.

h. Idle circuit noise levels: From 0 to +92 dBrnc with an accuracy of ±1 dB with a flat frequency response from 300 to 4 kHz, 300 to 15 kHz, and 300 to 4 MHz (-3 dB points).

i. Impulse noise count:

(1) Threshold range: 0 to 90 dBrn, selectable in 1 dB steps.

(2) Time period: 5 to 60 minutes, adjustable in 1 minute steps.

(3) Weighting: C-message or flat, switch selectable.

j. Crosstalk: From 0 to -80 dBm with an accuracy of ±1 dB.

k. Stability of net loss: Gain variation of up to ±10 dB over 24 hours with an accuracy of ±0.5 dB.

l. Intermodulation distortion: With two independently adjustable test tones, adjustable in amplitude from -40 to +15 dBm and adjustable in frequency from 300 Hz to 4 kHz, the frequency and amplitude of each intermodulation distortion product with an accuracy of ±1 dB in amplitude and ±1% in frequency. The range of measurement shall be -90 to +10 dBm, from 20 Hz to kHz.

m. Harmonic distortion: With a single test tone, adjustable in amplitude from -40 to 15 dBm and adjustable in frequency from 300 Hz to 4 kHz, the frequency and amplitude of each harmonic with an accuracy of ±1 dB in amplitude and ±1% in frequency, 20 Hz to 50 kHz.

n. Single tone interference: Without test tone or signal, amplitude of interference tone from -90 to +10 dBm with an accuracy of ±1 dB and frequency of interference tone from 20 Hz to 50 kHz with an accuracy of ±1%.

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o. Phase jitter: 0 to 30 degrees accurate to the nearest degree at a measurement frequency within the range of 1004 Hz to 1020 Hz.

p. Return loss: 0 to 35 dB with an accuracy of ±1 dB.

q. Specrum display: Visual, amplitude from -90 to +10 dBm, frequency range from 300 Hz to 4 MHz, scan width adjustable from 3 kHz to 4 MHz.

r. Eye pattern: Visual display of eye pattern for modem with audio output in the band of 300 Hz to 4 kHz.

s. Comparison of two signals: Visual comparison of two signals containing components up to 1 MHz with storage capability for time periods up to 20 minutes.

7.1.2.2 Instrument requirements:

a. Impedance (input): Input impedance shall be such that connection of the instrument will not affect the accuracy of measurement of the parameters defined above.

b. Signal levels (input): Overrange signal levels of 500 V magnitude shall not damage the instrument.

c. Longitudinal balance (input): The input longitudinal balance shall be such that connection of the instruments will not affect the accuracy of measurement of the parameters defined above.

d. Surge protection: (See 2.2).

e. Mounting: All test units to be cabinet mounted with all controls and indicators on the front panel.

f. Graphical records: Equipment shall provide permanent records of results of tests listed below:

(1) Tests listed in 7.1.2.1a, c, and k (details of required X-Y recordings are given in 7.1.2.2 and 7.1.2.2.c.

(2) Level measurements (7.1.2.1f) carried out over a 15-minute interval, allowing events 400 ms apart to appear not less than 2.5 cm (1 inch) apart.

7.1.2.3 Jack modules.

a. Interbay trunk modules.

(1) Interbay trunk and jack modules shall contain at least 24 jack and lamp sets. An example of a module with 26 lamps and jacks is shown in figure 23. A functional wiring diagram is shown in figure 25.

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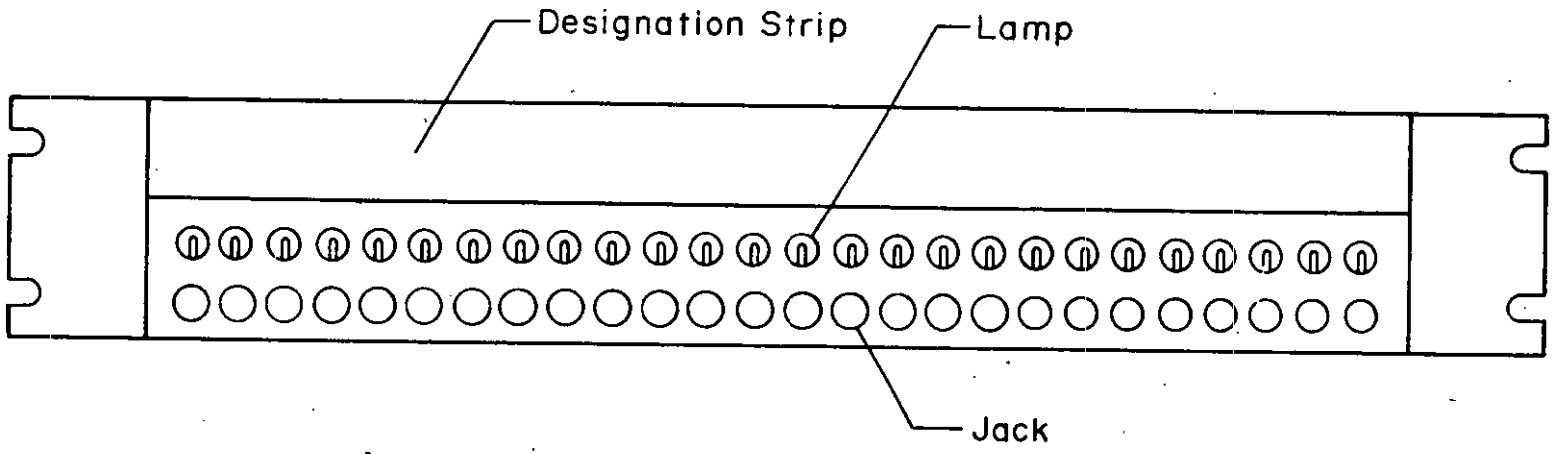


Figure 23. Interbay Trunk Terminations. (QA Test Center)

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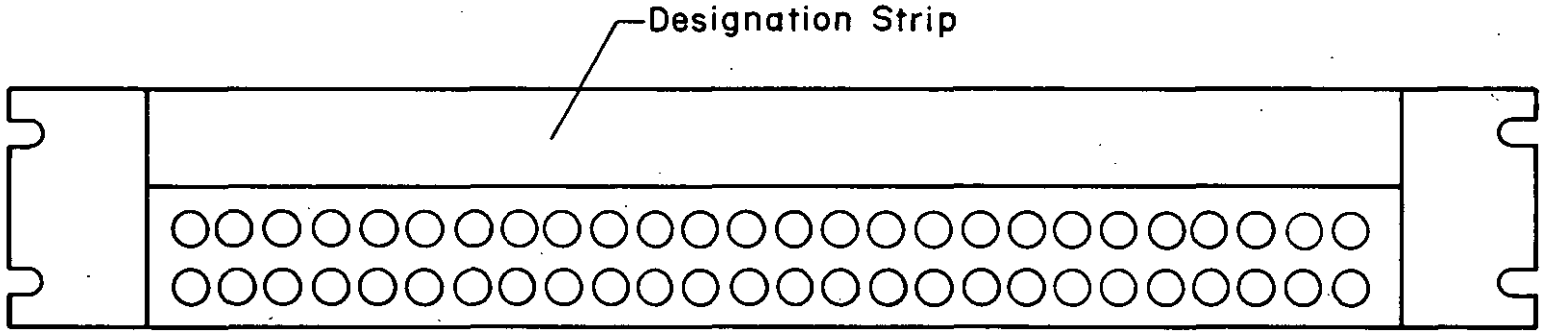


Figure 24. Test Equipment Jacks Module. (QA Test Center)

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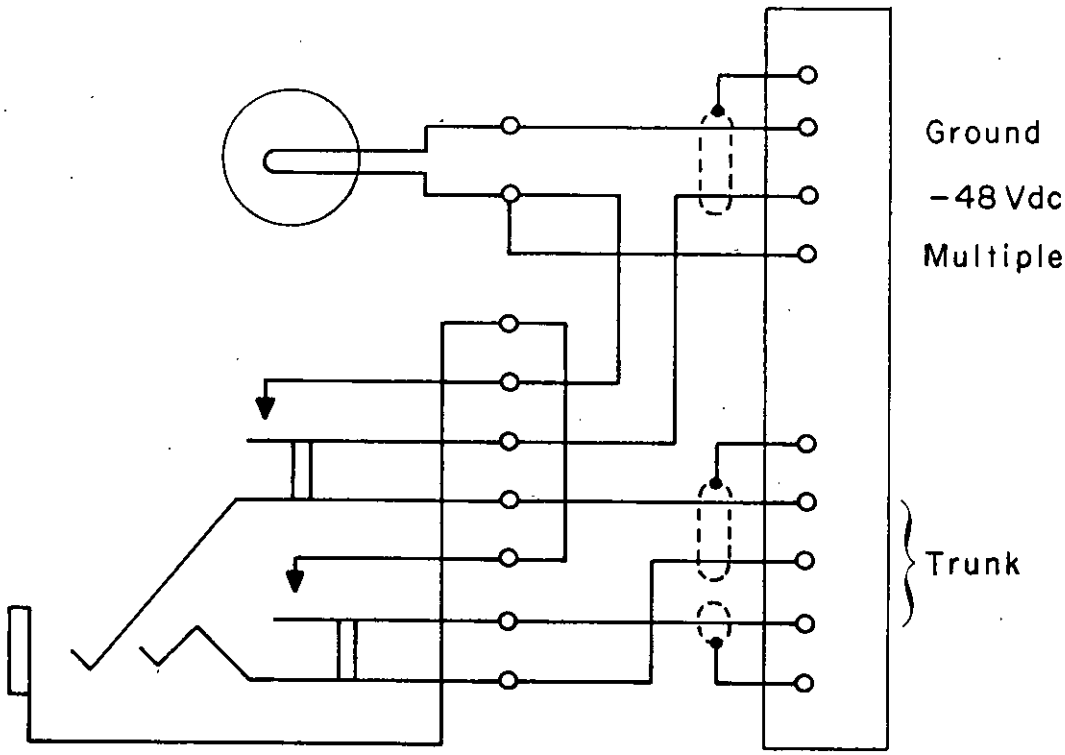


Figure 25. Jack Wiring of Interbay Trunks. (QA Test Center)

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(2) Jacks shall be compatible with 310-type telephone plugs.

(3) Module shall contain a terminal block for teminating all external wiring.

b. Test equipment jack module.

(1) Jack arrangement: Two rows of at least 24 jacks. An example of 26-jack arrangement is illustrated in figure 24. The jacks are wired to allow access to and from the test equipments.

(2) Jack type: Jacks shall be compatible with 310-type telephone plugs.

7.2 CABLE TEST BAY. The cable test bay consists of the following equipment:

a. Four-cycle interrupter.

b. Signal test set.

c. Wheatstone bridge.

d. Cable fault locator.

e. Jack modules.

7.2.1 Application. The cable test bay is used to make cable fault measurements for the purpose of detection of grounds, shorts, opens and crosses, impedance mismatches and the determination of conductor resistance. Equipment is also included for the generation of accurate standard signaling pulses and tones.

a. The four-cycle interrupter is used for generating low-frequency ac voltages in conjunction with Murray loop tests for locating open-cable faults. The interrupter reverses battery voltage eight times per second and simultaneously reverses the polarity of the Wheatstone bridge.

b. The Wheatstone bridge is used in the Murray, Varley, Hilborn and other loop tests to identify faulty wires in telephone cable; to locate grounds, shorts and crossed pairs; to locate open circuits by capacitance methods in conjunction with the four-cycle interrupter; and to measure conductor resistances.

c. The signaling test set is used for determining the quality of dial and supervisory signaling on receive circuits and for generating accurate standard signaling pulses.

d. The cable fault locator is a time domain reflectometer used for fault location and fault analysis on the outside plant cable system.

e. Two types of jack modules are included. One type is wired for interbay trunk use and provides 52 lamp/jack sets; 26 VF and 26 DC trunks and other.

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7.2.2 Equipment characteristics. The equipment in the cable test bay shall conform to the following characteristics: (7.2.1.1)

7.2.2.1 Four-cycle interrupter.

a. Function: The interrupter shall reverse battery voltage eight times per second and simultaneously reverse the polarity to the Wheatstone bridge.

b. Power requirements: 1.5 watts maximum (See paragraph 2.3).

7.2.2.2 Signaling test set.

a. Measurement capability.

(1) Percent break: From 0 to 100.

(2) Signaling speed: From 6 to 25 pulses per second.

b. Test signals.

(1) Generated from dial mechanism.

(2) Generated from internal oscillator and pulsing mechanism with selectable pulse rates from 6 to 20 pps and selectable percentage break from 15 to 80.

c. Type of signaling (selectable): E&M, loop and all commonly used combinations of battery, ground and open lead conditions.

d. Surge protection: (See 2.2).

e. Power requirements: 11 watts maximum (See paragraph 2.3).

7.2.2.3 Wheatstone bridge.

a. Capabilities.

(1) Identification of grounds, shorts and crossed pairs in a telephone cable by the Murray, Varley, Hillborn and other loop tests.

(2) Locate opens by capacitance methods when used with four-cycle interrupter.

(3) Measure conductor resistance (0.1 ohm to 1 M ohm minimum range).

b. Bridge accuracy: $\pm$0.15% +0.01 ohm.

c. Galvanometer sensitivity: 1/3 μA per mm of deflection.

7.2.2.4 Cable fault locator.

a. Type of presentation: Cathode ray tube and compatible with a standard oscilloscope camera.

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b. Range: 0 to 10 miles minimum in not less than four sweep ranges.

c. Sweep ranges (full-scale values): One, Two, five and ten miles (minimum requirement). Sweep display shall contain delay circuit to allow display of short sweep sections anywhere within each range.

d. Transmitter pulse length: Continuously variable from 0.2 to 10 us.

e. Transmitter pulse output level: Continuously variable from 5 to 125 V peak.

f. Accuracy.

(1) Sweep calibration: 3% of full scale.

(2) Sweep linearity: 2% of full scale.

g. Additional requirements: Equipment shall be capable of testing either energized or out-of-service cable pairs and shall contain coupling devices required to provide this function.

h. Power requirements: 60 watts maximum (See paragraph 2.3).

7.2.2.5 Jack modules.

a. Interbay trunk module (two each).

(1) Jack arrangement. At least 24 lamps and jacks per module. An example of a 26 lamp and jack set module is shown in figure 26 with functional wiring as depicted in figure 27.

(2) Jack type: Jacks shall be compatible with 310-type telephone plugs.

b. Test equipment jack module.

(1) Jack arrangement. Two rows of at least 24 jacks. An example of a 26-jack arrangement is shown in figure 28. The jacks shall be wired to allow access to and from the test equipments.

(2) Jack type: Jacks shall be compatible with 310-type telephone plugs.

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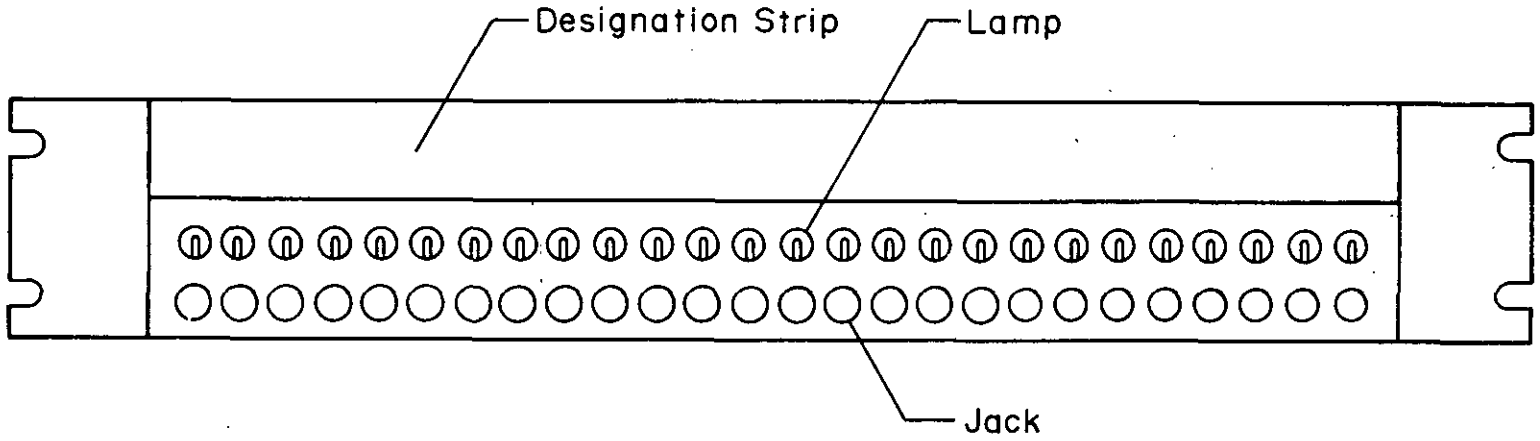


Figure 26. Interbay Trunk Terminations. (Cable Test Bay)

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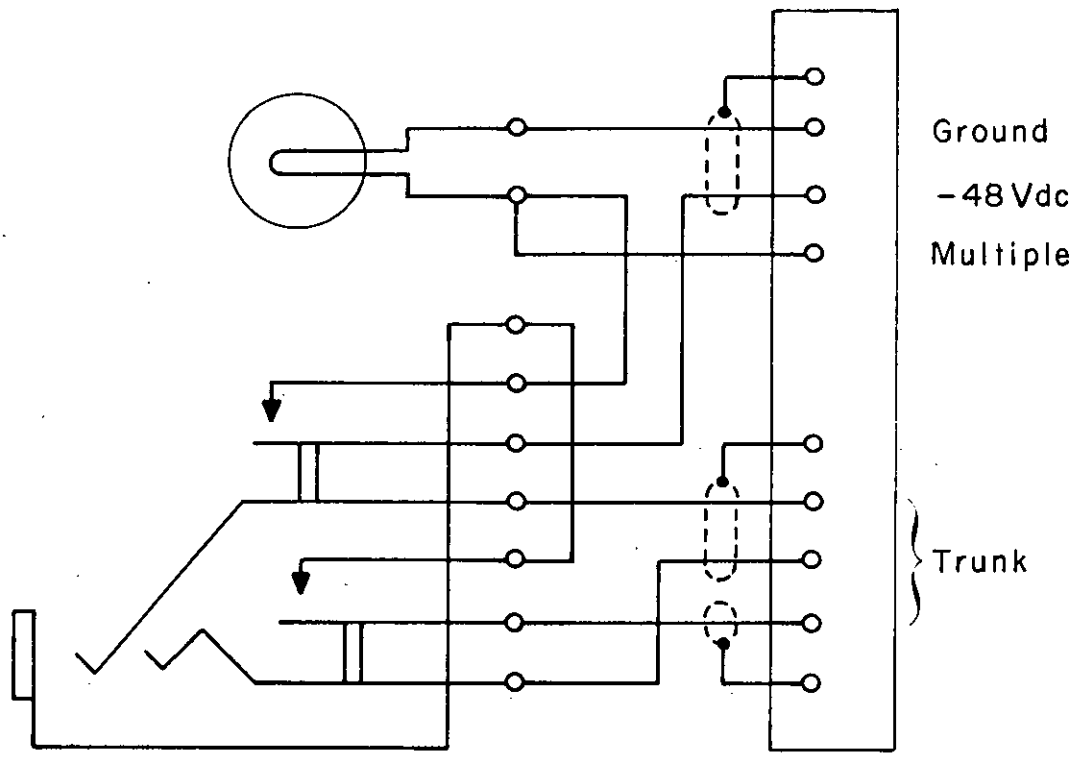


Figure 27. Jack Wiring of Interbay Trunks. (Cable Test Bay)

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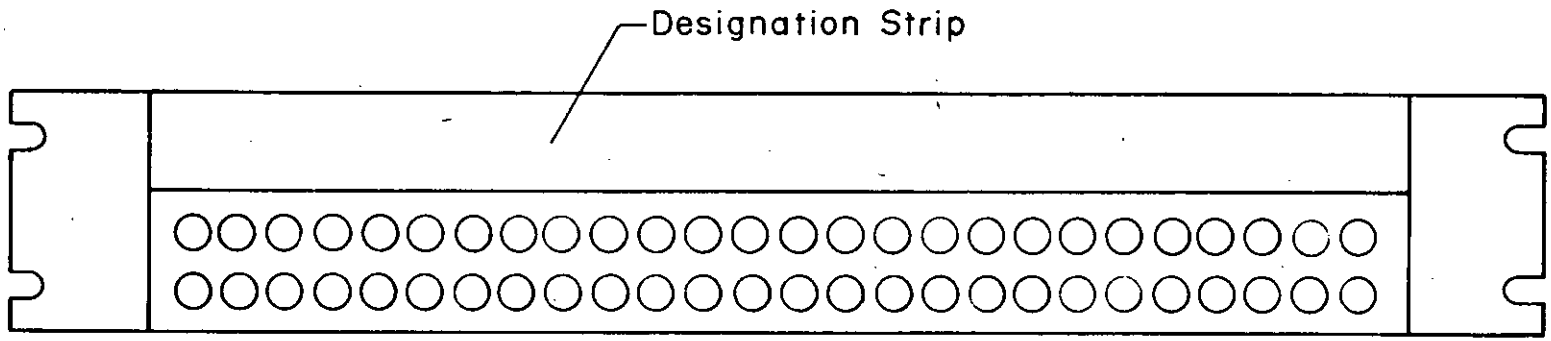


Figure 28. Test Equipment Jack Module. (Cable Test Bay)

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Chapter VIII

8. PATCH FACILITIES. The patch facilities consist of the following major units:

a. DC patch bay, includes primary DC patch bay and circuit DC patch bay.

b. Audio patch bay, includes Primary VF patch bay and circuit VF Equal Level patch bay.

c. Wideband patching module.

8.1 DC PATCH BAY.

8.1.1 Application. The dc patch bay is used to control, reroute, monitor, terminate, and test various types of digital signals. It is designed to accept unbalanced inputs and outputs. All dc circuits entering and leaving TCF which are not defined as standard low level, or requiring regeneration, are routed through the DC primary patch. The DC circuit patch contains patching facilities necessary to interface the transmission media at the standard equal level. The equipment consists of the following modular components:

a. Send jack set patching module (3 each).

b. Receive jack set patching module (3 each).

c. Miscellaneous jack set patching module (1 each).

d. Analog-type voltmeters (2 each).

8.1.1.1 The send and receive jack set modules provide for normal-through operation between equipment and line. Line and equipment jack are provided as well as a shunt-type, high-impedance monitoring jack for each circuit. The modules can accommodate either low-level or high-level signals, by changing the value of load resistors. Cut-key circuits are provided by which receiving and sending devices may be disconnected and visual indication (red light) of such action is provided. Examples of DC send and receive jack sets are shown in figure 29 and 30.

8.1.1.2 The miscellaneous jack set patching module performs the various patch functions defined in figure 31 to 37.

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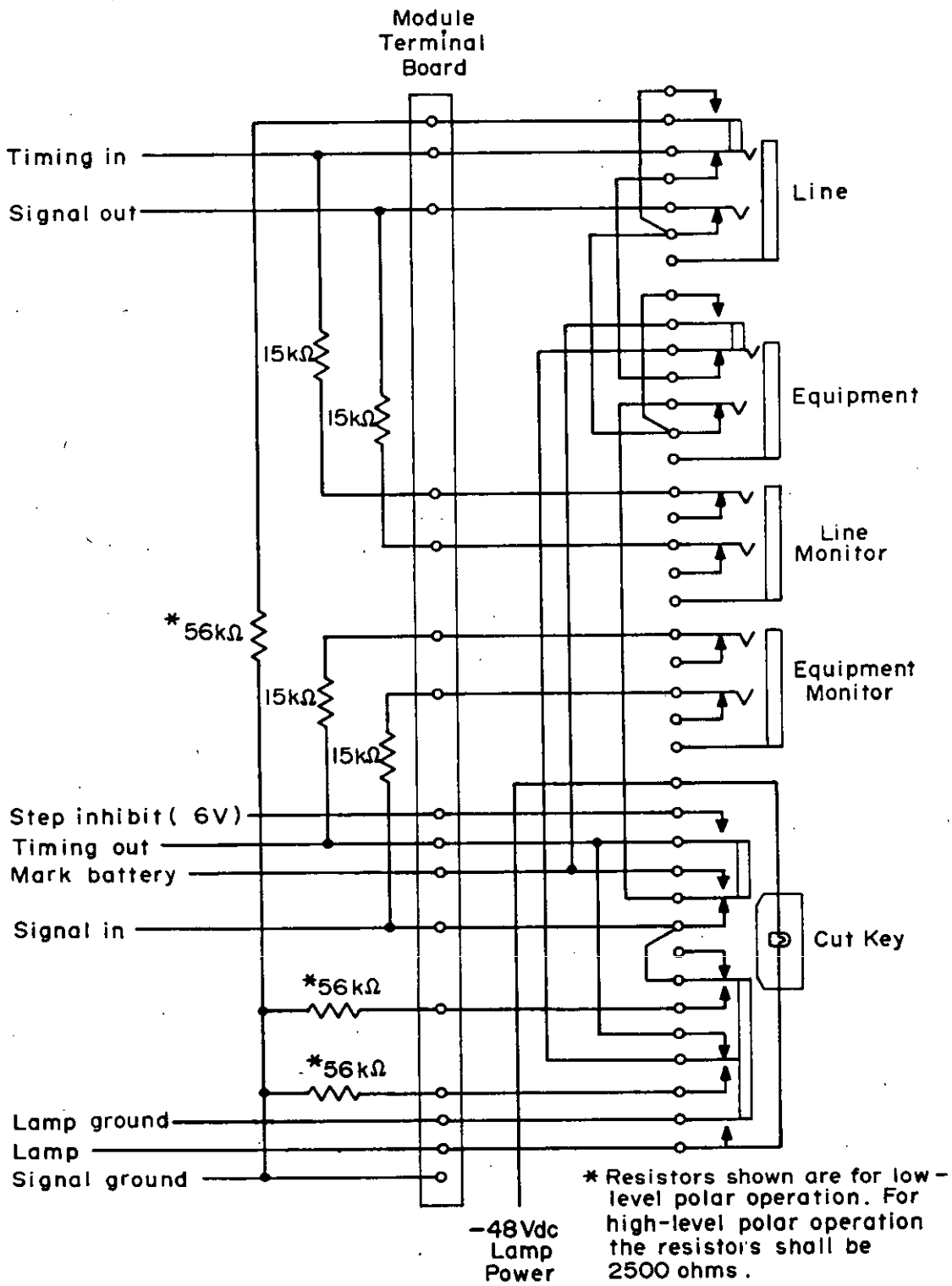


Figure 29. Send Jack Sets, Circuit Diagram.

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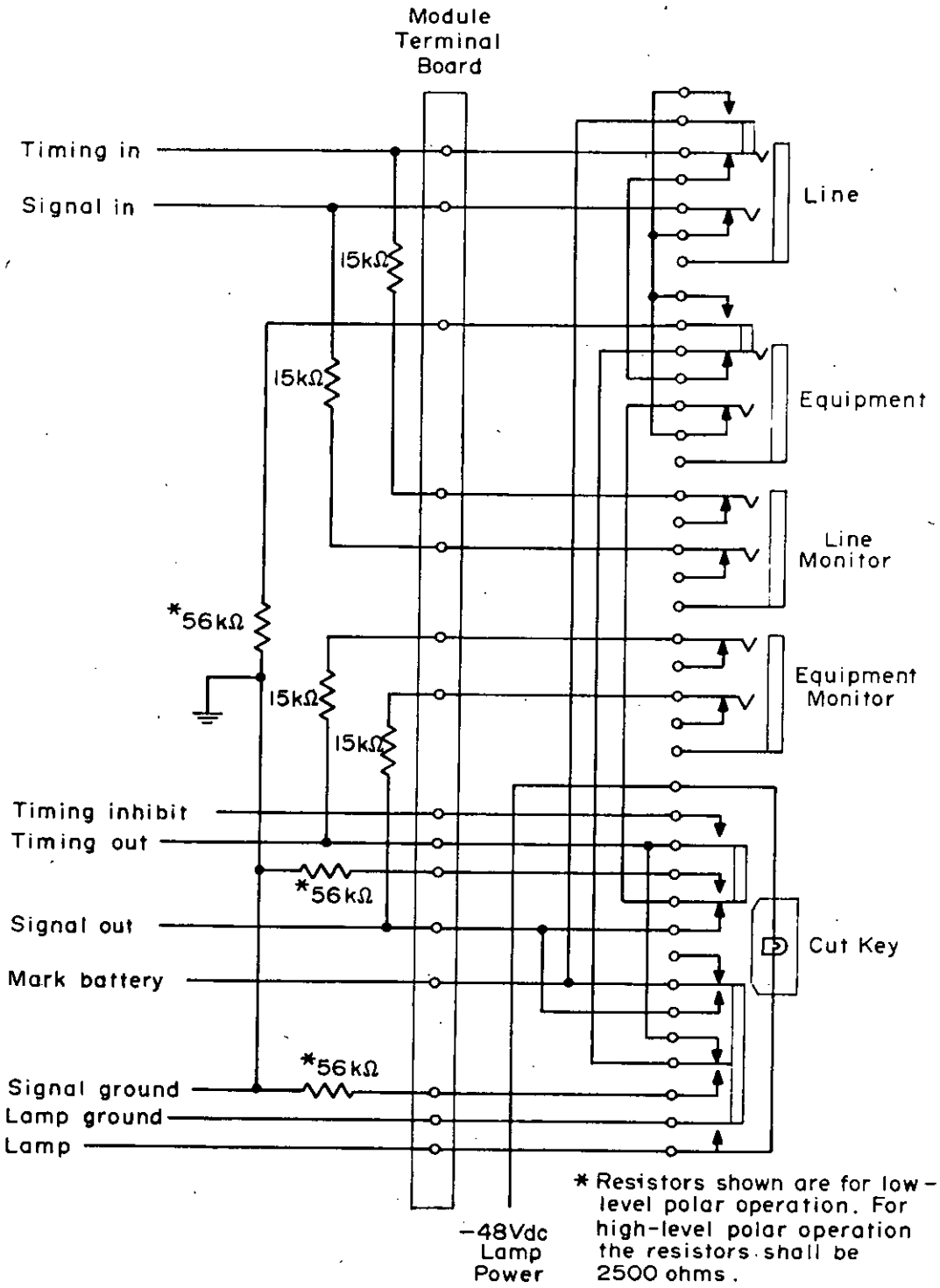
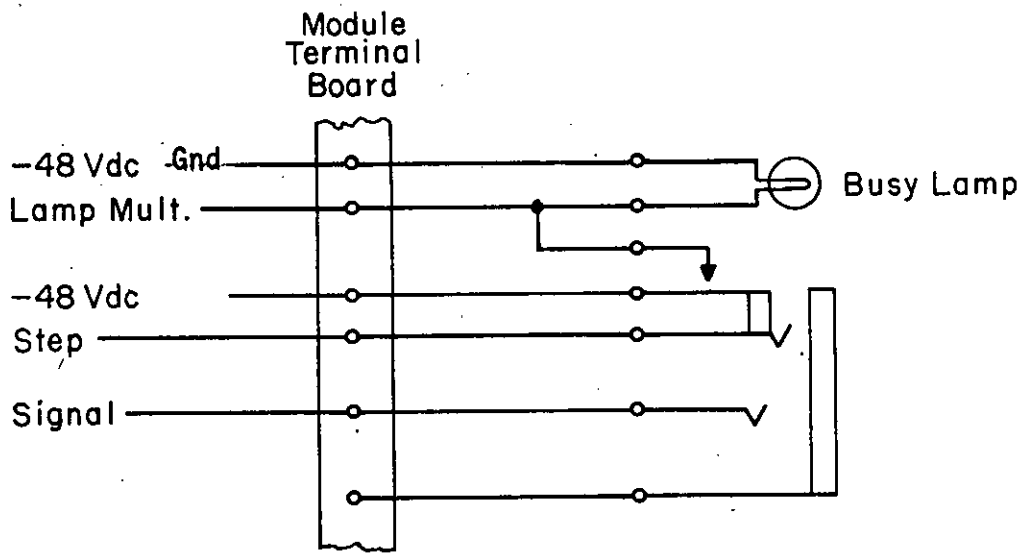
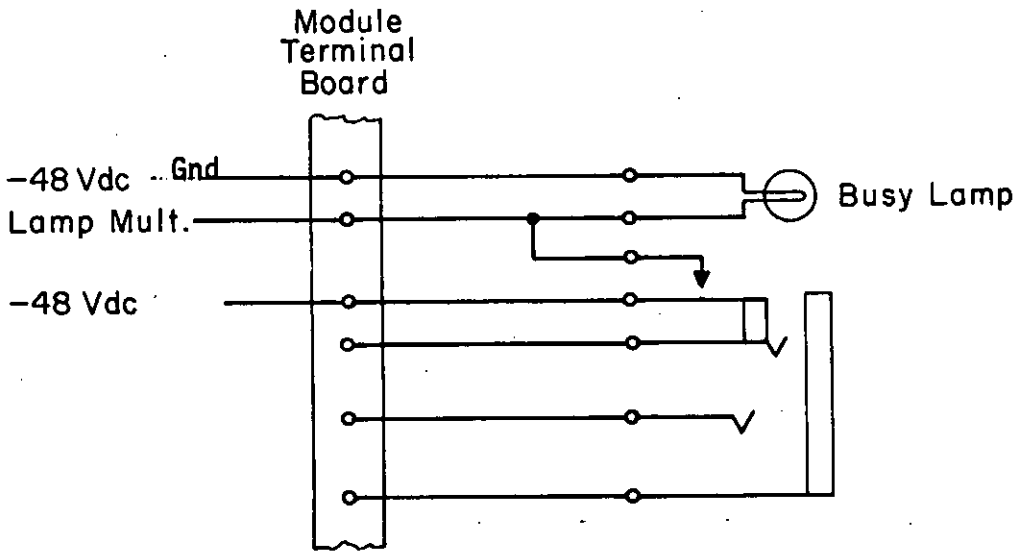


Figure 30. Receive Jack Sets, Circuit Diagram.

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(a) Low-level pattern generator output jack set



(b) Interbay trunk jack set

Figure 31. Low-Level Digital Pattern Generator Output and Interbay Jack Sets.

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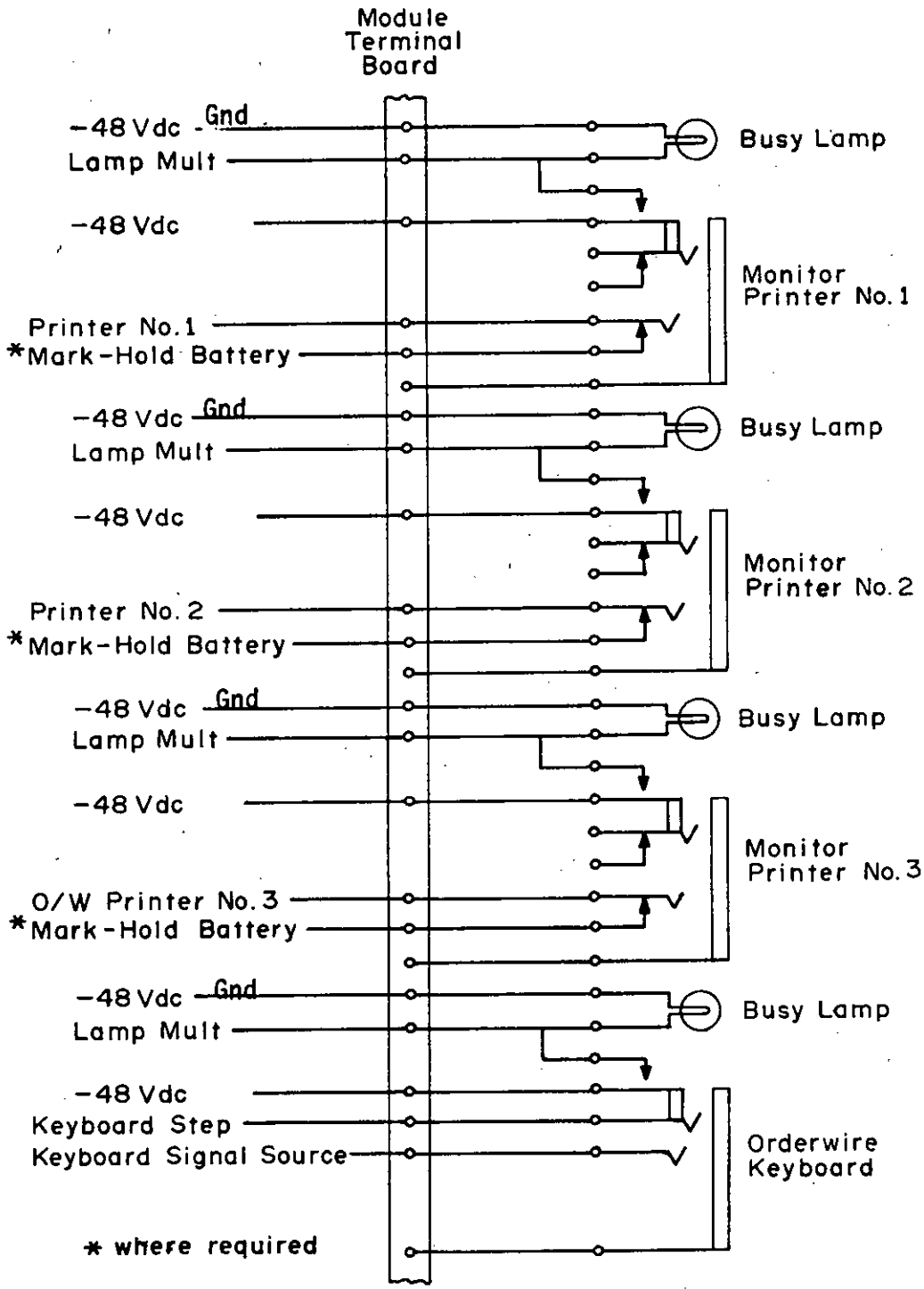
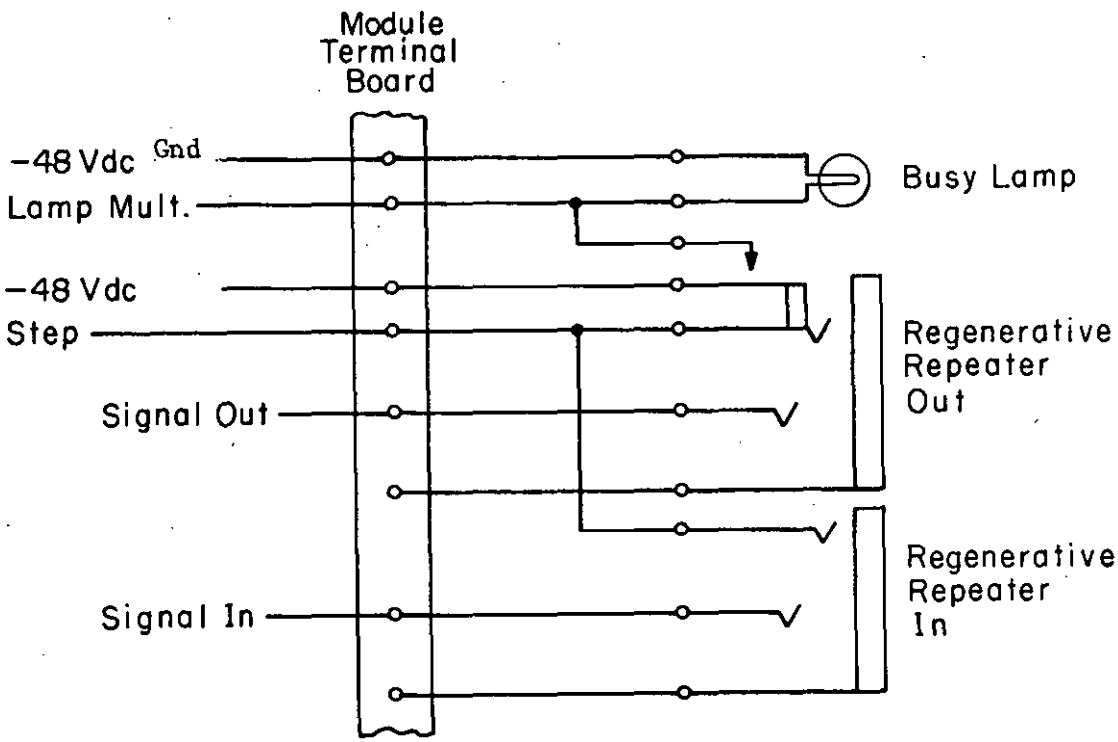
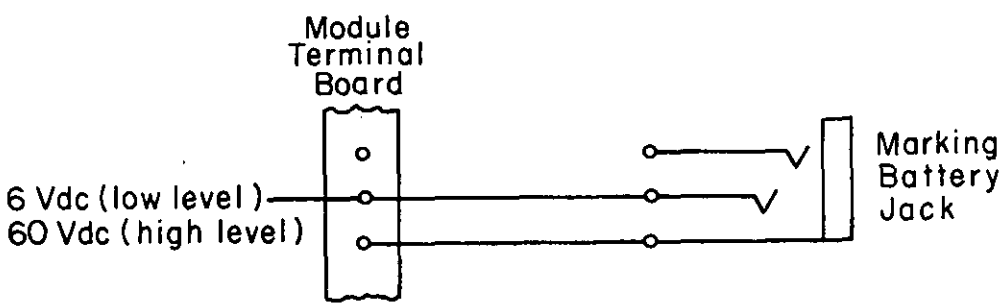


Figure 32. Teletypewriter Monitor Console Jack and Lamp.

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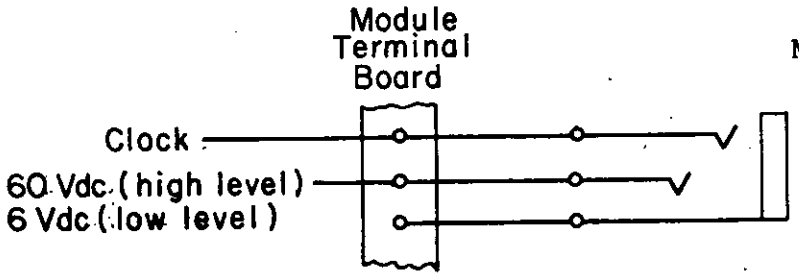
(a) Regenerative repeater jack set



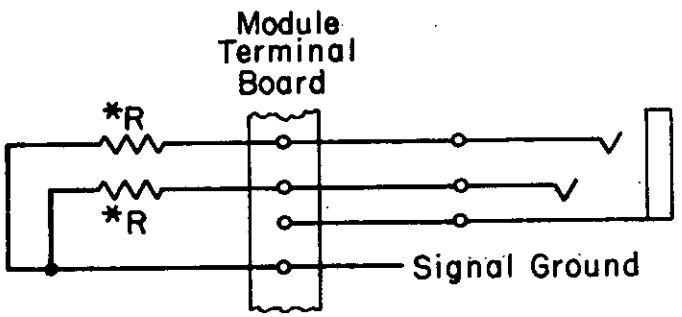
(b) Marking battery jack set

Figure 33. Regenerative Repeater and Marking Battery Jack Sets.

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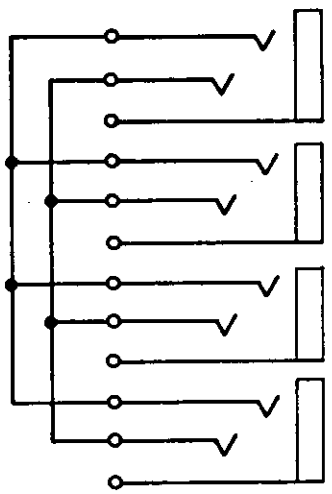


(a) Spacing battery jack set



*Note: R is 56kΩ ±10 percent for low-level signal.
and 2500Ω ±10 percent for high-level signal.

(b) Termination jack set



(c) Parallel jack set

Figure 34. Spacing Battery, Termination, and Parallel Jack Sets.

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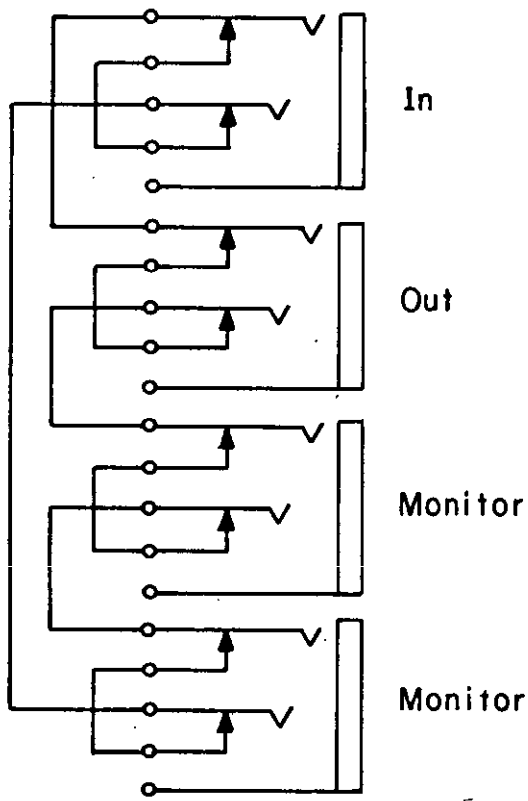
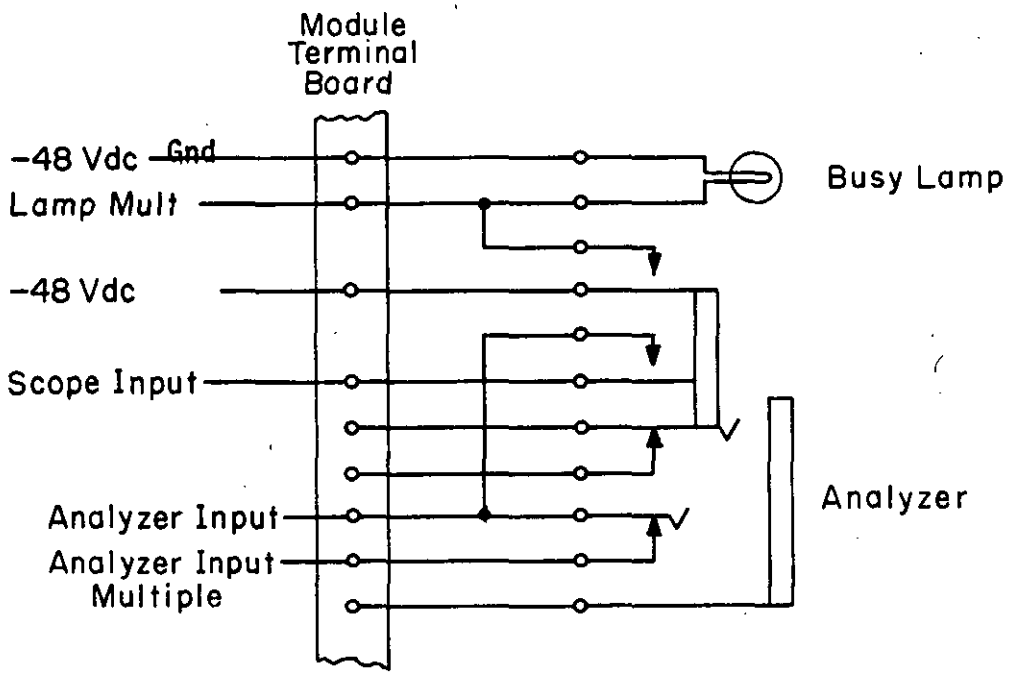
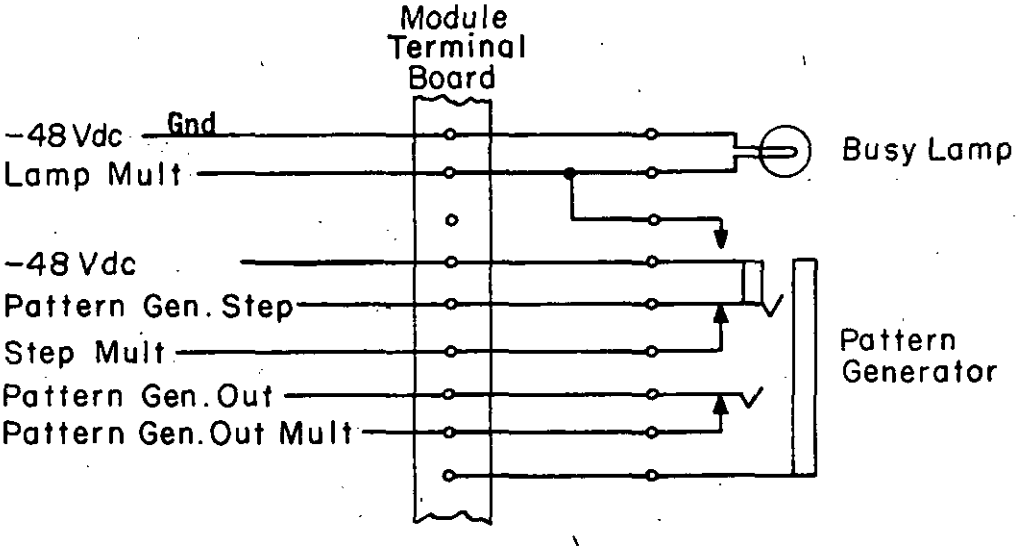


Figure 35. Looping Jack Set.

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(a) GGM-16 distortion analyzer jack set



(b) GGM-16 pattern generator output jack set

Figure 36. GGM-16 Distortion Analyzer and Pattern Generator Output Jack Sets.

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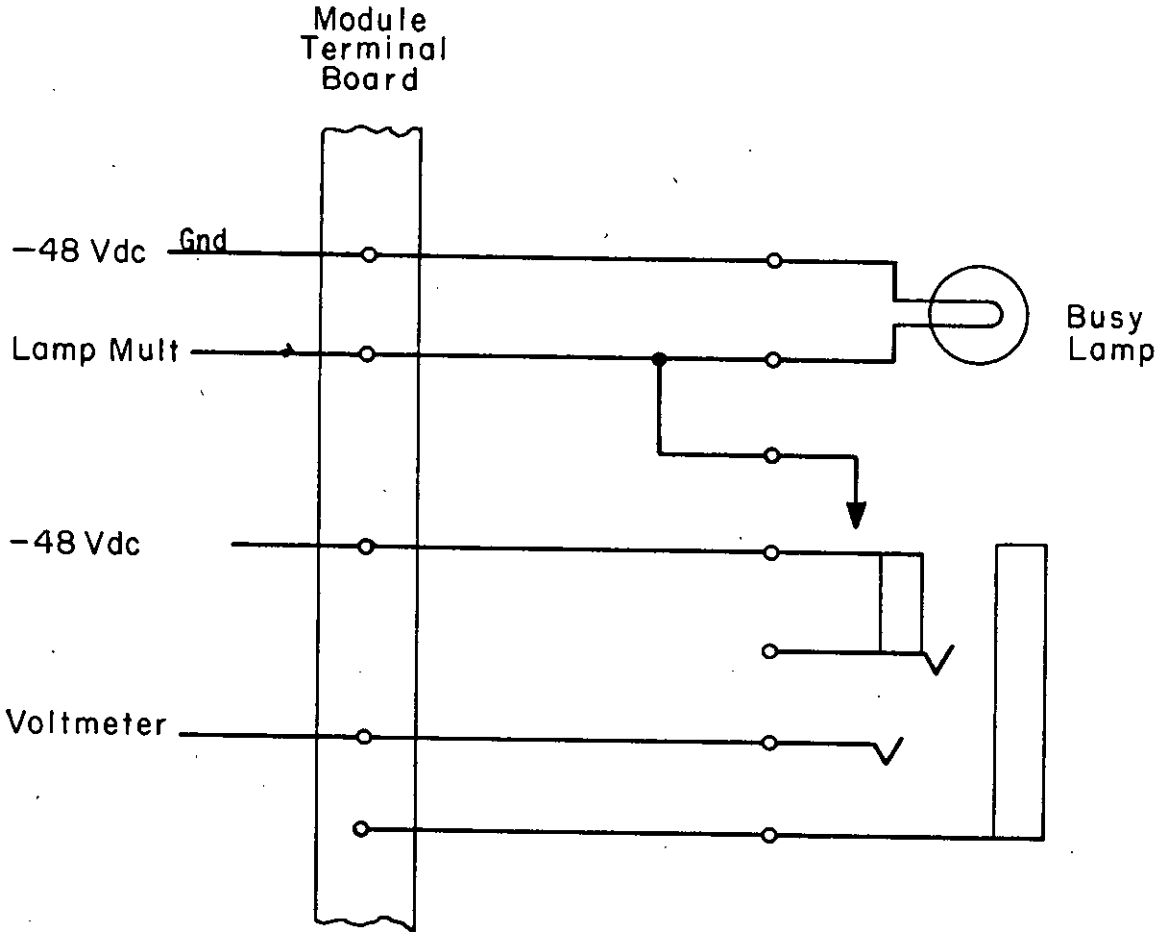


Figure 37. Voltmeter Jack Sets.

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8.1.1.3 The voltmeters are supplied for circuit testing.

8.1.2 Equipment characteristics. The dc patch bay shall conform to the following characteristics:

a. Signal types: By strapping to change values of terminating resistors, the equipment shall be capable of accommodating either high- or low-level signals as defined below.

(1) Low level: ±6 V polar unbalanced in accordance with MIL-STD-188-114 (5.2.2.1).

(2) High level: ±60 V polar unbalanced in accordance with MIL-STD-188-100 (4.3.1.4.1 and 4.3.1.4.2).

b. Terminating resistance: Values of all terminating resistors supplied by the patching operation shall be as follows:

(1) 56 K ohms ±10% for low-level circuits.

(2) 2.5 K ohms ±10% for high-level circuits.

c. Configuration: The jack field mounted in the front of the cabinet shall be arranged so that a send jack set patching module shall occupy the topmost portion of the jack field. A receive jack set patching module shall be mounted directly below the send jack set module. The send and receive groups, when assembled together, form three groupings of at least 24 duplex circuits. The miscellanous jack set module shall be mounted below the send and receive modules.

d. Patch cords: Sufficient patch cords compatible with the jacks shall be furnished.

8.1.2.1 Send jack set patching module. An example of a DC send jack set is shown in figure 8-1.

a. Circuit and jack-field configuration: At least 24 send jack sets shall be combined with an associated module terminal board, supporting framework and covers to constitute a patching module. Each of the send jack sets shall be arranged in a vertical row and shall appear in the following order from top to bottom: Line patch, equipment patch, line monitor, equipment monitor, and cut-key. The patching module shall be provided with all internal wiring completed.

b. Jacks: Tip, ring and sleeve type conforming to MIL-J-641.

c. Lamps: All lamps shall operate from a nominal 48 V dc source. Red lenses shall be used on cut-key circuits and white lenses on other circuits.

d. Cut key: The red lamp shall illuminate when switch is in "disconnect" position. The cut key shall be capable of disconnecting the sending device from the line and of switching in necessary circuit termination.

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8.1.2.2 Receive jack set patching module. An example of a DC receive jack set is shown in figure 30.

a. Circuit and jack-field configuration: At least 24 receive jack set shall be combined with an associated module terminal board supporting framework and covers to constitute a patching module. Each of the receive jack sets shall be arranged in a vertical row and shall appear in the following order from top to bottom: Line patch, equipment patch, line monitor, equipment monitor, and cut key. The patching module shall be provided with all internal wiring completed.

b. Jacks: Same as in paragraph 8.1.2.1b.

c. Lamps: Same as in paragraph 8.1.2.1c.

d. Cut key: The red lamp shall illuminate when switch is in "disconnect" position. The cut-key shall be capable of disconnecting the receiving device from the line and of switching in necessary circuit terminations.

8.1.2.3 Miscellaneous jack set patching module.

a. Jack-field configuration: Jack sets shall be configured in rows of at least 24 each. Busy lights and jacks in each row shall be mounted with the same vertical and horizontal center distances as those in the send and receive modules. The top row shall consist of at least 24 interbay jack sets of the type illustrated in figure 31. Other jack sets specified in b. below shall be mounted below the interbay sets. The module shall have a total capacity of seven rows of 24 jack sets each. The module shall be provided with a terminal block and with internal wiring of required jack sets complete.

b. Required jack sets.

(1) At least 24 interbay jack sets as illustrated in figure 31(b) (see a. above for mounting position in module).

(2) Ten low-level digital pattern generator output jack sets (see figure 31(a)).

(3) Two teletypewriter monitor console jack sets (see figure 32).

(4) Three regenerative repeater jack sets (see figure 33(a)).

(5) Quantity required of marking battery jack sets (see figure 33(b)).

(6) Quantity required of spacing battery jack sets (see figure 34(a)).

(7) Quantity required of termination jack sets (see figure 34(b)).

(8) Quantity required of parallel jack sets (see figure 34(c)).

(9) One looping jack set (see figure 35).

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(10) One GGM-16 pattern generator output jack set (see figure 36(b)).

(11) One GGM-16 distortion analyzer input jack set (see figure 36(a)).

(12) Two analog voltmeter jack sets (see figure 37).

8.1.2.4 Analog voltmeter.

a. Configuration: Two each installed at the top of the patch bay.

b. Range reading arrangement: 0 to ±6 V in one range for low-level operation and 0 to ±60 V in one range for high-level operation.

c. Voltmeter sensitivity shall be 10,000 ohms per volt minimum.

d. Accuracy: ±2% of full scale.

8.2 AUDIO PATCH CABINET. This patch bay may contain the following major units depending on the operational requirements.

a. DC multimeter.

b. Noise and level measuring set.

c. Type I patching module.

d. Miscellaneous jack module.

e. Interbay trunk patch module.

8.2.1 Application. The function of the audio patch cabinet is to provide a means of patching various lines and equipments to achieve the desired voice and data traffic and signal routing. The unit enables monitoring of all incoming and outgoing circuits and provides a means of connecting to either the line or equipment side for test purposes. It enables patching of channels and terminal equipment for maintenance purposes. The audio patch unit is used for both primary and equal-level patching. The VF primary Patch Bay contains the patching facilities to interface associated signal and conditioning equipment to VF user circuits entering and leaving the TCF that are not at VF standard equal level. The VF circuit patch bay contains the facilities necessary to interface the transmission media at an equal level standard.

8.2.1.1 The type I patching module can accommodate two-, four-, or six-wire (four-wire plus E&M leads) circuits. When two type I modules are included in the audio patch cabinet, eight-wire (four-wire plus E, M, echo suppressor control and PMB leads) circuits can be accommodated.

8.2.2 Equipment characteristics. The equipment items shall conform to the following characteristics:

8.2.2.1 Digital multimeter capabilities:

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a. DC voltage: 150 mV to 150 V with input resistance of 60,000 ohms minimum and accuracy better than 1% of reading.

b. DC current: 150 mA to 0.5 A with accuracy better than 1% of reading.

c. DC resistance range: 0.1 ohm to 50 megohms with accuracy better than 1% of reading.

d. Other multimeter features: Three-digit readout, automatic ranging, overrange indicator, automatic polarity indicator and overrange protection circuitry.

e. Power requirements: 15 watts, maximum.

8.2.2.2 Digital noise and level measuring set capabilities.

a. Tone level measuring range: -50 to +10 dBm.

b. Tone level measuring accuracy: $\pm$0.1 dB for frequencies from 300 to 3400 Hz when line impedance is 600 or 900 ohms.

c. Impedance, input: 600, 900 ohms or bridging.

d. Frequency response: $\pm$0.1 dB from 300 to 3400 Hz.

e. Noise weighting: C-message.

f. Noise measuring range: 0 to 92 dBrnc.

g. Noise measuring accuracy: $\pm$0.5 dB for noise with a Gaussian probability density distribution.

h. Three-digit readout with readings in dB and polarity indicator.

i. Surge protection: (See 2.2).

j. Power requirements: 40 watts maximum (See 2.3).

8.2.2.3 Type I patching module (six-wire):

a. Type of patching jacks: All jacks shall accommodate a telephone plug conforming mechanically and the 310-type telephone plug.

b. Jacks and terminal board: The module shall contain at least 96 jacks (24 by 4) and a terminal board as follows:

(1) The top two rows shall contain at least 48 three-conductor (tip, ring, and sleeve) patching jacks with all three conductors wired in a normal through arrangement and terminated on the module terminal board. An example of a (26 by 4) Type I jack module is shown in figure 38. A functional example of jack wiring is illustrated in figure 39.

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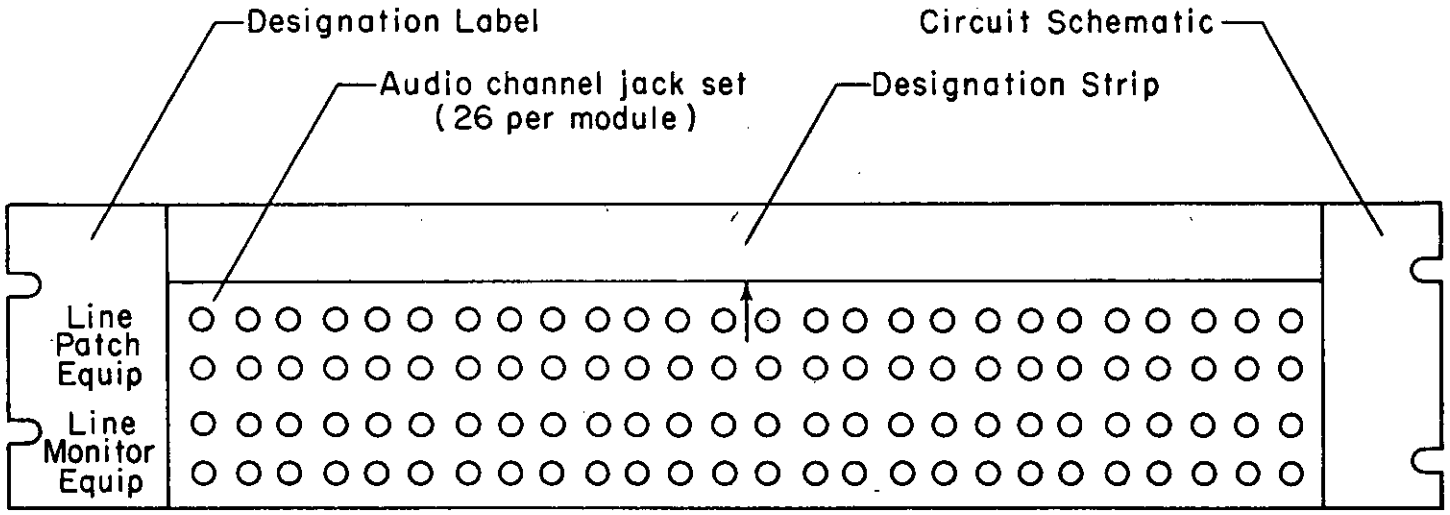


Figure 38. Type I Jack Module.

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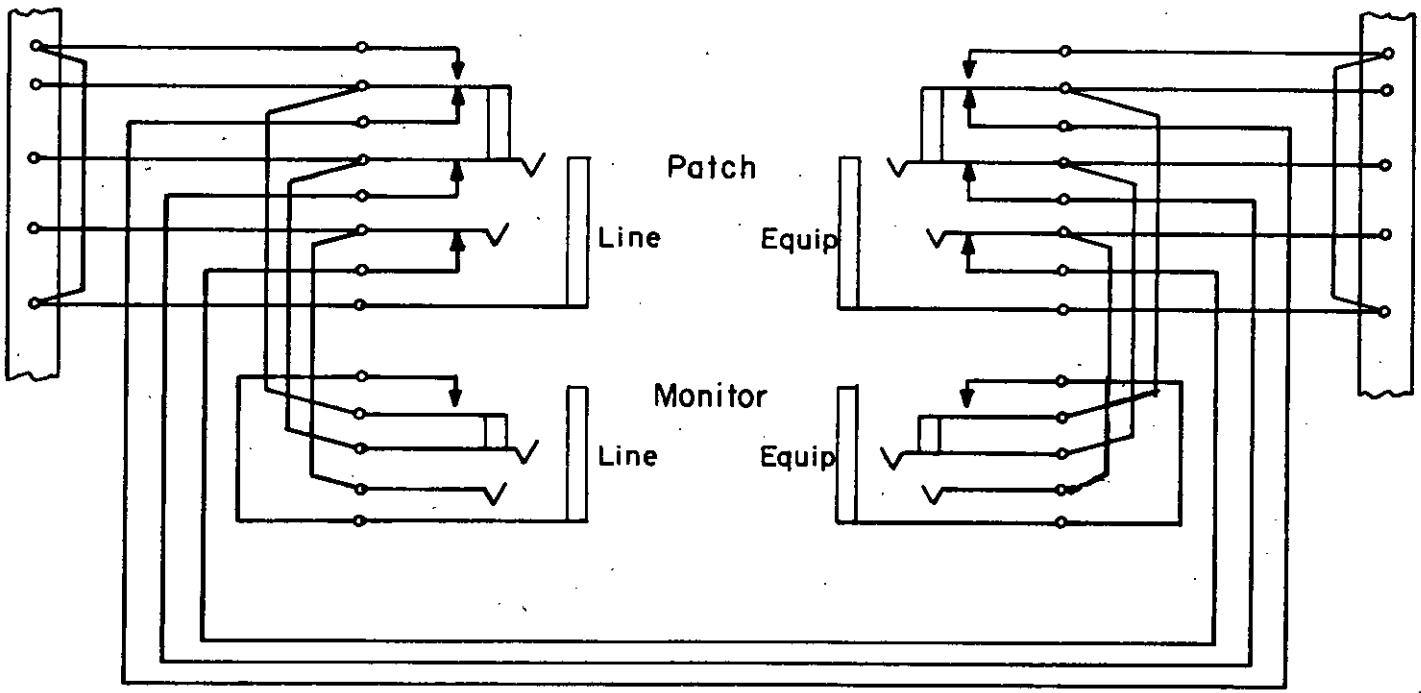


Figure 39. Wiring for Audio Channel Jack Module Send and Receive (26 per Module).

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(2) The bottom two rows shall contain at least 48 jacks (equivalent number to that described in (1) above) used for monitoring purposes. Wiring for the monitoring jacks is functionally illustrated in figure 40.

(3) The module terminal board shall be capable of accommodating all patching jacks from this module.

8.2.2.4 Miscellaneous jack module. The module shall contain at least 48 jack sets. An example of a 52 jack set module is illustrated in figure 40.

a. Access circuits: The module shall provide, but not be limited to the following jack circuits:

(1) One each as illustrated in figure 41(a).

(2) Ten each as illustrated in figure 41(b).

(3) Seventeen each as illustrated in figure 41(c).

b. Jack arrangements: At least 24 jacks and lamp sets. An example of a 26 jack and lamp arrangement is illustrated in figure 42, with wiring as depicted in figure 43.

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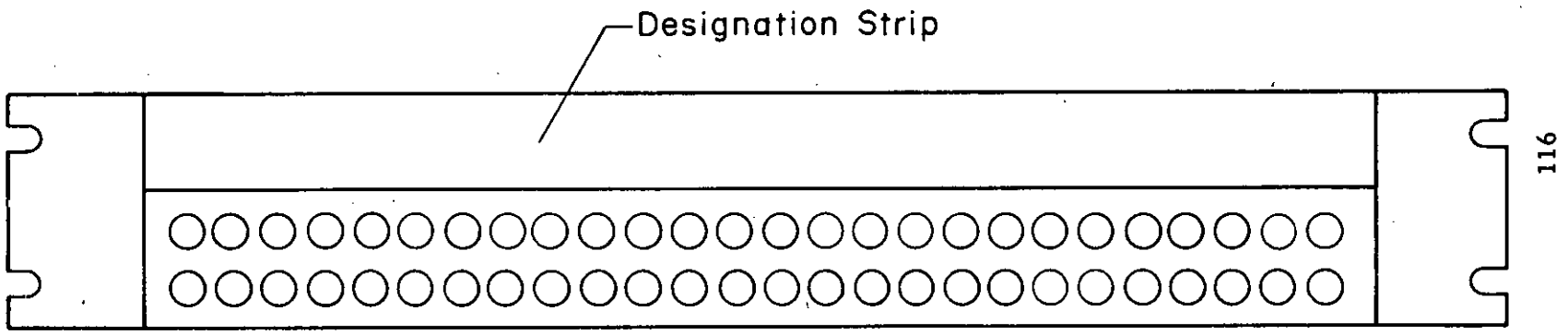
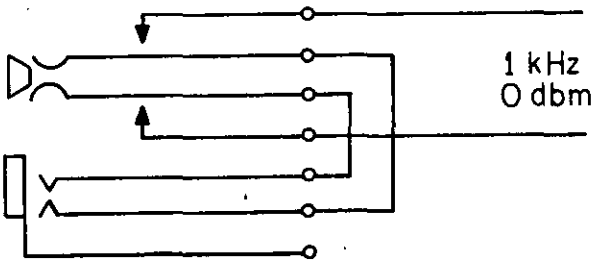
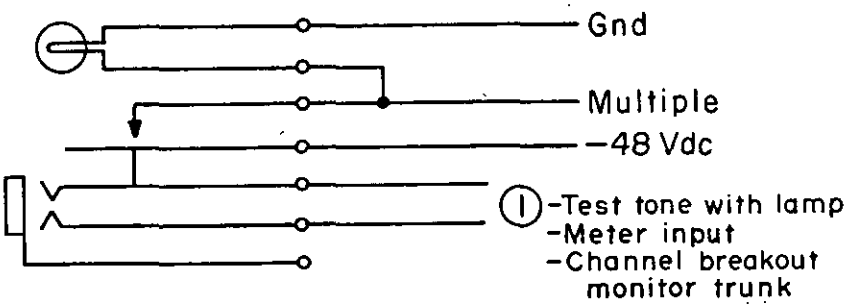


Figure 40. Miscellaneous Jack Module.

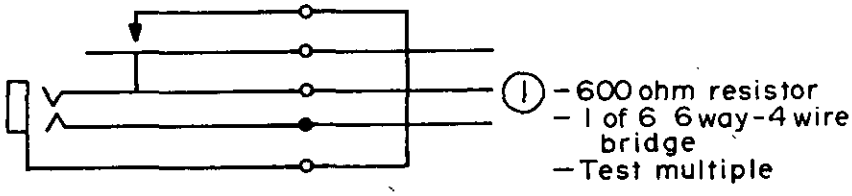
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(a) Test tone with key



(b) Jack with lamp



(c) Wire jack

Figure 41. Test Tone with Key, Jack with Lamp, and Wire Jack.

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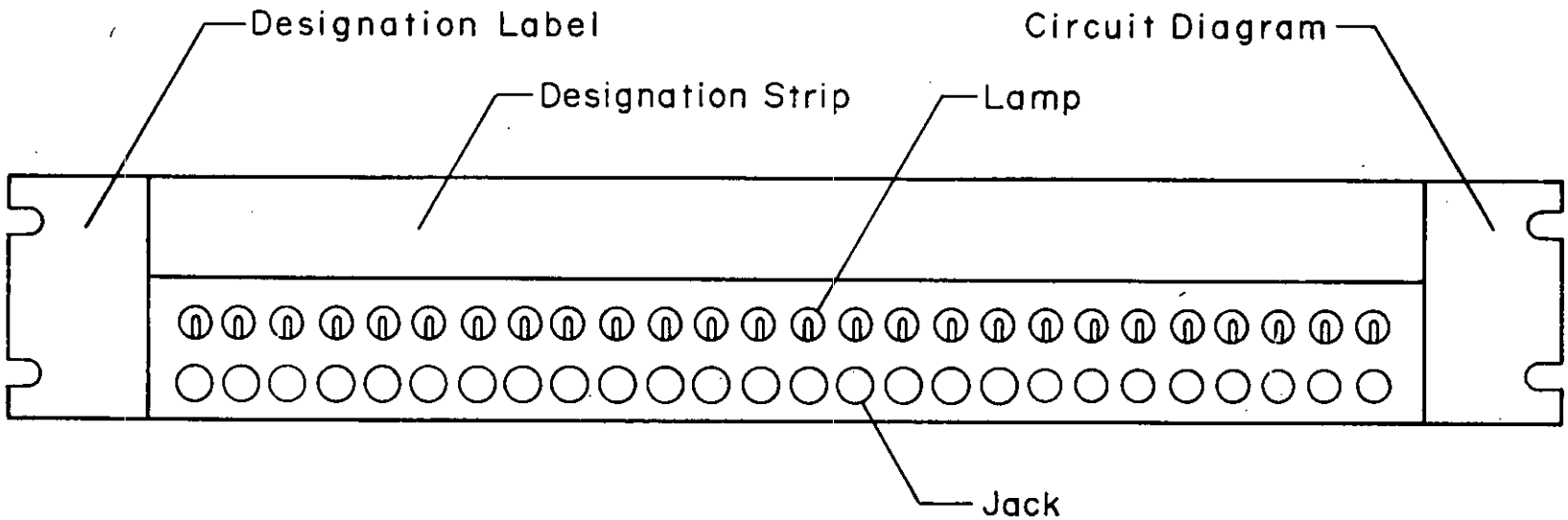


Figure 42. Interbay Trunk Module (Audio Patch Cabinet).

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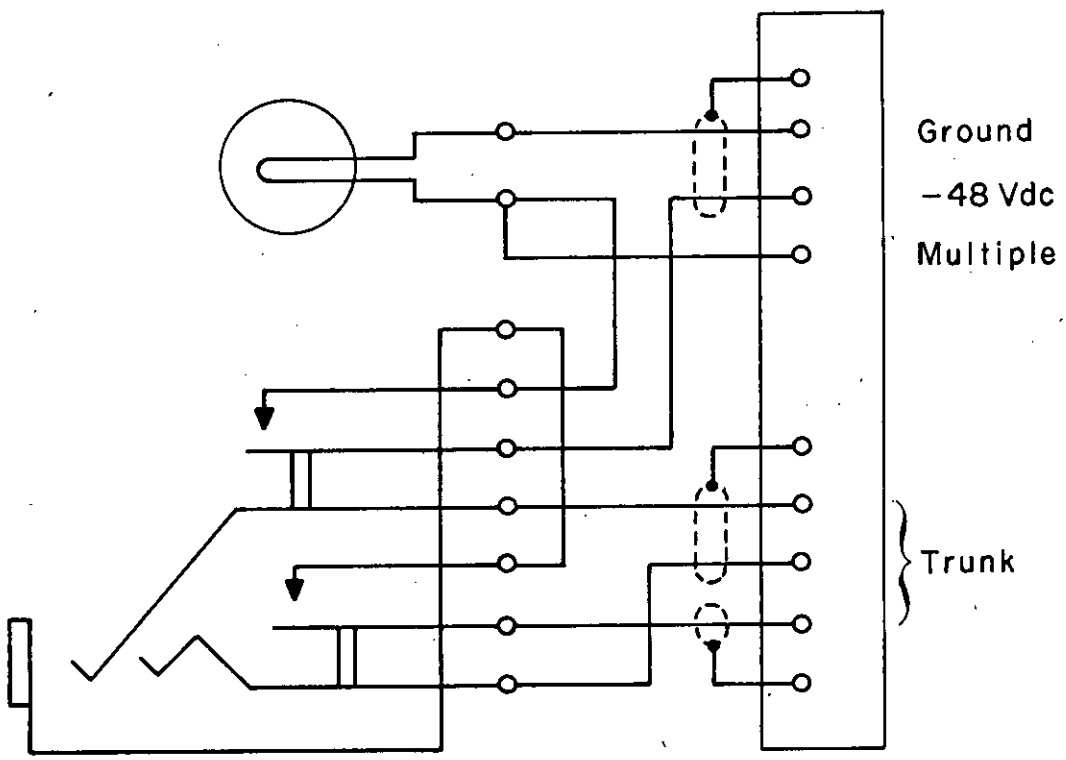


Figure 43. Jack Wiring of Interbay Trunks (Audio Patch Cabinet).

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8.2.2.5 Interbay jack module.

a. Type of jacks: Same as Type I module.

b. Jack arrangements: Twenty-six jacks and lamp sets arranged and wired as shown in figure 42 and 43.

8.3 WIDEBAND PATCHING MODULED. Wideband patching modules are classified according to the following three types:

a. Type I patching module.

b. Type II patching module.

c. Type III patching module.

8.3.1 Application. The wideband patching modules are used for controlling, routing, monitoring and testing of multiplex group, supergroup and baseband signals. The three types of modules are designed for use in the BLACK portion of a communication complex. The type I patching module accommodates the patching and monitoring of radio multiplex baseband and supergroup channels (frequency range from 3 kHz to 3 MHz). The type II patching module accommodates the patching and monitoring of multiplex group channels (frequency range from 60 kHz to 108 kHz), attenuation not to exceed $\pm$0.25 db throughout the frequency range. The type III patching module accommodates the patching of multiplex group (frequency range from 60 kHz to 108 kHz, attenuation not to exceed 0.1 dB), ancillary equipment. The signal levels present in the channels for all three types range from -50 to +16 dBm referenced to one (1) millwatt.

8.3.1.1 Tye type I patching module consists of coaxial and triaxial jacks arranged as shown in figure 44. For supergroup patching, coaxial jacks are mounted on the left side of the panel in ten groups of four each. For baseband patching, 16 triaxial jacks are mounted on the right side (these are represented by the larger circles in figure 44). Figure 45 shows the electrical connections between the rear coaxial and the front coaxial jacks. The connector and jack are actually made into one coaxial unit and no wiring is required. The coaxial jacks and connectors, and thus the coaxial shields, are insulated from the equipment chassis.

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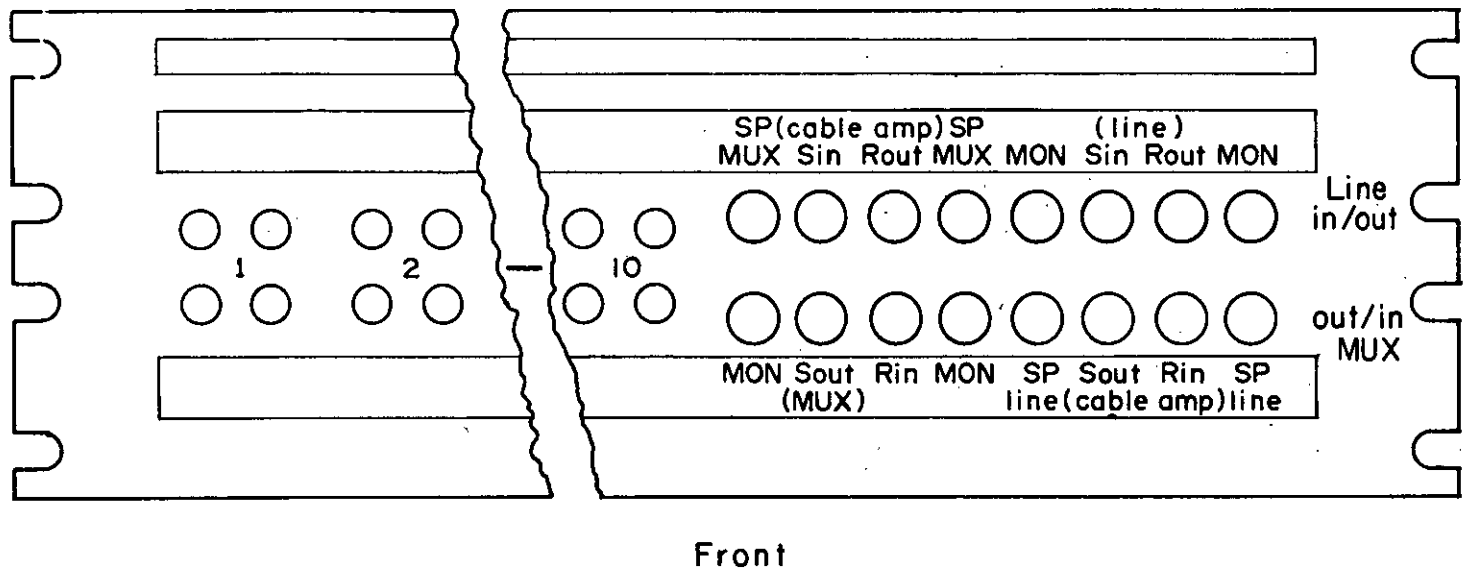


Figure 44. Wideband Patching Module.

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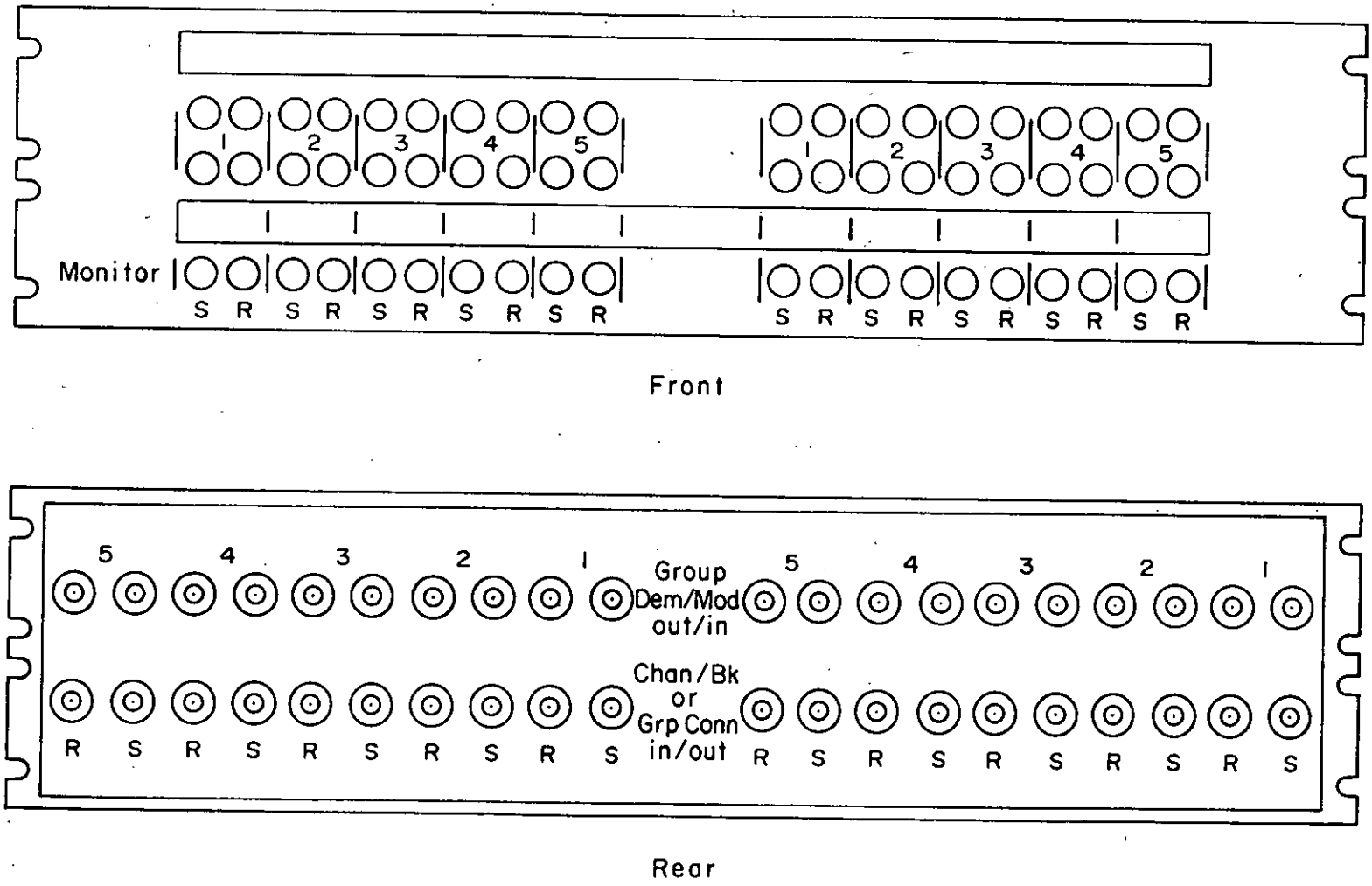


Figure 45. Type II Wideband Patching Module.

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8.3.1.2 The triaxial connectors in the type I module are made in a similar manner (connectors and jacks in one piece) to the coaxial ones. The only difference between the two is an additional through connection on the triaxial arrangement for the extra cable shield connection. Both shield connections are insulated from each other and from the module chassis. (see figure 46)

8.3.1.3 The type II module provides jacks and connectors arranged as shown in figure 8-18. The front panel is made up of two groups of 30 jacks each arranged in subgroups that contain six jacks each. Each subgroup is a duplex line and equipment patching arrangement that includes two associated monitor jacks. The rear panel contains 40 concentric connectors that take twinaxial cable. Figures 47 and 48 show wiring diagrams for receive and send jack arrangements. Patching jacks are tip-ring-sleeve types. All connectors and jacks are insulated from the module chassis.

8.3.1.4 The type III module has 52 tip-ring-sleeve jacks on the front panel arranged as shown in figure 49. Fifty-two concentric connectors for twinaxial cable are mounted on the rear of the module. Figure 50 shows how the jacks are connected to the twinaxial connectors. All connectors and jacks are insulated from the module chassis.

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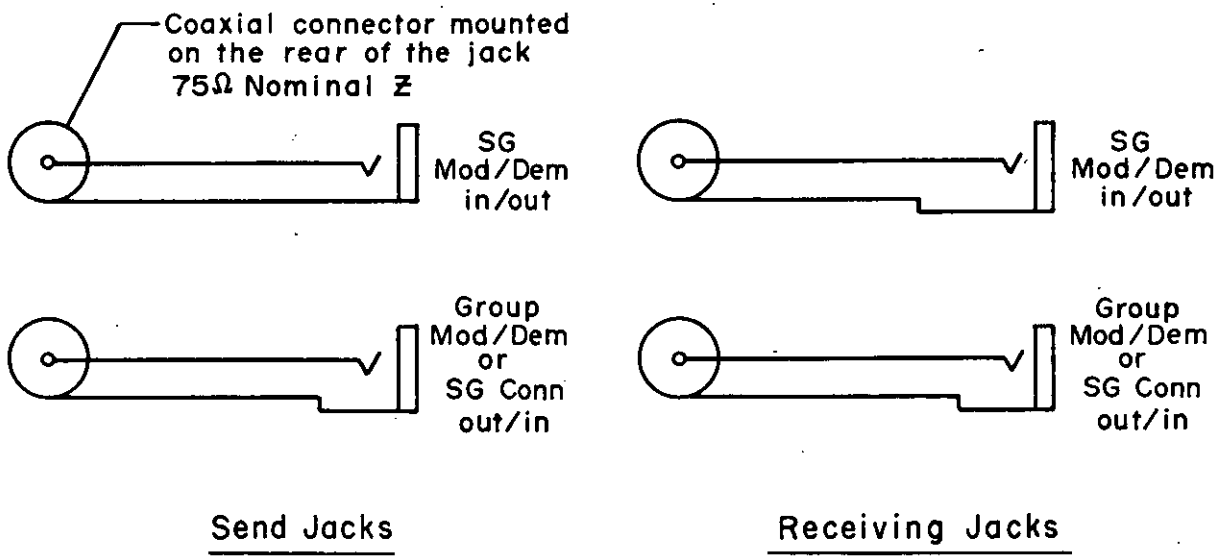


Figure 46. Type I Patching Module Jack Arrangement.

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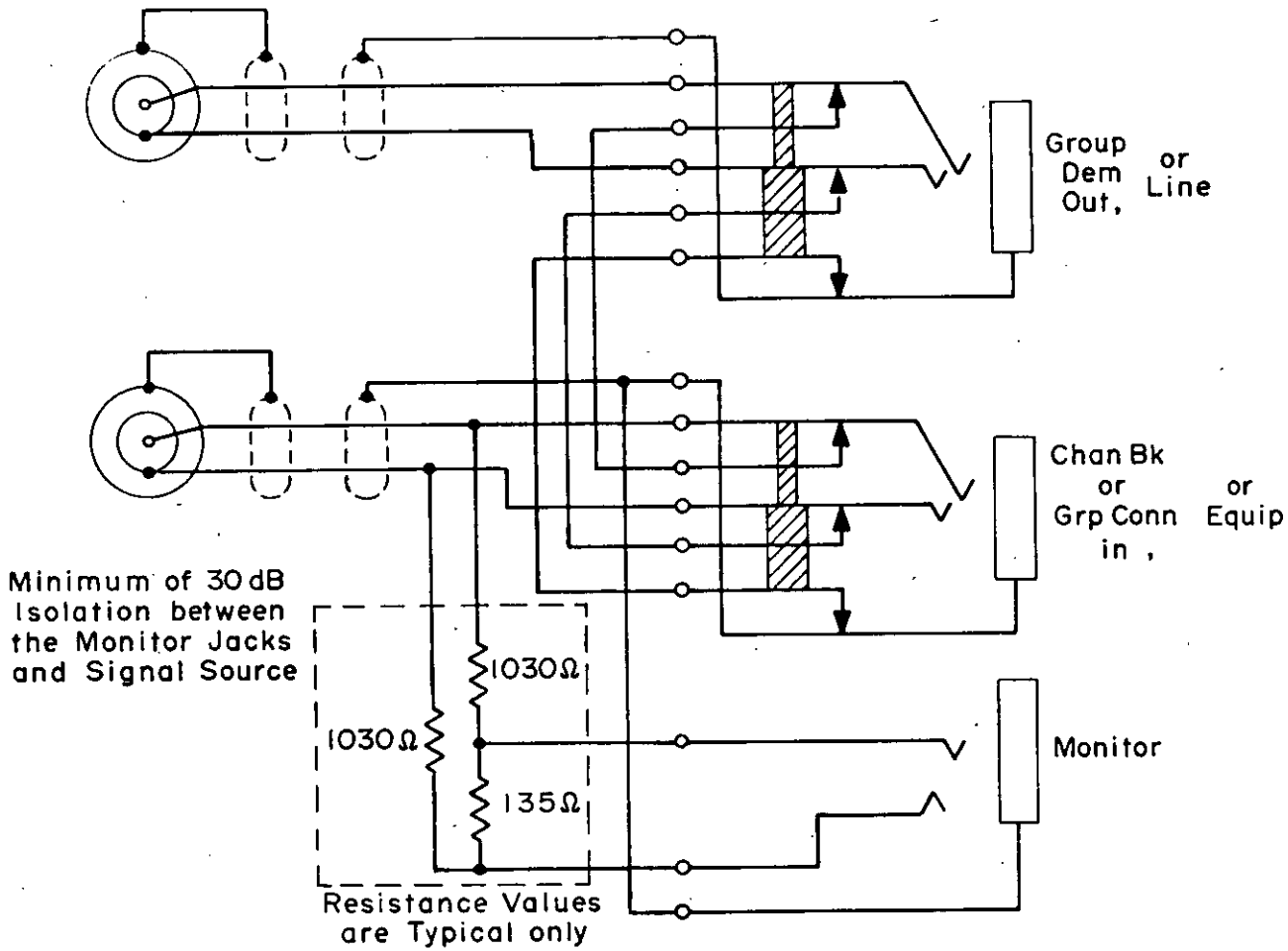


Figure 47. Type II Patching Module Receive Jack Arrangement.

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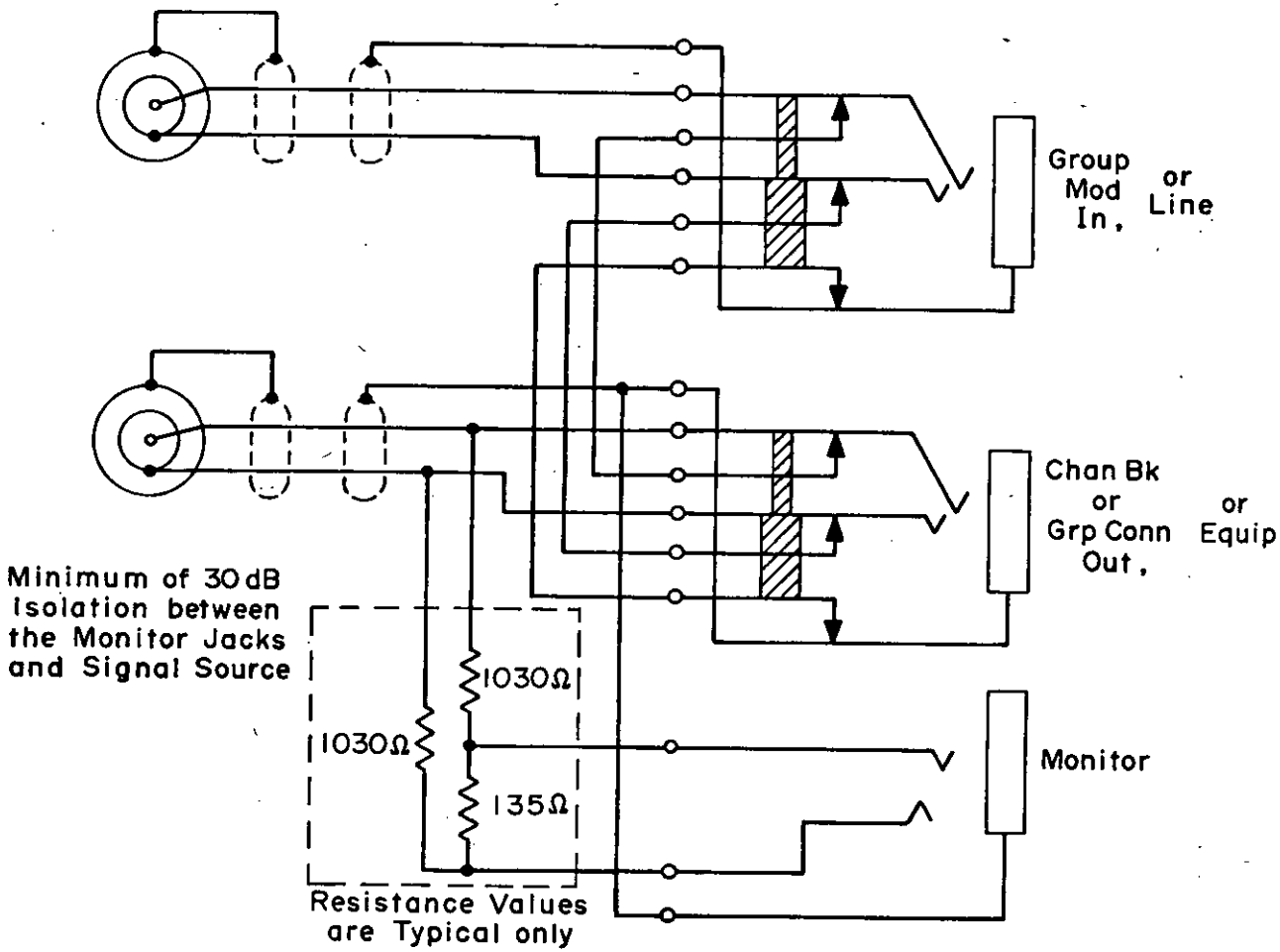


Figure 48. Type II Patching Module Send Jack Arrangement.

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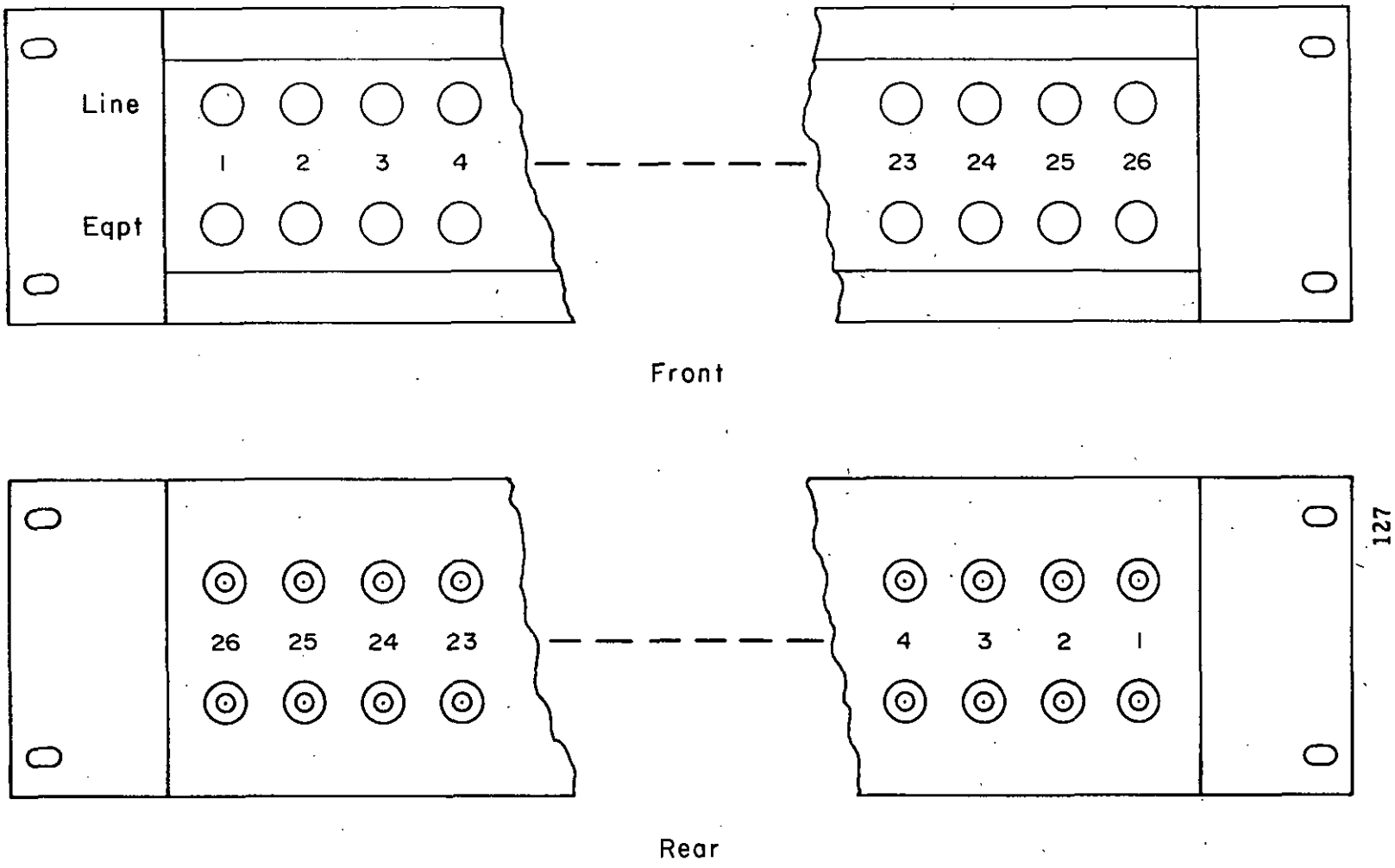


Figure 49. Type III Wideband Patching Module.

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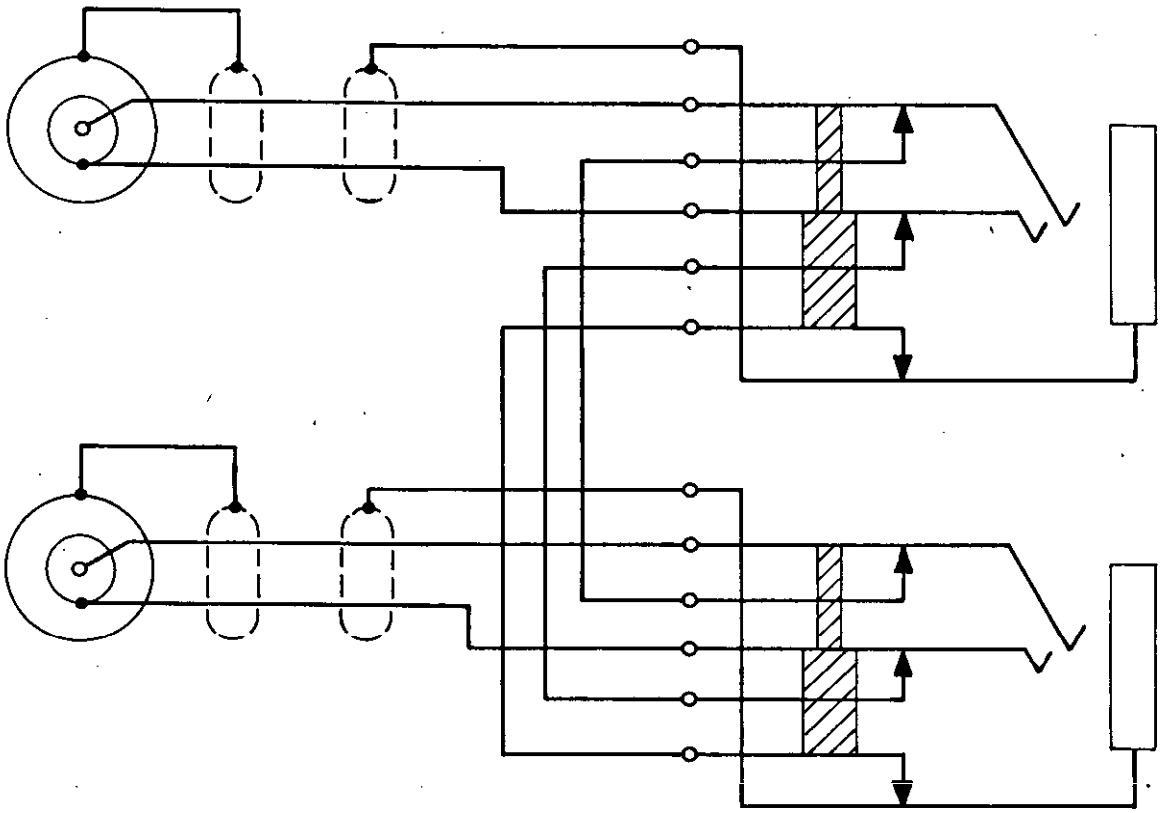


Figure 50. Type III Wideband Patching Module Jack Arrangement.

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8.3.2 Equipment characteristics. The wideband patching modules shall conform to the following characteristics:

8.3.2.1 Type I patching module.

a. Circuit type: 75 ohm unbalanced to ground for the radio multiplex baseband and multiplex supergroup circuits. The baseband jacks may also be used for 124-ohm circuits.

b. Frequency response: Maximum attenuation shall be 0.25 dB at any frequency in the range of 10 kHz to 3 MHz.

c. Return loss: 26 dB minimum from 10 kHz to 3.0 MHz with the output of the normal-through circuit terminal in 75 ohms ±1%.

d. Equipment configuration: The Type I patching module shall consist of coaxial and triaxial jacks arranged as shown in figures 44 and 45. The baseband jacks shall be compatible with PL2-type RF patch plugs. They shall be mounted on 0.62-inch centers. The supergroup jacks shall be compatible with PL3-type RF miniature patch plugs. Supergroup jacks will be mounted in group of four on 0.5-inch centers with 0.62-inch spacing between adjacent groups.

8.3.2.2 Type II module.

a. Circuit type: 135 ohm balanced for multiplex group circuits.

b. Frequency response: Maximum attenuation shall be 0.25 dB for any frequency in the range of 60 kHz to 108 kHz.

c. Return loss: 26 dB minimum from 60 kHz to 108 kHz with the output of the normal-through circuit terminated in 135 ohms ±1%.

d. Isolation: 30 dB minimum between patching jacks and monitor jacks.

e. Equipment configuration: They type II wideband patching module shall consist of two groups of 30 each tip-ring-sleeve jacks arranged as shown in figures 46, 47, and 48. This module shall provide ten duplex line and equipment jack circuits with associated monitor jacks. These shall be in subgroups of six jacks each. Each subgroup of four send and receive, line and equipment jacks and two monitor jacks shall be mounted on 5/8-inch centers with the associated monitor jacks mounted 1 inch below the equipment jacks. Adjacent subgroup of jacks shall be separated by a center-to-center spacing greater than 5/8 inch to aid in identifying associated jacks.

f. The line, equipment and monitor jacks shall be mechanically and electrically compatible with a 310-type telephone plug and shall include the switching logic shown in figures 47 and 48. The connectors on the rear of the module shall be compatible with PL-75 twinax connectors. All jacks and connectors shall be insulated from the module chassis. Sufficient patch cords and connectors compatible with jacks shall be provided.

8.3.2.3 Type III patching module.

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a. Circuit type: 135 ohm balanced for multiplex group ancillary circuits.

b. Frequenc response: Maximum attenuation shall be 0.25 dB for any frequency in the range of 60 kHz to 108 kHz.

c. Return loss: 26 dB minimum from 60 kHz to 108 kHz with the output of the normal-through circuit terminated in 135 ohms ±1%.

d. Equipment configuration: They type III wideband patching module shall consist of 52 each tip-ring-sleeve jacks arranged as shown in figures 49 and 50. This module shall provide 26 line and equipment jack circuits for patching and testing. Each pair of line and equipment jacks shall be mounted on 5/8-inch centers (vertical spacing). Horizontal spacing of adjacent jacks shall be 5/8-inches center-to-center. The line and equipment jack shall be mechanically and electrically compatible with a 310-type telephone plug and shall include the switching logic shown in figure 50. The connectors on the rear of the module shall be compatible with type PL-75 twinax connectors. All jacks and connectors shall be insulated from the chassis. Sufficient patch cords and connectors compatible with jacks shall be provided.

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Chapter IX

9 MISCELLANEOUS EQUIPMENT. The miscellaneous equipment list includes the following:

a. Alarm system.

b. Power supplies.

c. Orderwire ringdown and intercom facility.

d. Signal timing clock.

9.1 ALARM SYSTEM. There are two major units comprising an alarm system. They are:

a. The on-line monitor unit.

b. The station common alarm unit.

9.1.1 On-line monitor unit.

9.1.1.1 Application. The on-line monitor unit is used to automatically evaluate the operating efficiency of a teletypewriter or data transmission media and associated terminal equipment. The unit provides alarm signals for visual and audible indication of telegraph signal quality when the following conditions exist:

a. An "open" data line condition occurs.

b. A "no-transition" condition in the data occurs.

c. The number of distortion hits exceeds, during a preset period, those selected on a hit-to-alarm control. (A hit is a distortion value that exceeds a pre-determined threshold.)

d. The distortion level of a circuit removed from service improves to a preset level of acceptance.

9.1.1.1.1 Alarm signals under conditions of 9.1.1.1(a) or 9.1.1.1(b) also control a step-release pulse to interrupt the send circuitry. The alarm signal is completely isolated from the step pulse. Functional Description: The on-line monitor unit consists of the following modules:

a. Monitor module(s). One monitor module is used per data line to provide output alarm signals for input telegraph signal quality conditions. The number of these modules used in the on-line monitor unit will depend on the requirements of the station.

b. Time base generator. The time base generator provides all necessary timing

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functions required by the on-line monitor unit.

c. Status indicator panel. the status indicator panel provides the visual and audible alarm indication. Reverse logic switches are provided to allow the on-line monitor to provide an alarm when a circuit removed from service improves to a preset level of acceptance. Fan out circuit capabilities for controlling a number of remote alarm indications are also provided for by the panel.

9.1.1.2 Equipment Characteristics. The on-line monitor unit shall conform to the following characteristics:

a. Monitor input characteristics:

(1) Input signal types: asynchronous and synchronous binary codes, capable of accepting all binary code formats.

(2) Input signal levels: Low-level ±6 V dc polar, in accordance with MIL-STD-188-114, 5.2.1.2. High-level (signal power external) neutral or polar by option. Neutral 130 V dc at optionally available 20 or 60 mA. Polar ±60 V dc at 20 mA in accordance with MIL-STD-188-100, 4.3.1.4.

(3) Method of input signal sampling: Bridging mode.

(4) Impedance: 20,000 ohms minimum.

(5) Modulation rate range: Monitors signals at 37.5, 45.45, 50, 61.12, 75, 150, 300, 600, 1200, 2400, and 4800 baud. Signal rates shall be selectable by means of a front panel switch.

b. Alarm functions: Provides visual and audible alarms for conditions as described in 9.1.1.1(a) through 9.1.1.1(d).

c. Alarm function characteristics.

(1) Distortion (telegraph) threshold: Adjustable from 5 percent to 45 percent in discrete steps of 5 percent.

(2) Accuracy of distortion threshold (hit threshold): The unit shall be capable of sensing precent distortion in the range of -2 to +2 from the trus value.

(3) Hit counter (hit-to-alarm control): Alarm after 1 to 10 hits, selectable.

(4) Hit period: Hit counter resets to zero after a 3-minute period.

(5) Alarm response (for all functions): After 300 milliseconds, maximum.

(6) Alarm output (for each funtion): Form-C relay contacts.

(7) Alarm fan-out circuitry per status panel: 10 circuits, minimum.

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d. Step-release pulse interrupt function: Alarm signals under conditions of 9.1.1.1(a) through 9.1.1.1(d) also control a step-release pulse to interrupt the send circuitry. The alarm signal is completely isolated from the step pulse.

e. Timing signal characteristics.

(1) Internal clock: Capable of producing modulation rates in 9.1.1.2(a) long term, 30-day accuracy and stability of 1 part in 10^4; short term, 24-hour accuracy and stability of 1 part in 10^5.

(2) Synchronization capability: Capability shall be provided for synchronizing the internal clock at 96 kHz, minimum frequency, from an external reference source; input interface requirements shall be low-level ± 6 V dc in accordance with MIL-STD-188-114, (5.2.2.1 and 5.2.2.2).

(3) External clock: The on-line monitor unit shall be capable of deriving its timing information from an external clock with 1 Hz of clock output per bit output and with input interface requirements given in (2) above.

f. Power requirements for one monitor module, one time base generator and one status indicator panel: 18 watts, maximum (see 2.3).

9.1.2 Stations common alarm unit.

9.1.2.1 Application. The station common alarm unit is used to provide remote visible and audible indication of communications equipment and circuit malfunctions. The unit consists of a visual alarm section and an audible alarm section. The visible alarm consists of 12 red and 12 amber lights. The audible section consists of an audible alarm assembly which includes a manual reset button. Each audible alarm section shall possess the capability to service two visible alarm sections for a total of 24 circuits. The quantity of visible and audible alarm sections required is therefore facility dependent. A contact closure to ground on the input to an alarm circuit will cause a red light to illuminate and an audible alarm to be heard. When this occurs, depressing the manual reset button will cause the red light to extinguish, illuniation of an associated amber light, and silence the audible alarm. When the alarm is cleared (open contact to ground), will be extinguished.

9.1.2.2 Equipment Characteristics.

a. Types of alarms:

(1) Audible: Tone (variable level).

(2) Visual: Lamps, red and amber.

b. Signals input: Input signals shall be in the form of contact closures to ground.

c. Number of assemblies per alarm unit:

(1) Audible: 1.

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(2) Visual: 12 red and 12 amber, minimum.

d. Alarm release: manual.

9.2 POWER SUPPLIES. There are two major power supply systems as follows:

a. The 48 V dc power supply.

b. The 6 V dc power supply and distribution facility.

9.2.1 The 48 V dc power supply. The power supply consists of the following major components when operated as an uninterruptible floating battery system:

a. Floating type silicon-controlled rectifier units (minimum of two each).

b. Counter EMF (CEMF) cell(s).

c. Battery (24 cells) minimum.

d. 48 V dc to 120 V ac inverter.

9.2.1.1 Application. The 48 V dc power supply is engineered to operate either as a conventional supply utilizing the rectifier units only or as an uninterruptible floating battery system utilizing all equipment listed in 9.2.1.

9.2.1.1.1 The uninterruptible power system is designed to rectify incoming ac power to 48 V dc via solid-state rectifiers and to maintain continuity of electrical power to electrical power to electronic equipment by battery and inverter in an event of failure or deterioration of the ac input power source. Under normal (no power failure) conditions the uninerruptible system functions as follows:

a. The battery which is floated across the rectifier output maintains full charge.

b. 48 V dc is supplied to the electronic equipment from the rectifier output.

c. 48 V dc is supplied to the inverter input.

d. 120 V ac from the inverter output is supplied to the electronic equipment.

Under conditions of power failure the system functions as follows:

e. The battery supplies 48 V dc to the electronic equipment.

f. The battery supplies 48 V dc to inverter input.

g. The inverter supplies 120 V ac to the electronic equipment.

9.2.1.1.2 The CMEF cell(s) is used to regulate the dc voltage output during equalizing charge and float conditions of the battery.

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9.2.1.2 Equipment Characteristics. Each power supply component shall conform to the following characteristics:

9.2.1.2.1 Rectifiers.

a. Input power requirements: All single phase rectifiers shall operate satisfactorily with the input voltage nand frequency given in 9.2.1.2((a)1). All three-phase rectifiers shall operate from power sources listed in 9.2.1.2.1(a)(2) through 9.2.1.2.1(a)(4).

(1) 120 or 240 V ac ±10 percent, 47 to 63 Hz, single phase.

(2) 208 V ac ±10 percent, 47 to 63 Hz, three phase.

(3) 220 or 380 V ac ±10 percent, 47 to 63 Hz, three phase.

(4) 240 or 416 V ac ±10 percent, 47 to 63 Hz, three phase.

b. Battery charging current: Automatically controlled by the rectifier unit.

c. Current limit: Current limiting devices shall be incorporated to protect the rectifiers and prevent erroneous tripping of the ac power input line contractor and dc output circuit breakers. The current limiting feature shall be adjustable from 70 percent to 110 percent (minimum range) of the rectifier rated output.

d. Efficiency and power factor: The rectifier full load efficiency shall not be less than 80 percent. Power factor at full load shall not be less than 80 percent.

e. Regulator: Each rectifier output voltage shall not exceed ±1 percent maximum variation with load ranging from 0 to 100 percent of capacity rating under steady state ac input, transient variation of ±10 percent, and line frequency variation from 47 to 63 Hz. Each rectifier shall maintain a float charge of 2.15 volts per cell and an equalizing charge of 2.33 volts per cell.

f. Load capacity: The capacity of each rectifier unit shall be sufficient to supply the dc electronic equipment load, internal losses of the battery bank, dc input to inverter, and have sufficient extra capacity to bring the battery bank to a full charge.

g. Charging rate: The charging rate should be 1/4, minimum of the average hourly discharge rate for the uninterruptible supply.

h. Modes of operation with a minimum of two units:

(1) Parallel operation: Adjustable load sharing circuit(s) and device(s) shall be incorporated in each rectifier for sharing the dc current within 10 percent of each other when operating in parallel. Upon failure of one unit, this unit shall be automatically switched out of the circuit with the other unit assuming full load.

(2) One unt on-line and one unit in "cold standby": Upon failure of the on-line unit, the defective unit shall be automatically switched out of the circuit and the standby unit shall be automatically switched into the system.

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i. DC filters: Filters shall be furnished with each rectifier to limit audio noise at the output terminals. Ripple content shall not exceed 100 millivolts rms.

j. Controls and meters: All meters, indicators and controls provided as part of the system assembly shall be mounted on the rectifiers in such a manner as to be easily accessible and visible. The rectifiers shall include as a minimum the following:

(1) DC voltmeter.

(2) DC ammeter.

(3) Float-equalize toggle switch.

(4) Float adjust rheostat.

(5) Equalize adjust rheostat.

(6) AC control toggle (on-off auto) switch.

(7) DC output circuit breaker or fused knife switch.

(8) Panel lights (as fail and no output/charge with as AC fail, no DC output, no charge, etc.).

(9) Relays with dry Form C contacts for remote and local alarm functions.

(10) Relays and terminal boards wired to ac toggle switch for local and remote automatic and manual control of the rectifier(s).

9.2.1.2.2 CEMF cell(s): An automatically controlled and switched dry type counter EMF cell(s) shall be provided to maintain the 46 to 52 volts required by the communication electronic equipment during equalizing charge and float conditions. This CEMF cell shall allow 2.33 volts per cell for equalize charging while allowing electronic equipment to function within the voltage range noted above.

9.2.1.2.3 Battery.

a. Capacity: Without the rectifier unit in operation, the battery shall be capable of supplying power for the complete technical load for a period of one hours minimum without the dc voltage dropping below the specified requirements of the electronic equipment. In addition, this requirement shall be met without harmful discharge of any of the cells.

b. Type: Float service type with excepted 20-year service life. All cells shall conform to FED SPEC W-B134C, Class 2, Style A.

9.2.1.2.4 DC-AC Inverter.

a. DC input voltage: 42 V dc to 56 V dc.

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b. Transient voltage (input): 65 V peak maximum for duration of 2 ms minimum shall not cause deviation from specifications or damage equipment.

c. Output and regulation: Inverter output voltage shall be 120 V ac ±4 percent from no load to full load with power factor of 0.75 lag to 1.0 in the temperature range of from 0° to 40° C.

d. Frequency (output): 60 ±2 percent Hz from zero to full load.

e. Freqency stability: ±1 percent.

f. Capacity (volt-amperes): 1 to 10 KVA depending on facility requirements.

g. Distortion, total harmonic: 26 dB minimum for all loads.

h. Efficiency: 75 percent minimum at 50 V dc input and at full rated load.

i. Circuit protection: Each inverter shall contain the following:

(1) DC input circuit breaker.

(2) Fuses to protect internal control circuit.

(3) AC output circuit breaker or fuse(s).

j. Short circuit protection. Inverter shall have automatic current limiting devices rated at 100 percent current rating to protect inverter from short circuit and overloads.

k. Start up. Start up provisions such as manually operated dc input circuit breakers shall be provided. Start up shall be within 1/2 cycle to provide full output voltage when the dc voltage is applied to input.

l. Meters (2 percent accuracy): Each inverter shall contain:

(1) AC ammeter.

(2) AC voltmeter.

(3) DC input ammeter.

m. Alarms: Visual lamps to indicate inverter failure.

n. By-pass switch: A by-pass switch shall be provided which allows commercial power to be connected to the load when the inverter is shut down for maintenance or due to inverter failure.

(o) DC filters: A dc input filter shall be furnished at the input of the inverter to limit audio noise. The ripple content shall not exceed 100 mV rms.

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9.2.2 Six-volt dc power supply assembly and distribution facility. The power supply system consists of the following major units:

a. Six-volt dc power supply assembly.

b. Switching and indicator unit.

c. Ballast lamp distribution module.

9.2.2.1 Application. The 6-volt dc power supply assembly and distribution facility furnishes positive and negative power simultaneously for data, clock timing, and mark hold applications where the maximum current available to any single loop is less than 1 mA. The system is of solid-state design and derives the ±6 V dc by rectifying 120 V ac single-phase power. The distribution of the ±6 V dc to external communications devices is accomplished via protective fuse or ballast lamps or both fuse and ballast lamp. Other features include standby power capability for both positive and negative voltages, meters for indicating voltage and current output, and visual and audible alarms for indicating failure of a dc supply or fuse. The facility is designed for operation in a RED area and is usable in a BLACK area but cannot power both simultaneously. Power supplies used as a signal source must be isolated from clock and equipment power supplies.

9.2.2.2 Equipment Characteristics.

9.2.2.2.1 Six-volt dc power supply assembly. the assembly consists of four 6 V dc power supplies normally utilized as follows:

a. One supply furnishes primary positive-to-ground voltage.

b. One supply furnishes primary negative-to-ground voltage.

c. The other two supplies are used for positive and negative stand-by operation.

9.2.2.2.2 Individual power supplies. Each power supply shall conform to the following characteristics:

a. Output voltage range: 6 V dc continuously adjustable with a minimum range of 5 to 7 V, and a maximum range of 4.5 to 7.5 V.

b. Output voltage adjustment resolution: Adjustable with a resolution of 30 millivolts or better.

c. Output current capacity: 4 amps, minimum, at 50° C, 3.5 amps, minimum, at 60° C.

d. AC power input: 120 or 240 V ac ±10 percent, 47 to 63 Hz, single phase.

e. Output voltage regulation: 0.05 percent, maximum, for input voltage variation as in (d) and for load current variation from no load to 105 percent of rated load.

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f. Response time: 50 μS maximum, for conditions stated in 9.2.2.2.2(e).

g. Ripple and noise: 250 μV rms maximum, with either positive or negtive terminal grounded. The output positive or negative terminal grounded. The output shall be free of rectifier switching spikes for load current and voltage variations defined in above 9.2.2.2.2(c).

h. Output impedance: 0.5 milliohm, maximum, when the load is keyed at 185 Hz.

i. Temperature coefficient: Output voltages shall not change more than 0.015 percent per degree C, for ambient (room) temperature changes of 0° to 50° C.

j. Stability: After 60-minute warm-up, a maximum voltage drift of 0.025 percent in 8 hours.

k. Meters: Each individual power supply unit shall have one dc voltmeter and one dc ammeter accurate to ±2 percent of full scale.

l. Protective devices: Devices shall be provided which protect the individual units under the following conditions:

(1) Primary power supply failure: Automatically switches from primary to standby power supply unit when primary unit malfunctions. Includes individual power supply activation of remote alarm and alarm lamp on the assemby front panel. Relay contact rating for alarm activation shall have 0.25 amps, minimum, at 120 V dc.

(2) Over temperature: Thermostat with automatic reset when over temperature condition is removed.

(3) Internal failure: AC power circuit breaker or fuses.

(4) External overload: Adjustable automatic electronic current limiting.

m. Polarity: The supply shall be designed to permit the selection of either the positive or negative pole for connection to signal ground.

9.2.2.2.3 Switching and indicator unit. The power supply switching and indicator unit serves as a focal point for the interconnection of all components common to the overall assembly, i.e., alarm panel, total current and voltage indicating devices, power supply isolation diodes, and circuitry for power supply switching. The unit shall comply with the following performance requirements:

a. Positive bus: The unit shall be capable of accepting the outputs of two positive power supplies and shall provide for interconnection of the positive power to the ballast lamp distribution modules. Isolation diodes shall be provided between the primary and standby power supply outputs and the input to the positive bus to prevent power feedback to a faulty rectifier.

b. Negative bus: The unit shall be capable of accepting the outputs of two

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negative power supplies and shall provide for interconnection of the negative power to the ballast lamp distribution modules. Isolation diodes shall be provided between the primary and standby supply outputs and the input to the positive bus to prevent power feedback to a faulty rectifier.

c. Ground bus: The unit shall provide for an insulated ground bus conforming to MIL-HDBK-232 requirements for the termination of the return leads of requirements for the termination of the return leads of the positive and negative power supplies. The ground bus shall also provide for the termination of a number 2 insulated external ground wire and other common ground return leads within the unit.

d. Voltage and current monitoring: The unit shall provide for two each of 0 to 10A full-scale meters and two each 0 to 10 V full-scale meters, accurace of 2 percent full scale or better.

e. Alarm: The unit shall accept the external alarm contacts from the two positive and two negative power supplies and shall provide audible failure indicating circuitry to indicate the failure of any one of the power supplies. Upon failure of a power supply the faulty unit shall be automatically removed from the common bus by its internal failure circuitry, a visible indication shall appear on the power supply and a contact closure shall be provided at the power supply terminals. This contact closure shall be used to activate the audible alarm circuitry of the switching and indicating panel. A reset button shall be provided on the switching and indicating panel to disable the audible alarm; however, the visible alarm on the faulty power supply shall remain in an alarm condition until the trouble is corrected.

9.2.2.2.4 Ballast lamp distribution module. The ballast lamp distribution module shall provide for the multiple mounting on slide base type lamp holders and ballast lamps. The unit shall provide the ballast protected distribution of positive and negative power for up to 30 external equipments.

a. Fuse and fuse holders: The ballast distribution module shall be equipped with four horizontal rows of three each miniature fuse holders and fuses of the type which gives a visual indication and activates alarm contacts when a fuse is blown. The visual indication shall be by means of a colored plastic flag in a different color for each fuse ampere rating. The fuses provided with the panel shall be rated at 180 mA.

b. Alarm lamps: One positive and one negative alarm lamp shall be provided. Each lamp holder shall be equipped with a 6-volt lamp.

c. Ballast lamp: Thirty positive and 30 negative ballast lamps shall be provided. The lamp holders shall accept the telephone slide base lamp. Lamps shall be provided for 60 lamp positions. The lamp shall be a type 6c lamp or equivalent.

d. Alarm relay assembly: Separate relay circuity shall be provide for both positive and negative fuse alarm buses. The relay assemblies provided shall be capable of performance from voltage sources of 6, 24, 48, 60, 80 and 120 V dc. A resistance network shall be provided to permit strapping the resistors for any of the above voltages. The relay windings shall be protected by a fuse. The relays shall be equipped with two sets of "C" contacts. One set shall be used to energize the front panel alarm lamp and the other

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set shall be terminated on the terminal block of the module for external connection to a normally open or normally closed alarm circuit. The contacts shall be rated at 250 mA and shall be protected from pitting by use of diodes in the contact circuit.

9.3 INTERCOMMUNICATIONS AND ORDERWIRE EQUIPMENT.

9.3.1 Application. This equipment comprises the terminal equipment for a voice intercommunications (intercom) and orderwire network. The equipment provides technical control and maintenance personnel with a communications capability necessary for effective control of the DCS. A brief description of the intercom and orderwire systems contained in the network follows.

9.3.1.1 Network systems.

a. Intercom system. The intercom system provides necessary communications to permit technical coordination between operating elements within the Technical Control Facility (TCF) and to support operating and maintenance elements located within the vicinity of the TCF. Figure 51 shows a typical DCS intercom system.

b. Orderwire system. The orderwire system provides for voice communications for the exchange of information regarding circuit and facility status, circuit order work, maintenance and trouble conditions, and to satisfy the objectives specified in DCAC 310-50-6. The orderwire system is configured into four independent categories of circuits or networks, although the equipment is to be provided at selected TCFs and PTFs (Patch and Test Facilities) to interconnect the separate networks on a temporary basis. The classification of orderwire circuits as "Local," "Link," "Express," and "System" is based on requirements of DCAC 310-50-6 as reflected below.

(1) Local orderwires. Local orderwires interconnect the serving DCS, TCF and PTF with local users accessing the DCS facility. Typical of such users are the DCS common-user terminals (AUTOVON and AUTODIN), command posts, Tactical Air Control Centers (TACC), Combat Operations Centers (COC), special security elements (NSA and USAFSS), supply computer centers, weather computer and relay centers, and other special operations as required. Local orderwire circuits will also be used to outlying DCS station supporting elements such as transmitter-receiver sites and commercial government cable plants. Local orderwires are designed for two-way automatic signaling between the two end instruments, with manual conferencing to other local, link, express, and system orderwires provided at the servicing DCS, TCF or PTF. A typical local orderwire arrangement is shown in figure 52.

(2) Link orderwires. Link orderwires interconnect terminal PTFs to the major serving TCF and connect major TCFs at both ends of a series of tandem radio relay transmission links. In those instances where the technical control function is absent, where the patch and test operations are extremely small, or where the prime function is maintenance, the maintenance channel may, in fact, be used as the link orderwire channel. Link orderwires are configured as selective calling party-line circuits with intermediate locations (between terminal TCFs) provided with waystation operator access panels. Outstations connected to the link orderwire call specific TCF functional areas, or floor positions at remote TCFs through the use of a third-digit following the two-digit station selector code. The manual conferencing of link orderwires to other link, local system and express orderwires is provided at the master control panel at each serving TCF. A typical link orderwire layout is depicted in figure 53.

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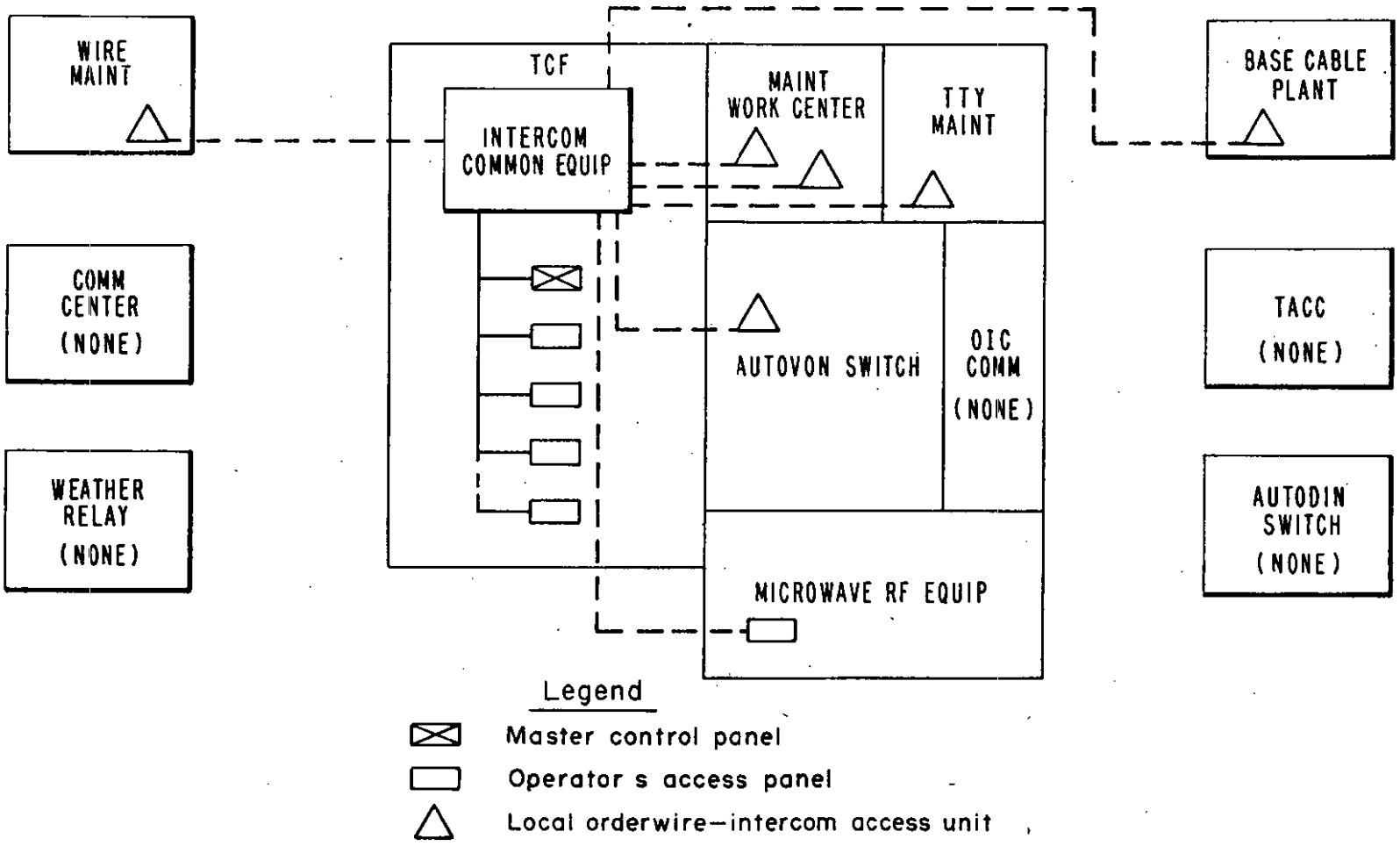


Figure 51. Typical DCS Intercom Layout.

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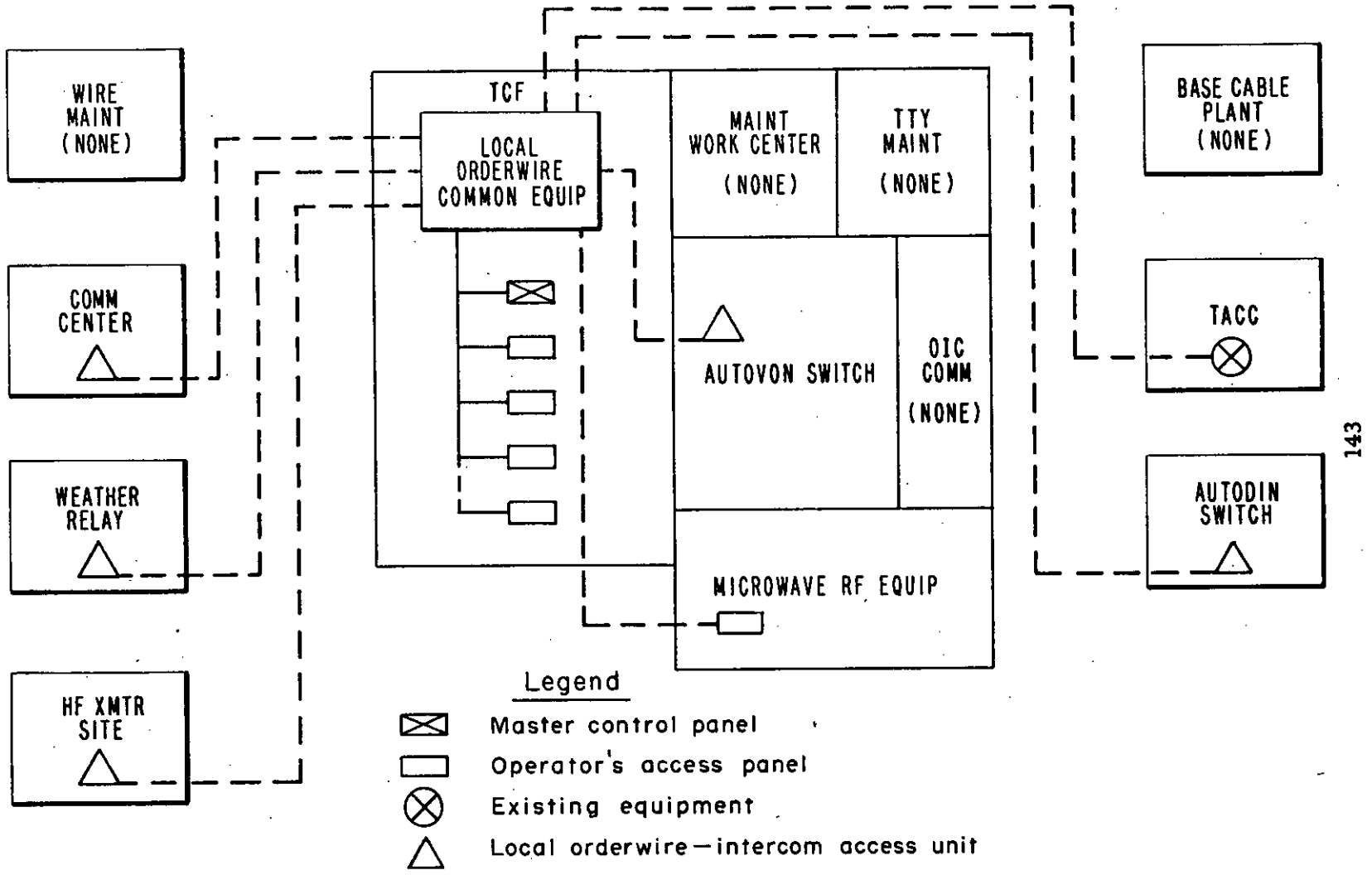
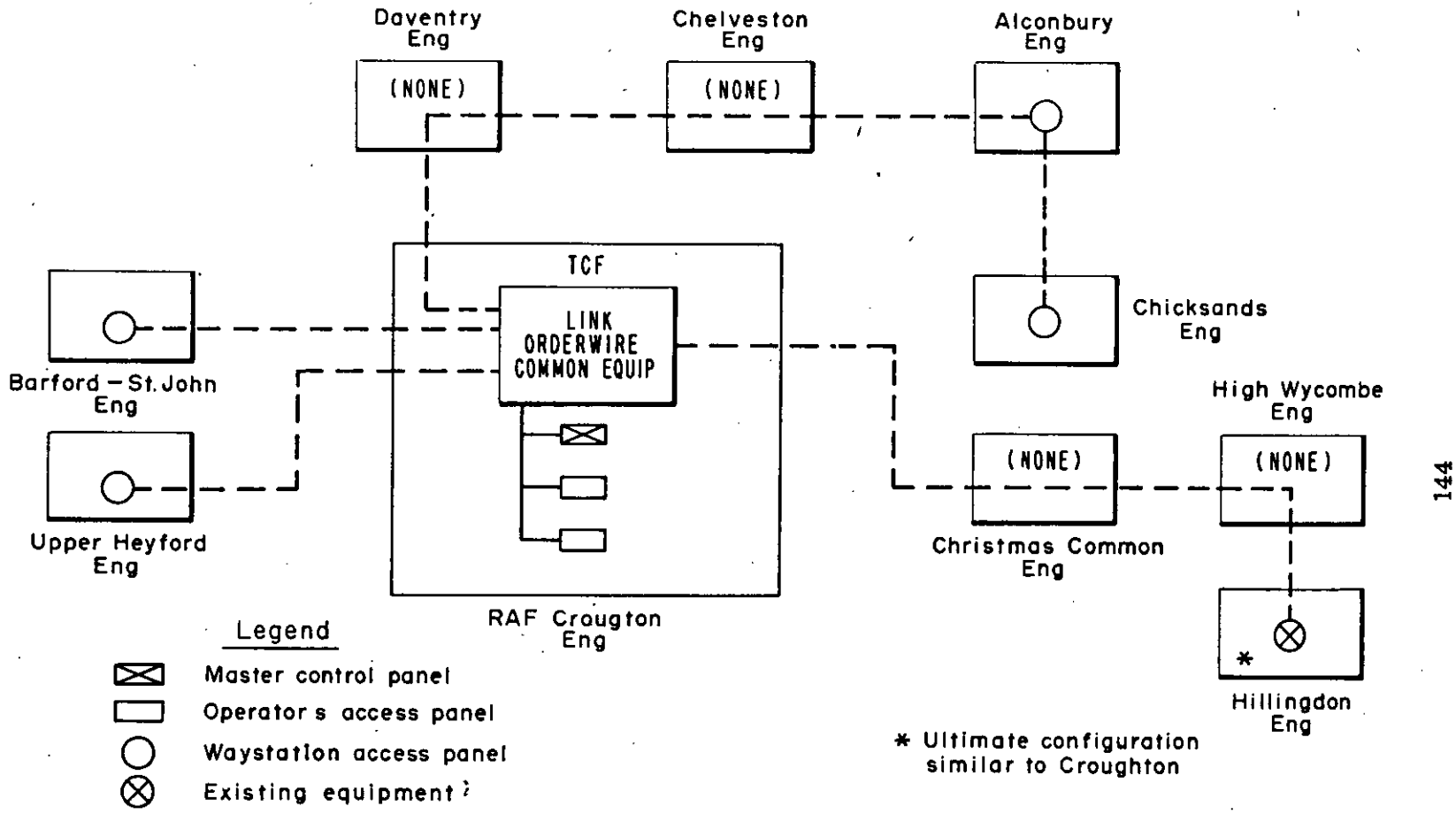


Figure 52. Typical DCS Local Orderwire Layout.

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Figure 53. Typical DCS Link Orderwire Layout.

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(3) Express orderwires. Express orderwires interconnect terminal and nodal DCS stations serving as facility control office (FCO), circuit control office (CCO) or intermediate control office (ICO) which have manual multiplex responsibilities and similar control office responsibilities. Express orderwires will normally inconnect only two nodal points, but are configured as selective calling party-line circuits to accommodate major multiplex breakout points that may exist between the nodal TCFs. In addition to the manual conferencing (see (1) above), the express orderwire is capable of being automatically extended through the nodal TCF to another express orderwire terminated therein. This automatic tandem interconnection of two pre-programmed express orderwires is activated by a discrete two-digit connect code and is deactivated by a specific operator panel on either express orderwire. The selective calling of specific functional areas within a nodal TCF is provided. A typical express orderwire is depicted in figure 54.

a. TCF common equipment.

b. TCF master control panel.

c. TCF operator's access panel.

d. Waystation access panel.

e. Local orderwire and intercom access unit.

(1) The TCF common equipment contains switching and distribution components and other common elements necessary to provide the intercom and orderwire service. It serves as the central terminus for connection to the master control panel, the operator's access panels, the local orderwire-intercom access unit and two- and four-wire circuits to remote locations.

(2) The TCF master control panel is the sole point withn the TCF where the status of the access to all intercom, orderwire, AUTOVON and base PABX circuits are provided.

(3) The TCF operator's access panel provides technical control personnel with access to base PABX, intercom and local, link, express and system orderwire circuits.

(4) The waystation access panel provides remote site operator access to four-wire orderwire circuits. The panel is designed to operate independently of other station equipment and only requires the termination of the four-wire party-line orderwires and access to a source of power. A typical application of the panel is shown in figure 56.

(5) System orderwires. System orderwires interconnect major terminal and nodal TCF's assigned extensive FCO, CCO, and ICO responsibilities within a given geographical area. These voice and data communications circuits provide direct communications for coordination of control office problems between TCF's outside the geographical area served by the express orderwire network. A typical system orderwire layout is depicted in figure 55.

9.3.1.2 Major components. The equipment consists of the following major components together with such other ancillary items as are required to provide a completely functional intercom and orderwire system. The quantity of equipment to be provided will vary with different installations. Typical quantities that may be required are tabulated in table IV.

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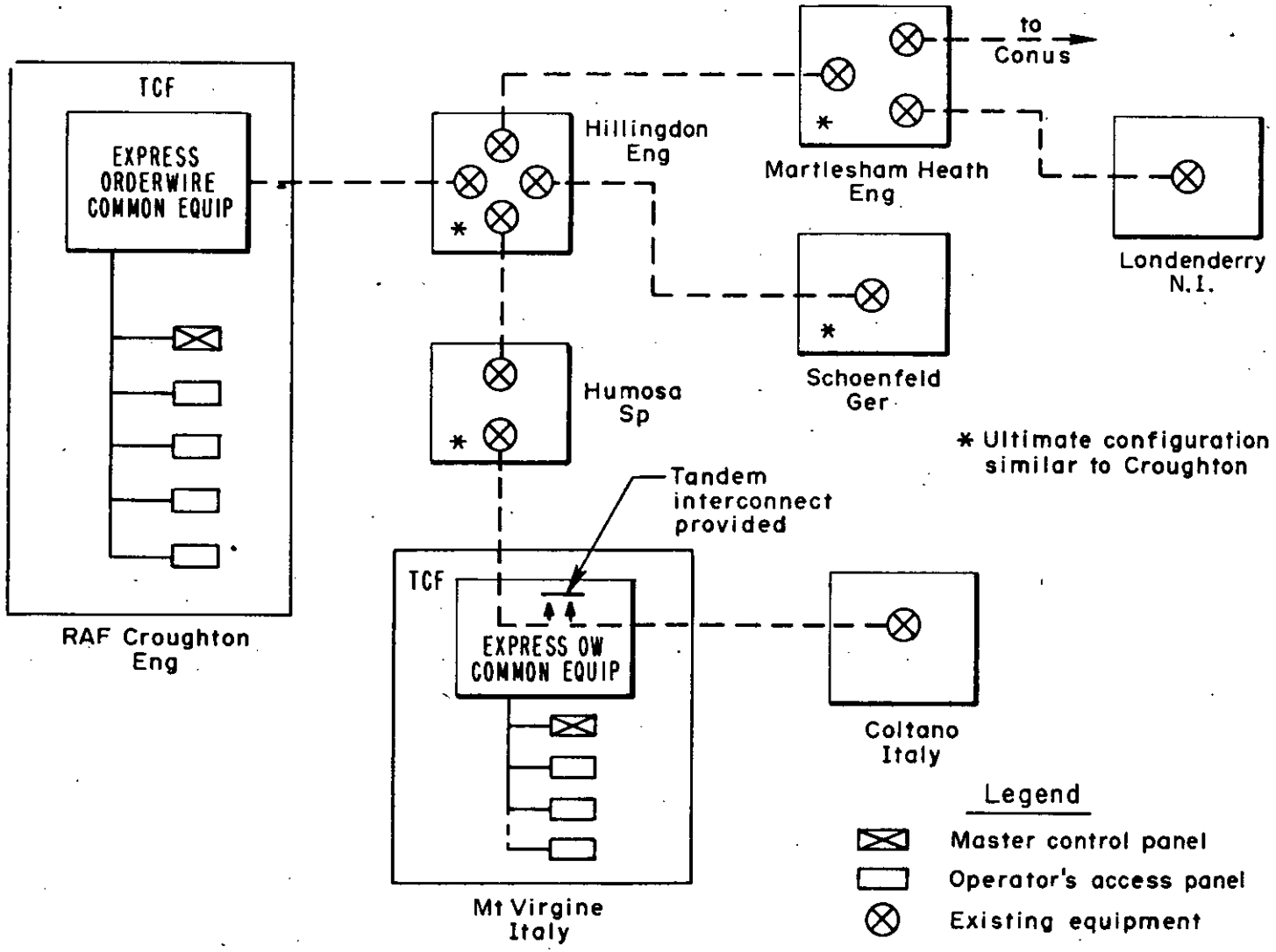


Figure 54. Typical DCS Express Orderwire Layout.

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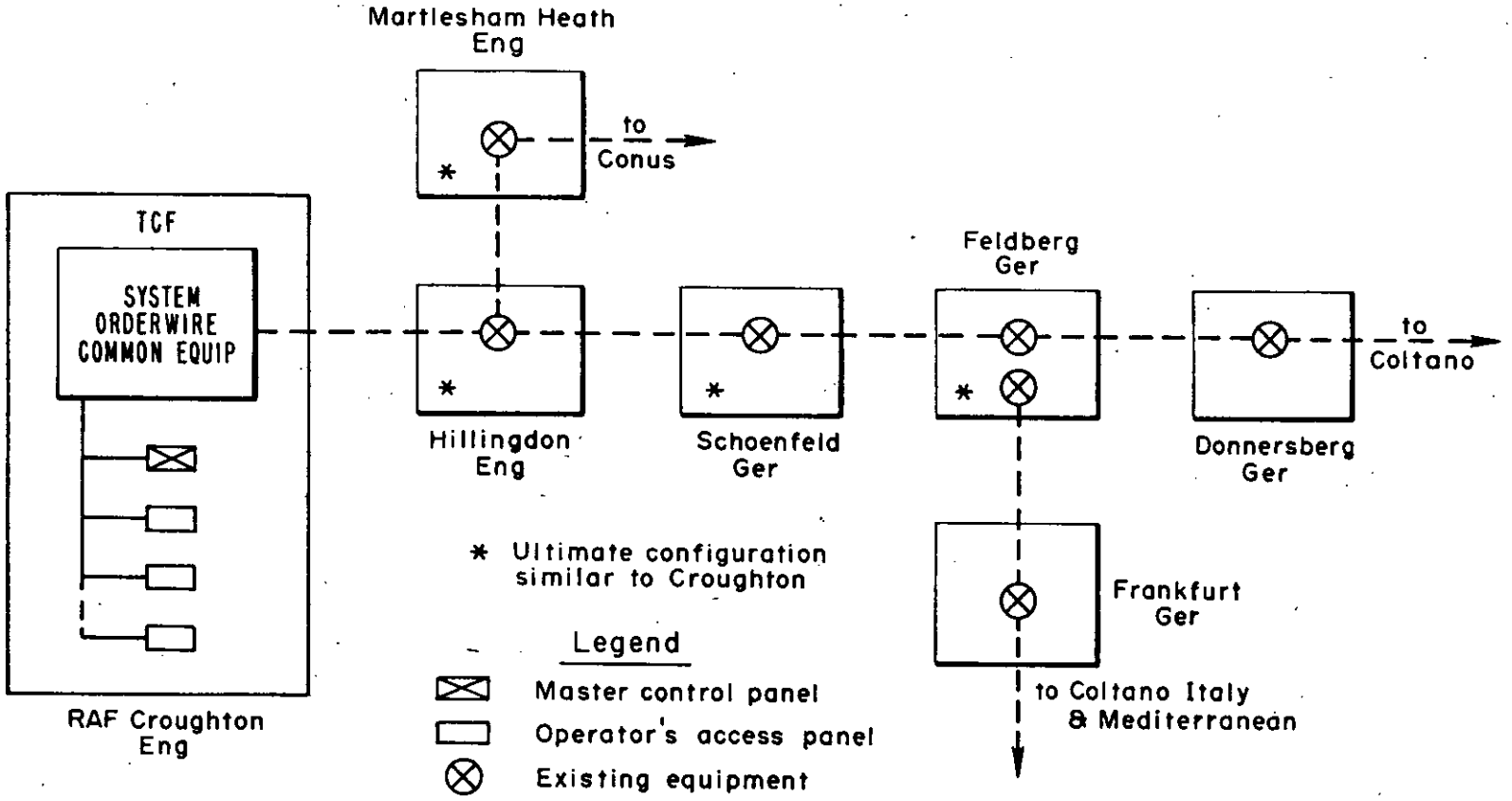
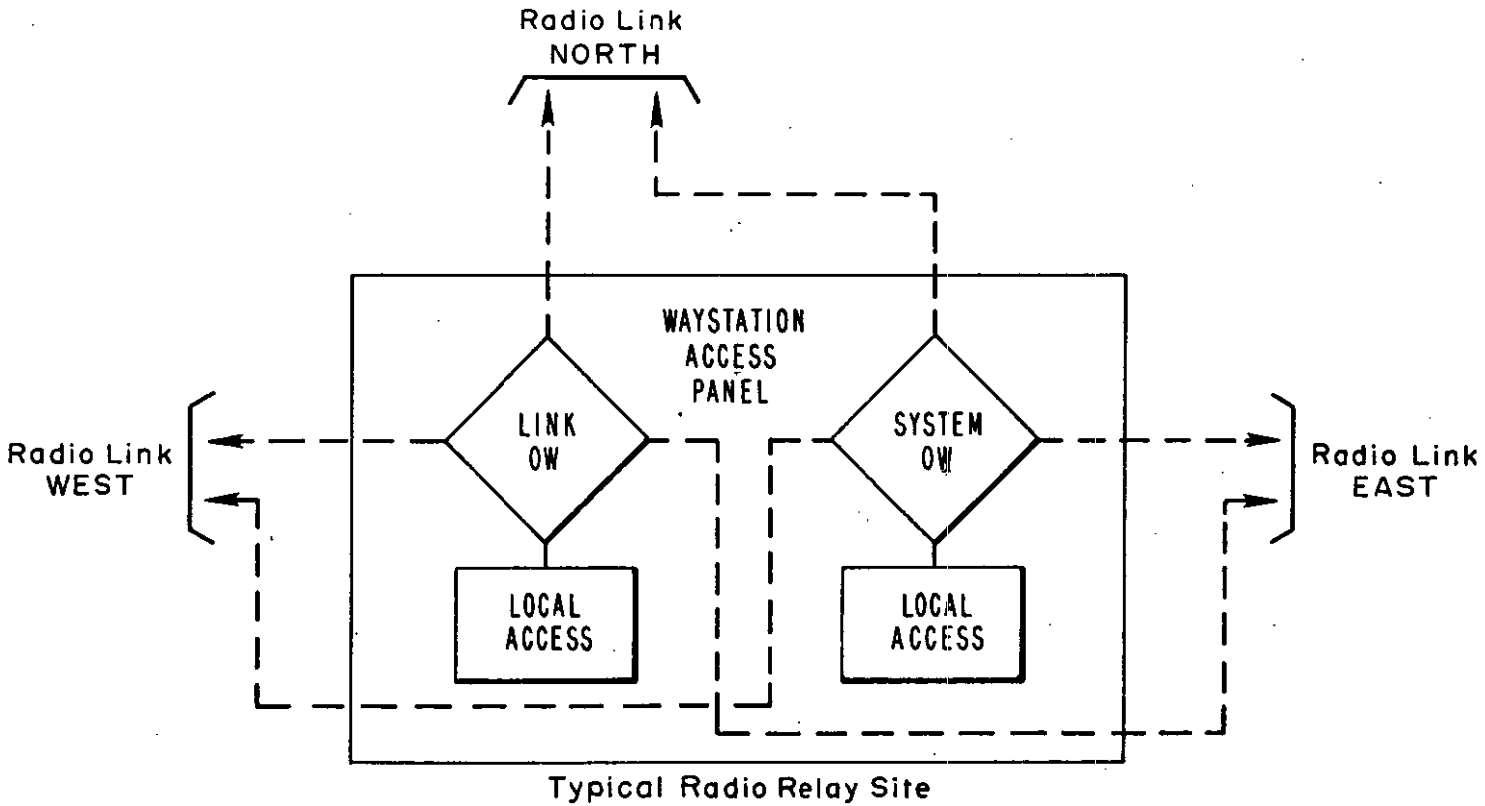


Figure 55. Typical DCS System Orderwire Layout.

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Figure 56. Typical Application of Waystation Access Panel.

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TABLE IV. TYPICAL QUANTITATIVE REQUIREMENTS

STATION / EQUIPMENT		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TCF COMMON EQUIPMENT	SIZE A		X			X																	X	
	SIZE B	X		X			X	X	X		X					X	X	X						
	SIZE C				X					X		X	X	X	X				X	X	X	X		X
	DTMF ORDERWIRE (CKTS)	12	27	10	5	18	14	9	8	7	12	4	8	5	4	3	5	11	3	3	4	5	20	3
	LOCAL ORDERWIRE (CKTS)	9	22	7	12	17	19	18	18	10	10	10	9	12	12	18	20	16	12	9	12	7	8	4
	TANDEM INTERCONNECT (EA)	4	7	0	0	5	0	0	2	0	4	0	1	0	1	0	2	3	0	2	0	0	5	0
MASTER CONTROL PANEL (EA)		1	2	2	1	2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
OPERATOR'S ACCESS PANELS (EA)	10 ORDERWIRE CAPACITY																		6	11	9	16		9
	20 ORDERWIRE CAPACITY	17		18	8					14	22	6	18	9	6	6	15	22						
	30 ORDERWIRE CAPACITY		26			23	26	26	17														26	
WAYSTATION ACCESS PANEL (EA)		2	18	0	7	14	4	12	8	2	10	3	7	5	0	2	2	12	2	6	0	3	10	4
LOCAL ORDERWIRE/ INTERCOM ACCESS UNIT (EA)	DESK MOUNT	9	22	0	4	16	2	12	10	3	2	3	3	4	3	5	8	7	4	2	5	3	3	3
	WALL MOUNT	8	7	0	4	2	9	9	0	2	6	5	3	4	2	5	6	6	5	5	1	2	2	3
	RACK MOUNT	0	2	8	0	4	3	4	3	4	0	0	0	0	0	3	4	2	2	1	3	2	4	0

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(5) The local orderwire and intercom access unit provides lcoal users and supporting elements of the DCS, TCF and PTF with access to either the local orderwire or intercom system, or both. The units are provided only for those locations wherre the subscriber's existing telephone instruments, keysets, or communications console equipment cannot be used to access the local orderwire and intercom circuits.

(6) Details of operation of the major equipment components are specified in section 9.3.2.

9.3.1.3 Guiding Principles or order concepts:

a. All orderwire circuits entering or leaving a DCS station will appear at a common location at the same transmission level within the TCF.

b. The capability will be provided to interconnect any type orderwire circuit appearing in the TCF with any other similar type orderwire circuits appearing at the same location. Interconnect of orderwire circuits will be under the control of the DCS station at the point of interconnection.

c. The link orderwire will be configured for dual appearance in both the Radio Frequency (RF) equipment area and the TCF or PTF.

d. Sufficient quantities of orderwire terminal instruments will be provided to make use of all orderwire circuits terminated in the TCF commensurate with the operational requirement and personnel capability of the TCF.

e. The link, express, and system orderwires serving a specific area will be provided the capability of selectively signaling all TCFs or PTFs served by the circuits.

f. Orderwires will be equipped with visual and audible signaling indicators capable of being remoted.

g. Orderwire circuit performance standards will be in accordance with the DCS technical schedules for the minimum grade of service required.

h. Any equipment required to subdivide the orderwires from the "mission" traffic shall be provided with redundant multiplex and power supplies to increase reliability.

9.3.2 Equipment characteristics. The intercommunications and orderwire equipment shall conform to the following characteristics:

9.3.2.1 Individual equipment requirements.

9.3.2.1.1 TCF common equipment.

a. Arrangement of components: The arrangement of components and equipment between the common equipment and the control and access panels shall be such that intra-TCF cabling is minimized.

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b. Expansion capability: The common equipment shall be modular in contruction to permit the future expansion of the quantity of orderwires in single-station increments up to a maximum of 40 local and 30 DTMF party-line orderwires for size "A" TCFs, a maximum of 20 local and 15 DTMF party-line orderwires for size "B" TCFs, and a maximum of 12 local and 8 DTMF party-line orderwires for size "C" TCFs. Such expansion shall be capable of being implemented at a future date through insertion of plug-in modules or cards into the through insertion of plug-in modules or cards into the pre-wired and fitted common equipment, master control panel, and operator's access panels. The common equipment also shall be initially equipped to accommodate 30 intercom, 1 AUTOVON, and 2 PABX circuits.

c. Flexibility in assignment of circuits: The common equipment shall provide for the flexible assignment of intercom, orderwire, and PABX dial trunks to the various master, operators, and local orderwire or intercom access units at the option of local station operating personnel. Each master control panel and operator's access panel shall be individually cabled to the TCF common equipment to satisfy the requirements for flexibility and the assignment of circuits in multiple to various intra-TCF panels. The requirement for the multiple appearance of orderwire circuits at several panels within the TCF shall not result in a condition where a busy condition at one TCF access panel prohibits simultaneous and full access to the orderwire circuits from any other access panel within the TCF.

d. Tandem interconnect capacity: The common equipment shall be modularly designed to accommodate a maximum of 10 tandem interconnects per station (see 9.3.2.1.2(h)(3)).

9.3.2.1.2 TCF master control panel.

a. Multiple connection: Connection in multiple, one additional panel at the larger size TCFs, shall be provided. For those locations where two master control panels are to be provided, it is sufficient that the conferencing capability specified in 9.3.2.1.2(c) be provided at only one of the two master control panels.

b. Expansion capability: The panel shall be modular in construction and shall be expandable, without the use of special tools or additional wiring, to accommodate a maximum of 30 intercom, 40 local orderwires, and 30 DTMF party-line orderwires. In addition, the panel shall accommodate one AUTOVON and two base PABX telephone access circuits.

c. Conferencing capability: The panel shall be equipped with one each conferencing circuit to permit the manual conferencing of from one to six orderwire, PABX, or AUTOVON circuits. The conferencing shall be accomplished by the master control panel operator through the use of appropriate push-button switches. Patch cords shall not be used. Entry into the conference and lister-only loudspeaker monitoring shall be available at the master control panel. The means shall be provided to permit the master control panel operator to manually release orderwire subscribers from the conference on an individual basis. The release of PABX and AUTOVON subscribers from the conference shall be automatic upon their return to an on-hook condition.

d. Circuit-hold: The panel shall provide circuit-hold for the AUTOVON, base

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PABX, and intercom circuit appearances.

e. Signaling: The panel shall include the standard 12-button DTMF pushbutton keyset and the AUTOVON precedence pushbuttons, and shall also interface with PABX systems which use rotary dial, DC pulse-type signaling. DTMF signaling shall be used on AUTOVON, link, express, and system orderwires. Rotary dial signaling shall be used on intercom and base PABX circuits.

f. Handset: The panel shall be equipped with one push-to-talk noise-cancelling-type handset and its associated cradle. The handset cord shall be of the coiled type with an unextended length of 3 feet maximum, and an extended length of twelve feet minimum. The cord shall be permanently attached to the handset, and the other end of the cord shall terminate with a standard plug. The panel front shall be equipped with a compatible handset access jack which shall also be compatible with and permit the use of the operator's access panel headset specified in 9.3.2.1.2(g) below. When used at the master contol panel, the headset microphone shall be controlled by the footswitch specified in (g) below. The handset shall comply with the electrical-acoustical characteristics specified in Chapter II, of DCAC 370-V165-1.

g. Speaker-microphone: The panel shall be equipped with a speaker-microphone to permit hands-free operation. A handset speaker-microphone switch shall be provided to select the desired mode of operation. A foot-operated switch shall be provided to activate the microphone when the speaker-microphone mode is used. Adjustable microphone sensitivity and speaker output level shall be provided.

h. Capacity: The panel shall accommodate 30 intercom, 40 local orderwires, 30 DTMF party-line orderwires. In addition, the panel shall accommodate one AUTOVON and two base PABX telephone access circuits.

(1) Monitor-talk switch: A monitor-talk switch shall be provided to permit the operator to listen only in the "monitor" position and to permit full operation in the "talk" position.

(2) Types of status indicators: The visual status indicators shall be a steady "on" indication for circuit busy (off-hook), a flashing on-off indication for an incoming ringing signal, and a "wink" flash for a circuit hold condition. An audible indication of an incoming call shall be provided for the AUTOVON, base PABX, local orderwires, and for those link, system and express orderwires where the third digit specifically designates the master control panel. It is required that upon receipt of two or more simultaneous incoming-call visual signals together with the audible ringing signal at a master control or operator's access panel the operator must be able to visually determine which of the incoming calls is specifically directed to that panel. A 5-second "however" type audible signal shall be provided upon receipt of a "ringback while off-hook" signal, accompanied by the visual flashing indication, for all orderwires appearing in the station. The panel shall include a switch to permit the operator to disable the audible signals.

(3) Status indication of tandem interconnect circuits. The panel shall provide a visible indication of the status of up to ten (10) each tandem interconnect circuits provided for the system and express orderwires within the station. A momentary

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(nonlocking) switch associated with each indicator shall provide the operator with the means to manually disconnect the tandem interconnection.

9.3.2.1.3 TCF operators access panel.

a. DTMF requirements: Each panel shall be assigned to programmable DTMF decoders which shall recognize as an incoming DTMF orderwire call a discrete third-digit number specifically assigned to the TCF operating area which the panel serves, i.e., the quality assurance test center, the audio equal level patch area, the wideband patch area, etc. Each panel, and multipled panels, within a given operational area of the TCF will therefore be assigned the same third-digit for each link, express or system orderwire terminated within the panel. The third-digit decoder shall be programmable by local operating personnel.

b. Multiple requirements: Capability shall be provided to operate ten panels connected in muliple within a single operational area of the TCF as defined in paragraph 9.3.2.3.4(a).

c. Panel capacity: The panel shall accommodate one intercom, one base PABX, and thirty orderwire circuits. The panels shall initially be provided in three incrementally-size units to accommodate 10, 20, and 30 orderwires. In addition, the panel shall accommodate one base PABX and one intercom circuit.

d. Circuit-hold: The panel shall include the circuit-hold for the base PABX and intercom circuits.

e. Signaling requirements: The panel shall include the standard 12-button DTMF keyset for signaling on the link, system and express orderwire circuits and shall also interface with rotary dial DC pulse-type signaling on the intercom and base PABX circuits.

f. Handset requirements: The panel shall be equipped for handset operation as specified in A-9.3.2.1.2(f). One each handset shall be provided for use with each operator's access panel. The handset shall be mechanically and electrically interchangeable with the handsets provided in accordance with 9.3.2.1.2(f).

g. Headset requirements: Hands-free operated headsets, equipped with a swivel-type microphone transmitter, shall be provided for use with each operator's access panel. The headset cord shall be extensible and equipped for plug-in interchangeability with the handset specified in 9.3.2.1.2(f). A locking switch to activate or deactivate the microphone shall be provided as a part of the headset assembly. The headset shall comply with the electrical-acoustical characteristics specified in DCAC 370-V165-1, Chapter II.

h. Status indication and operator access: The panel shall be provided with a switch and lamp field to provide operator access to and a status indication of the intercom, base PABX and each orderwire circuit routed to the panel. Operator listen-only, listen-talk options shall be available on a pushbutton basis using either handset or headset option. The visual and audible indication shall be as specified in 9.3.2.1.2(h) above for the base PABX intercom, local orderwire and those link, express and system orderwires which appear at the panel.

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9.3.2.1.4 Waystation access panel.

a. Panel capacity: The panel shall accommodate two party-line link, express or system orderwires, with each circuit accommodating up to three out-station, four-wire circuits.

b. Panel completeness: The panel shall include all required bridges, interface cards, amplifiers, pads, DTMF encoders and decoders, power supplies, busy-tone generators and detectors and ancillary equipment to permit operation without external equipment.

c. Signaling: The panel shall be equipped with a standard 12-button DTMF keyset. Waystation access panels shall respond to a two-digit DTMF selective calling code. The DTMF decoders provided within the panel shall include strapping operations to permit complete flexibility in the assignment of the two-digit station address codes.

d. Status indication and operator access: The panel shall be equipped with a switch and lamp field to provide operator access to, and a status indication of the two orderwires. The visual and audible status indications shall be as specified in 9.3.2.1.2(h) for circuit busy, circuit idle, ringing and "ringback" while "off-hook" signaling.

e. Handset operation: The panel shall be equipped for handset operation as specified in 9.3.2.1.2(f).

9.3.2.1.5 Local orderwire and intercom access unit.

a. Configuration and completeness: The unit shall be configured for either desk-top, wall-mounted, or rack-mounted use and shall be provided with all ancillary termination devices which are required to provide the required service at the subscriber's location without access ot other equipment or power supplies.

b. Circuit provisions: The unit shall include provision for access to termination of one, two-wire, automatic-ringdown, loop-signaling intercom circuit; and one, two-wire, dc dial, loop-signaling base PABX circuit.

c. Circuit hold: The unit shall include circuit-hold for the intercom and PABX circuits.

d. Dial: The unit shall include a standard rotary dial for use on the intercom and base PABX circuits.

e. Handset: The desk-top unit shall be equipped with a handset and cradle assembly with attached cord of the type specified in 9.3.2.1.2(f), except that the cord shall be permenently attached to the unit. The handset shall not be provided with the noise-cancelling or push-to-talk features. The wall-mounted and rack-mounted units shall be equipped with a plug-in handset and cradle of the type specified in 9.3.2.1.2(f) above.

f. Status indication and operator access: The unit shall be equipped with a switch and lamp assembly to provide operator access to and status indication of the intercom, base PABX, and local orderwire circuits. The visual and audible indications

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shall be as specified in 9.3.2.1.2(h).

9.3.2.2 Required network characteristics. The networks shall conform to the following characteristics:

9.3.2.2.1 Intercom.

a. Subscriber access and hold capability: The subscriber accesses the intercom system by operating the single intercom pushbutton appearance of the access unit. As the calling party comes off-hook, the intercom pushbutton lamp is steadily illuminated at the local panel and all multipled panels; and upon receipt of dial tone, the operator proceeds to dial the two-digit address code of the called party. If the called party's intercom drop is on-hook, the called party receives and audible and visual indication of an incoming call and the calling party simultaneously receives a ringback tone at the handset receiver. If the called party's intercom drop is off-hook (busy), a busy signal is returned to the calling party. When the called party answers the incoming call (off-hook), the ring and ringback signals are disabled, the talking path is established, and the intercom lamp is steadily illuminated at the local panel. If during the conversation either the calling or the called party desires to answer another incoming call at the panel, the circuit-hold pushbutton may be depressed to hold the party on the line and to cause a "wink" flashing indication at the intercom pushbutton and at all multipled panels. The call may be reaccessed and the hold released at either the local panel or any multipled panel.

b. TCF access, override and status display: Access to the intercom system from the TCF master control panel shall be as specified in 9.3.2.21(a) except that an override feature shall be provided at the master control panel permitting access to "busy" intercom circuits. In addition, the master control panel shall be provided with an indicator light field which shall display the "idle," "busy," "ringing," or "wink-hold" status of each intercom circuit (See 9.3.2.1.2(h).

c. Conferencing capability: The means to manually establish an intra-intercom conference call, accommodating one to six intercom subscribers, shall be provided under the control of any intercom subscriber. The conferencing shall be available on a dial-up, meet-me type of nonactive conference circuit. Release from the conference shall be automatic upon return to an on-hook condition by the intercom subscriber.

d. General system requirements: The intercom system shall be a 30% internal trunking, switched system providing PABX-type service among 30 subscriber's stations.

9.3.2.2.2 Local orderwire.

a. Type of service: The local orderwire shall provide two-way automatic signaling type service between intra-base communications users and the TCF, i.e., the seizing of the circuit at the TCF or at the user's station shall automatically ring the other end location. Whenever possible, existing user communications consoles, keysets, and telephone instruments shall be used to provide user access to the local orderwire.

b. User access, "busy" indication and ringing: The user accesses the local orderwire by operating the appropriate local orderwire pushbutton at the master control, operator's access or local orderwire intercom units, and going off-hook. The calling party

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is, therefore, directly connected via the orderwire line unit to the orderwire, and the indicator light will indicate "busy" at the local position and at all multiple appearances of the orderwire. Automatic ringing is initiated and the calling party receives the ringback tone at his handset receiver. At the called position, the audible and visual ringing signals are activated; and when the operator goes off-hook, ringing is disabled.

c. Multiple connection capability: All local orderwires shall appear at the master control panel within the TCF. The multiple connection of each local orderwire at up to three operator's access panel areas and the enabling of audible ringing at one or more of the multiple appearances, through optional strapping, shall be provided. Re-ringing shall be inhibited, when, at the conclusion of a call, one party return on-hook prior to the other party returning on-hook.

d. Conferencing capability: The means to conference local orderwires shall be provided as specified in 9.3.2.3.2 below.

e. Common equipment compatibility requirement: Intra-building local orderwire cabling may be routed directly from the access units to the orderwire common equipment. The orderwire common equipment shall include the necessary components which are required to effect compatibility with subscriber's existing equipment utilizing the various types of signaling specified in MIL-STD-188-310.

9.3.2.2.3 Link orderwire.

a. Type of service: The link orderwire shall provide DTMF selective-calling, party-line service between subscribers which are interconnected via a common four-wire carrier channel.

b. Subscriber access, calling procedure, status lights, ringback tone: The subscriber accesses the link orderwire by operating the desired pushbutton at the master control, operator's access or waystation access panel. When the calling party goes off-hook, he is immediately connected to the orderwire and the line lamp will be illuminated, indicating a "busy" condition for that link orderwire at all subscriber panels, both local and remote, which have access to the orderwire. The calling party then proceeds to sequentially depress the desired two-digit DTMF code for the called station. This completes the calling procedure to those subscribers provided with waystation access panels and for all other subscribers not equipped for third-digit intra-station selection and recognition. For those TCFs equipped with master control panels and operator's access panels, the calling party proceeds to depress the third-digit to select the specific subscriber's position within the called TCF. The DTMF tone-pairs are transmitted over the orderwire where they activate the particular detectors which are set to receive them. Recognition of the two-digit station code at the called TCF equipped with master control and operator access panels causes an unassigned ringing visual indication at all TCF panels, while visual assigned ringing and audible signal is activated only at that panel designated through recognition of the third digit. A ringback tone is returned to the calling party until the called party answers the call by depressing the flashing pushbutton and going off-hook, at which time the ring and ringback tones are disabled and the indicator lamps return to a steady "busy" indication. If the called party fails to answer the incoming call (go off-hook), the ringing and ringback tone shall be automatically disabled (timed-out) after an interval of from 20 to 40 seconds. This interval shall be

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adjustable by local operating personnel.

c. Ringback-while-off-hook capability: The ability to re-ring an off-hook subscriber (ringback while off-hook) shall be provided. The ringback may be initiated by either the called or the calling party by transmitting the appropriate two- or three-digit code. The ringback shall cause the return of the visual and audible "howler" type ringback indications for a period of 5 seconds.

d. Conferencing capability: The conferencing of multiple subscribers on a single orderwire circuit, by successively transmitting the station calling codes of the subscribers to be conferenced, shall be provided. Other conferencing capability shall be provided as given in 9.3.2.3.2. The maximum interval required between successive station calling codes shall not exceed one second.

e. Supervisory tone capability: A 2600-Hz supervisory tone shall be used to provide a busy indication at all subscriber panels on the party-line orderwire circuits. A "tone-on busy" signaling sense, in addition to the capability to disable the supervisory tone generation and detection circuits, shall be provided as strappable options for each party-line orderwire. The 2600-Hz tone generation and detection equipment shall be arranged to enable transmission of the supervisory tone from the first subscriber (calling party) to go off-hook and to inhibit transmission of the supervisory tone from subscribers (called parties) who subsequently go off-hook on the same party-line network. The supervisory tone shall not interfere with normal speech on the orderwire.

f. The maximum inter-digit interval between successive digits of a station calling code shall not exceed five seconds. All DTMF decoders shall automatically reset to await receipt of the first digit after expiration of this five-second interval.

9.3.2.2.4 Express orderwire: The express orderwire shall provide service identical to that specified in 9.3.2.2.2(b) through 9.3.2.2.2(e). In addition, the expess orderwire system shall provide the following:

a. Equipment shall be provided to automatically connect, in tandem, two preprogrammed orderwires which terminate within the TCF. One such tandem interconnect shall be provided for each two orderwire circuits (up to a maximum of ten tandem interconects per station), enabling any subscriber on either orderwire circuit to effect the automatic connection through the transmission of a discrete two-digit interconnect code. Automatic disconnect shall be effected by the transmission of a discrete two-digit disconnect code by any subscriber on either orderwire. The equipment shall permit local operating personnel to change, on a semi-permanent basis, the tandem assignment of the orderwires as usage or requirements may dictate.

b. Status display and manual orderwire disconnect capabilities: The means to visually observe the status of, and to manually disconnect, each tandem interconnection shall be provided at the master control panel.

c. Controlled access capability: Access to a busy orderwire through automatic tandem interconnect shall be prohibited where the called orderwire utilizes the supervisory signaling option specified in 9.3.2.2.3(e). When the called tandem orderwire is busy, the calling orderwire subscriber shall be automatically routed to the master control panel

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for subsequent action.

9.3.2.2.5 System Orderwire. The system orderwire shall provide service identical to that specified for the express orderwire.

9.3.2.3 General requirements.

9.3.2.3.1 External access requirements:

a. Access to the AUTOVON system. Access to the AUTOVON system shall be provided at the master control panel. The existing four-wire AUTOVON line termination equipment located with the TCF shall continue to be used, with the net result being the substitution of the master control panel for the existing AUTOVON telephone instrument. The master control panel shall therefore include the necessary components required to permit handset and headset access directly to the AUTOVON system in accordance with the requirements of DCAC 370-V175-6.

b. Access to private branch exchange (PABX). Access to the local base telephone system shall be provided at the master control panel (two lines), the operator's access panel (one line), and the local orderwire intercom access unit (one line). The access shall be to two-wire dc dial loops routed through the TCF and interfaced with the local PABX in accordance with paragraph 5.1 of MIL-STD-188-310.

9.3.2.3.2 General conferencing requirements: With the exception of the capability specified in 9.3.2.2.1(c) and 9.3.2.2.3(d) the conferencing specified in 9.3.2.1.2(c) shall be manually accomplished at the master control panel. A request for a conference call may originate via AUTOVON, PABX or orderwire circuits and is directed to the controller at the master control panel. The controller will then successively call each desired conferencing circuit until all desired parties are connected thereto. The controller may participate in the conference via the loudspeaker. Capability shall be provided for the subscriber who requested the conference to recall the controller and request that the conference be disconnected at its conclusion.

9.3.2.3.2 General selective requirements.

a. DTMF selective signaling. Dual-Tone Multi-Frequency (DTMF) address signals shall be used for selective signaling on the link, express and system party-line orderwire networks and for AUTOVON signaling. Details are defined below:

(1) Signaling codes: The transmitted DTMF signaling codes shall conform to the CCITT nominal 4 x 4 multi-frequency code as follows:

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	Digit	Frequency - Hz
	1	697 + 1209
	2	697 + 1336
	3	697 + 1477
	4	770 + 1209
	5	770 + 1336
	6	770 + 1477
	7	852 + 1209
	8	852 + 1336
	9	852 + 1477
	0	941 + 1336
	-	941 + 1209
	A	941 + 1477
AUTOVON ONLY	FO	697 + 1633
	F	770 + 1633
	I	852 + 1633
	F	941 + 1633

(2) Signaling-tone frequency: The signaling tones shall not vary more than 1.3 percent from the assigned frequencies.

(3) Signaling-tone level: Each signaling tone shall be injected on the line at a level of -16 dBm0 $\pm$0.5 dB (composit level of -13 dBm0 $\pm$1 dB for two tones).

(4) Tone receiver amplitude response: The DTMF receiver shall respond to tones over a voltage range of +5 to -22 dBV per frequency (across 600 ohms) with a maximum difference of 8 dB between the frequency pair.

(5) Tone receiver sensitivity: The receiver shall operate properly with a signal-to-noise ratio of 22 dB per frequency (thermal noise, flat weighting) with minimum allowable received signals and maximum allowable frequency error.

(6) Tone receiver frequency response: The receiver shall operate properly with tone pairs falling with $\pm$1.5 percent of the nominal frequencies.

(7) Tone receiver time response: The receiver shall respond to dual tone-pair signals havings a minimum duration and a minimum interpulse interval of 69 $\pm$9 ms (6.5 to 8.2 digits per second).

(8) Tone receiver decoder: The DTMF receiver decoder shall not be influenced by speech at normal levels and shall check that two, and only two frequencies are present for each digit; one from each group, high and low (a bonafide signal).

b. Supervisory (busy) signaling. A 2600 $\pm$5 Hz signal shall be used as a supervisory (busy) signal for the link, express, and systems party-line orderwire circuits. The equipment shall include a strappable option which will permit local operating personnel to select either the "tone-on busy" signaling sense or which will permit the disabiling of the supervisory tone on those links where existing subscriber equipments will not permit its use. The supervisory signal shall be injected on the line at a level of -20

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dBm0 ±1.5 dB.

9.3.2.3.4 Multiples, appearances and extensions. The multiples, appearances and extensions specified below shall be provided without degrading circuit quality.

a. Number of appearances: An appearance shall normally be provided at each of nine TCF operating areas for every link, express and system orderwire accessing the TCF. Anyone of the nine functional areas may be selectively called through the use of the third-digit code described in 9.3.2.2.3(b). Assignment of the third-digit to the TCF functional areas is standardized as follows:

Third Digit	TCF Functional Area
1	Coordinator Reporting Position
2	Quality Assurance Test Center
3	Audio Equal Level Patch Area
4	Audio Primary Patch and Test Area
5	DC Primary Patch and Test Area
6	Wideband Patch Area
7	DC Circuit Patch and Test Area
8	Station Common Equipment Area
9	Secondary Patch Facility (AUTOVON and AUTODIN)

b. Access panels and multiples: Operator's access panels shall be provided for all TCF functional areas except the coordinator reporting position where the master control panel shall be provided. The means shall be provided to selectively multiple two master control panels at the reporting position. A maximum of ten operator's access panels may be used in multiple within any one functional areas.

c. Local orderwire appearances: An appearance shall be provided at each master control panel for every local orderwire accessing the TCF. In addition, the means to selectively multiple each local orderwire at up to three additional operator's access panels shall be provided.

d. PABX appearances. An appearance shall be provided at each master control panel for two base PABX dial telephone circuits. In addition, the means to selectively multiple either of these PABX circuits to the operator's access panels of eight TCF operating areas shall be provided.

e. Waystation appearances: An appearance shall be provided at each waystation access panel for two party-line orderwire circuits accessing the site. The means to multiple two additional waystation access panels at a site shall be provided.

9.3.2.3.5 Alarms. The TCF common equipment shall include visual alarms for all significant or pivotal components to assist in isolating equipment malfunctions. All alarms shall also provide closures to permit remoting to the station common alarm bay.

9.3.2.3.6 Marking and Designation.

a. Visibility of scales, gradicules and functional markings: Readable from a distance of 3 feet with each pushbutton and lamp clearly distinguishable.

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b. Switch-lamp lenses: Switch-lamp lenses shall be color-keyed as follows:

Circuit	Lens Color
Local Orderwire	Amber
Link Orderwire	Green
Express Orderwire	Blue
System Orderwire	Red
Intercom	Clear
PABX	Clear
AUTOVON	Clear

Sufficient lenses shall be supplied with each access panel to permit any possible mix of orderwire circuits.

c. Intercom subscriber directory: A designation plate or directory shall be provided on each master control panel and operator's access panel to identify each of the 30 intercom subscribers and their assigned two-digit number. The design shall permit local operating personnel to make changes to the directory with only minor disassembly and reassembly of parts.

9.3.2.3.7 Interface requirements. The requirements given below are for interfaces where the intercom and orderwire system connect to transmission lines of the TCF. Interface parameters internal to the intercom and orderwire system are not specified.

a. TCF Common equipment.

(1) Interface: Audio equal-level test bay.

(2) Level: 0 TLP.

(3) Impedance: 600 ohms $\pm$10 percent, balance from 300 to 3400 Hz.

(4) Return loss: 26 dB minimum from 300 to 3400 Hz.

(5) Longitudinal balance: 40 dB minimum from 50 to 4000 Hz.

b. Waystation access panels.

(1) Interface: Directly with four-wire carrier circuits.

(2) Level: 0 to -16 TLP (transmit) and 0 to +7 TLP (receive) (adjustable or strappable).

(3) Impedance: 300, 450, 900 or 1200 ohms (strappable).

(4) Return loss: 26 dB minimum from 300 to 3400 Hz.

(5) Longitudinal balance: 40 dB minimum from 50 to 4000 Hz.

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c. General interface requirements.

(1) Sidetone: With limits of 35 to 40 dB and provision to disable sidetone internally within the equipment.

(2) PABX termination: Equipment shall accommodate the termination of PABX dial lines as specified in paragraph 9.3.2.1.2(c), 9.3.2.1.3(c), and 9.3.2.1.5(b).

9.3.2.3.8 VF channel characteristics. VF channels of equipment listed in 9.3.2.1.2 shall conform to the following characteristics:

a. Frequency response: ±0.25 dB from 300 to 3400 Hz referred to 1000 Hz.

b. Distortion, total harmonic: 40 dB minimum over the normal operating range of input and output levels and over the frequency range of 300 to 3400 Hz.

c. Noise output: -70 dBm maximum from 300 to 3400 Hz (flat weighting).

d. Surge protection: (see paragraph 2.2).

9.3.2.3.9 Power requirements.

a. TCF common equipment, master control panel and operator's access panels: 48 V dc ±10 percent with suitable overload protection (see paragraph 2.3).

b. Waystation access panel and orderwire and intercom unit: These units shall operate properly using any of the optional sources (by strapping) listed below.

(1) 120 V ac ±10 percent, single-phase.

(2) 230 V ac ±10 percent, single-phase.

(3) 48 V dc ±10 percent.

(4) 24 V dc ±10 percent, containing 100 mV, maximum peak-to-peak ripple and noise.

Note: Maximum noise and frequency range for 9.3.2.3.9(a), (1), (2), (3) are as defined in paragraph 2.3.

9.4 SIGNAL TIMING CLOCK.

9.4.1 Application. The signal timing clock is a crystal oscillator that provides high accuracy reference-controlled timing signals for all of the synchronous telegraph and data equipment. The clock is capable of automatically synchronizing to an input reference signal from an external master reference of greater accuracy (cesium, rubidium, or Loran C) and furnishing a regenerated output timing signal. There are three crystal oscillators in the clock system whose output signals are continuously compared in amplitude and frequency for determination of their accuracy. In case of failure of any of the signals to meet comparison tolerances to other signals an alarm will activate indicating operational

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degradation. If the primary oscillator fails, it is switched out automatically and a second oscillator assumes control. The device is completely self-contained, but of modular construction. It contains the following basic modules:

a. Crystal oscillator module.

b. Clock control.

c. Output timing drive.

The crystal oscillator module contains three identical crystal oscillators on separate cards. Switching out of defective oscillators is done by the clock-control module. The output timing module furnishes expanded output driving capability.

9.4.2 Equipment characteristics. The signal timing clock shall conform to the following characteristics:

9.4.2.1 Crystal oscillator.

a. Accuracy, long term: ± 5 parts in 10^{10} when referenced to a distant clock.

b. Stability (long term): One part in 10^{9} per day in the temperature range of 0° to 50°C with no reference data input.

c. Phase jitter: Less than $\pm$ 0.85 us.

d. Half-bit drift period: Minimum 24-hours at 2400 baud without reference signal.

e. Reference data speeds: 75, 150, 300, 600, 1200, 2400, 4800 or 9600 baud.

f. Reference input signal requirements: Either low-level or high-level as defined below:

(1) Signal levels: High-level, neutral 130 V dc at 20 mA or 60 mA or optional polar $\pm$60 V dc at 20 mA compatible with MIL-STD-188-100 (4.3.1.5.1 and 4.3.1.5.2), but with distortion values up to 48 percent. Low-level 6 V dc compatible with MIL-STD-188-114 (5.2.2.1), but with distortion values up to 48 percent.

(2) Input impedance: High-level 1500 to 3000 for polar signals: 1500 to 6500 ohms for neutral signals. Any value in ranges shall be available by adjustment of controls and strapping. Low-level in accordance with MIL-STD-188-100 (5.2.2.2).

(3) Sensitivity: In accordance with MIL-STD-188-100 (4.3.1.4.6 for high-level signals, in accordance with MILSTD188-100 (5.2.2.2) for low-level signals.

(4) Sense: Provisions shall be included to obtain synchronization on signals of either polarity.

g. Distortion (digital) output: When synchronized to a reference, the

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regenerated output shall be unaffected by reference signal input peak distortion of up to 48 percent.

h. Timing output speeds: 75×2^n over the range of 75 Hz to 614.4 kHz, where n is a positive interger, with one additional output frequency of 96 kHz.

9.4.2.2 Output timing drive module.

a. Output requirements: Each output timing drive module shall be capable of accepting up to four of the above (9.4.2.1h) timing output speeds. The output timing drive module shall provide up to four outputs for any selected input frequency.

b. Timing output mode: Low-level, polar square wave, and short circuit proof. Each line driver card should adapt to balanced or unbalanced load conditions. If an unbalanced load is applied to the driver output, the line driver logic should adapt to the unbalanced generator requirements of MIL-STD-188-114, (5.2.1). If a balanced load is applied, the driver logic should be capable of conforming to the balanced generator requirements of MIL-STD-188-114, (5.1.1). Either option shall be selectable without special programming.

c. Timing driver capability: 1.0 mA per selected speed for ten circuits.

d. Expansion of drive capability: Expandable in increments of 10 by use of plug-in cards.

9.4.2.3 Clock control module.

a. Switching: No phase discontinuity when switching from a primary oscillator to a standby unit and the module shall switch automatically if distortion exceeds a selected maximum (see 9.4.2.3(b)), or if primary reference is lost.

b. Distortion threshold, digital: Adjustable from 10 percent to 49 percent each channel.

9.4.2.4 Power requirements: 50 watts maximum from uninterruptible supply (see 2.3).

Custodian
Army - SC
Navy - EC
Air Force - 90

Preparing Activity:
Air Force - 90

Review Acitivities:
Army - SC, EL
Navy - AS, YD
Air Force - 1, 11, 13, 17, 80, 89

User Activities:
Army - SC, EL
Navy - YD
Air Force 90

Other Interest:
JSC - J6
NSA - NS
TRI-TAC

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